DEPARTMENT OF

**INDUSTRIAL ENGINEERING & MANAGEMENT**

# OPERATIONS MANAGEMENT

Question Bank with Model Answers

| | | | |
|---|---|---|---|
| **Course Title** | Operations Management | **Course Code** | IM352IA |
| **Semester** | V | **Units Covered** | Unit I to Unit V |
| **Compiled From** | CIE and SEE papers, 2019–2026 | **Questions** | Part A: 118 Part B: 102 |

**Notes:**

1. Questions are compiled from the CIE and SEE question papers of 2019–2026. The unit and the source paper are shown below each question.
2. Part A contains fill-in-the-blank and short-answer questions. Part B contains descriptive and numerical questions (4 to 20 marks).
3. Model answers are written to the depth expected for the marks allotted. Numerical answers show the complete working, and the final result is stated after **Answer:**.
4. Where a question repeats an earlier one, the answer refers to the earlier question (the reference is a link).
5. M: Marks; BT: Bloom's Taxonomy level; CO: Course Outcome. Numerical answers are worked with the formulas of the prescribed textbook.

| Sl. No. | Questions | M | BT | CO |
|---|---|---|---|---|
| | **PART – A** | | | |
| **1** | Operations ensure the efficient use of _____ to meet customer demands.<br>Source: Unit I \| CIE-I, 25/11/2024 | 01 | 1 | 1 |

**Answer:** resources

An operation is a group of resources (workers, managers, equipment, facilities, materials, land and energy) performing all or part of one or more processes. Operations management ensures that these resources are used efficiently to meet customer demand.

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| **2** | A supply chain includes the flow of materials, information, and _____ across organizations.<br>Source: Unit I \| CIE-I, 25/11/2024 | 01 | 1 | 1 |

**Answer:** services

Supply chain management is the synchronization of a firm's processes with those of its suppliers and customers to match the flow of materials, services and information with customer demand.

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| **3** | Effective supply chain management ensures timely delivery and reduces _____.<br>Source: Unit I \| CIE-I, 25/11/2024 | 01 | 1 | 1 |

**Answer:** cost (waste and unnecessary cost)

Every activity in the supply chain should add value to the preceding activities; waste and unnecessary cost are eliminated, so effective supply chain management gives timely delivery at lower cost.

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| **4** | Competitive priorities like cost, quality, and _____ determine how a company competes in the market.<br>Source: Unit I \| CIE-I, 25/11/2024 | 01 | 1 | 1 |

**Answer:** time (the fourth capability group is flexibility)

The nine competitive priorities fall into four capability groups: cost, quality, time and flexibility.

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| **5** | The Fourth Industrial Revolution integrates technologies like IoT, _____, and robotics into operations.<br>Source: Unit I \| CIE-I, 25/11/2024 | 01 | 1 | 1 |

**Answer:** Artificial Intelligence (AI)

The Fourth Industrial Revolution (Industry 4.0) integrates cyber-physical technologies into operations: the Internet of Things (IoT), artificial intelligence and machine learning, robotics and automation, big data analytics, cloud computing, additive manufacturing (3D printing) and digital twins.

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| **6** | Processes can be categorized as _____ or supporting based on their direct impact on the final product.<br>Source: Unit I \| CIE-I, 25/11/2024 | 01 | 1 | 1 |

**Answer:** core

Processes are of two main types: **core processes**, which deliver value directly to external customers (customer relationship, new service/product development, order fulfillment and supplier relationship), and **support processes**, which provide vital resources and inputs to the core processes.

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| **7** | Differentiate between order winner and order qualifier.<br>Source: Unit I \| CIE-I, 25/11/2024 | 02 | 2 | 2 |

| **Basis** | **Order winner** | **Order qualifier** |
|---|---|---|
| Meaning | A criterion that customers use to differentiate the services or products of one firm from those of another. | The minimal level required on a set of criteria for a firm to do business in a particular market segment. |
| Effect on sales | Wins the order against competitors; superior performance earns more business. | Does not ensure success; it only positions the firm to compete. Falling below the threshold disqualifies the firm. |
| Examples | Low price (supported by low-cost operations), superior quality, delivery speed, customization. | Consistent quality, basic safety and reliability standards, acceptable delivery time. |

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| **8** | A hospital is considering a new procedure to be offered at Rs. 200 per patient. The fixed cost per year would be Rs. 100,000, with total variable costs of Rs. 100 per patient. What is the break-even quantity for this service?<br>Source: Unit I \| CIE-I, 25/11/2024; SEE Jan/Feb 2021 (18IM54), Q1.2 | 02 | 2 | 2 |

**Answer:** Break-even quantity = 1,000 patients per year

The break-even quantity is the volume at which total revenue equals total cost.

$$Q = \frac{F}{p - c} = \frac{100{,}000}{200 - 100} = 1{,}000 \text{ patients}$$

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| **9** | When Amazon decides where to locate its fulfillment centers, it is making a _____ level decision in operations management.<br>Source: Unit I \| CIE-I, 06/11/2025 | 01 | 2 | 2 |

**Answer:** strategic

Locating fulfillment centers is a long-term, high-investment decision on the design of the supply chain (facility location and inventory placement). Such decisions are strategic-level operations decisions.

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| **10** | Decision-making models in operations include deterministic, probabilistic, and _____ models.<br>Source: Unit I \| CIE-I, 06/11/2025 | 01 | 1 | 3 |

**Answer:** uncertainty

Decisions are made under three situations: certainty (deterministic models), risk (probabilistic models, where event probabilities are known) and uncertainty (probabilities cannot be estimated).

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| **11** | The Fourth Industrial Revolution is driven by technologies such as IoT, AI, and _____.<br>Source: Unit I \| CIE-I, 06/11/2025 | 01 | 1 | 1 |

**Answer:** robotics (big data analytics or cloud computing are also accepted)

Industry 4.0 is driven by IoT, artificial intelligence, advanced robotics, big data analytics, cloud computing, additive manufacturing and cyber-physical systems.

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| **12** | When a factory balances machine utilization and customer delivery times, it is managing the _____ responsiveness trade-off.<br>Source: Unit I \| CIE-I, 06/11/2025 | 01 | 2 | 2 |

**Answer:** efficiency (efficiency–responsiveness trade-off)

High machine utilization lowers cost (efficiency) but leaves little spare capacity, while short delivery times need spare capacity and inventory (responsiveness). Balancing the two is the efficiency–responsiveness trade-off in supply chain design.

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| **13** | Walmart's use of RFID and data analytics to track inventory is an example of using _____ as a driver of supply chain performance.<br>Source: Unit I \| CIE-I, 06/11/2025 | 01 | 2 | 3 |

**Answer:** information

Information is a key driver of supply chain performance. Radio frequency identification (RFID) identifies items through radio signals from a tag attached to each item; Walmart shares daily sales information with suppliers such as Procter & Gamble to manage inventories and warehousing jointly.

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| **14** | Zara's ability to rapidly design and distribute new fashions demonstrates the competitive priority of _____.<br>Source: Unit I \| CIE-I, 06/11/2025 | 01 | 2 | 2 |

**Answer:** development speed (a time-based priority)

Development speed means quickly introducing a new service or product. Zara brings fashionable clothing designs from the runway to the market within a few weeks.

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| **15** | Differentiate between goods and services.<br>Source: Unit I \| SEE Feb/Mar 2026 (IM352IA), Q1.1 | 02 | 2 | 1 |

| **Basis** | **Goods (manufacturing)** | **Services** |
|---|---|---|
| Nature of output | Physical, durable, tangible product | Intangible, perishable act or performance |
| Inventory | Can be produced ahead and held in stock | Cannot be inventoried; capacity is lost if not used |
| Customer contact | Low; the customer is usually not in the process | High; the customer is often part of the process |
| Production and consumption | Separate in time and place | Simultaneous |
| Quality and demand | Quality is easy to measure; demand is easier to forecast | Quality is harder to measure; demand varies by time of day |
| Example | A car or a refrigerator | A haircut or a hospital visit |

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| **16** | What is meant by productivity in operations?<br>Source: Unit I \| SEE Feb/Mar 2026 (IM352IA), Q1.2 | 02 | 2 | 1 |

**Productivity** is a basic measure of how well resources (inputs) are being used to produce services and products (outputs). It is the ratio of the value of outputs to the value of the inputs used:

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}}$$

**Labor productivity** is the output per person or per hour worked (for example, insurance policies processed per employee per week). **Multifactor productivity** uses the value of all the resources (labor, materials, energy, capital) in the denominator. A rise in productivity means more output from the same resources, so managers track it to spot areas that need improvement.

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| **17** | Calculate the productivity for the following operation: three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week.<br>Source: Unit I \| SEE Jan/Feb 2021 (18IM54), Q1.1 | 02 | – | – |

Labor hours used in a week = 3 employees × 8 hours/day × 5 days/week = 120 hours.

$$\text{Productivity} = \frac{600 \text{ policies}}{120 \text{ labor-hours}} = 5 \text{ policies per labor-hour}$$

**Answer:** 5 policies per labor-hour (equivalently 200 policies per employee per week).

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| **18** | Distinguish between the terms 'competitive priorities' and 'competitive capabilities'.<br>Source: Unit I \| SEE Feb/Mar 2025 (IM352IA), Q1.1 | 02 | 2 | 1 |

| Basis | Competitive priorities | Competitive capabilities |
|---|---|---|
| Meaning | The critical operational dimensions (cost, quality, time, flexibility) that a process or supply chain must possess to satisfy its customers, now and in the future. | The cost, quality, time and flexibility dimensions that a process or supply chain actually possesses and is able to deliver. |
| Nature | What the customer wants (the target) | What the process can do (the current ability) |
| Relationship | Management selects the priorities that are most important. | When a capability falls short of its priority, management must close the gap or revise the priority. |

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| **19** | What is meant by 'nested process' in operations management?<br>Source: Unit I \| SEE Feb/Mar 2025 (IM352IA), Q1.2 | 02 | 2 | 1 |

A process can be broken down into subprocesses, and each subprocess can be broken down again into still smaller subprocesses. This idea of a process within a process is called a **nested process**.

Separating one part of a process from another helps because one person or department may be unable to perform all parts of the process, different parts may need different skills, and some parts can be designed for routine work while others handle special cases.

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| **20** | Differentiate between process and operation.<br>Source: Unit I \| SEE Apr/May 2024 (21IM53), Q1.1 | 02 | 2 | 1 |

| Basis | Process | Operation |
|---|---|---|
| Meaning | Any activity or group of activities that takes one or more inputs, transforms them and provides one or more outputs for its customers. | A group of resources (people, equipment, materials, facilities) performing all or part of one or more processes. |
| Focus | The work (activities and flows) | The resources that carry out the work |
| Example | Order fulfillment | The warehouse that performs picking and packing |

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| **21** | List any two supply chain business processes.<br>Source: Unit I \| SEE Apr/May 2024 (21IM53), Q1.2 | 02 | 1 | 1 |

Supply chain processes are business processes that have external customers or suppliers. Any two of:

- **Order fulfillment process:** takes the customer order and delivers the service or product (inventory, production, distribution).
- **Supplier relationship process:** selects suppliers, negotiates contracts and manages the flow of purchased inputs.
- **Customer relationship process:** identifies, attracts and builds relationships with external customers and facilitates the placement of orders.

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| **22** | Explain the term value-added.<br>Source: Unit I \| SEE Mar/Apr 2023 (18IM54), Q1.1 | 02 | – | – |

**Value added** is the increase in worth that an activity creates for the customer. Each activity in a process should add value to the preceding activities, and each process should add value for customers throughout the supply chain. Activities that do not add value (waste, delays, unnecessary cost) are eliminated.

In terms of output and input, the value added by a process is the value of its outputs less the cost of the inputs it consumes.

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| **23** | List the nine competitive priorities.<br>Source: Unit I \| SEE Mar/Apr 2023 (18IM54), Q1.2 | 02 | – | – |

| Group | Competitive priorities |
|---|---|
| Cost | 1. Low-cost operations |
| Quality | 2. Top quality; 3. Consistent quality |
| Time | 4. Delivery speed; 5. On-time delivery; 6. Development speed |
| Flexibility | 7. Customization; 8. Variety; 9. Volume flexibility |

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| **24** | Differentiate between internal and external customer with respect to process view.<br>Source: Unit I \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.1 | 02 | – | – |

| Basis | External customer | Internal customer |
|---|---|---|
| Who | End users or intermediaries (manufacturers, financial institutions, retailers) who buy the firm's finished services or products. | One or more employees or processes in the firm whose inputs are the outputs of earlier processes inside the firm. |
| Link to the process | Receives the final output of the firm's core processes. | Receives the output of an upstream process and uses it as input to the next process. |
| Example | A retailer buying from a manufacturer | The assembly department receiving parts from the machining department |

Every process and every person in an organization has customers and suppliers, so processes must be managed with the customer (internal or external) in mind.

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| **25** | What are the four core processes nested in operations management?<br>Source: Unit I \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.2 | 02 | – | – |

The four core processes nested in operations management (the processes that deliver value to external customers) are:

(1) **Customer relationship process:** identifies, attracts and builds relationships with external customers and facilitates the placement of orders.
(2) **New service/product development process:** designs and develops new services and products.
(3) **Order fulfillment process:** produces and delivers the service or product to the customer.
(4) **Supplier relationship process:** selects suppliers, develops contracts and manages the flow of purchased materials and services.

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| **26** | Define an operations strategy.<br>Source: Unit I \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.1 | 02 | – | – |

**Operations strategy** specifies the means by which operations implements the corporate strategy and helps to build a customer-driven firm. It links long-term and short-term operations decisions to the corporate strategy and develops the capabilities (cost, quality, time and flexibility) that the firm needs to be competitive. It is the pattern of decisions made in managing processes and supply chains so that they achieve the chosen competitive priorities.

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| **27** | Diagrammatically represent the integration between the three main functional areas of business, namely operations, finance and marketing.<br>Source: Unit I \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.2; SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.1 | 02 | – | – |

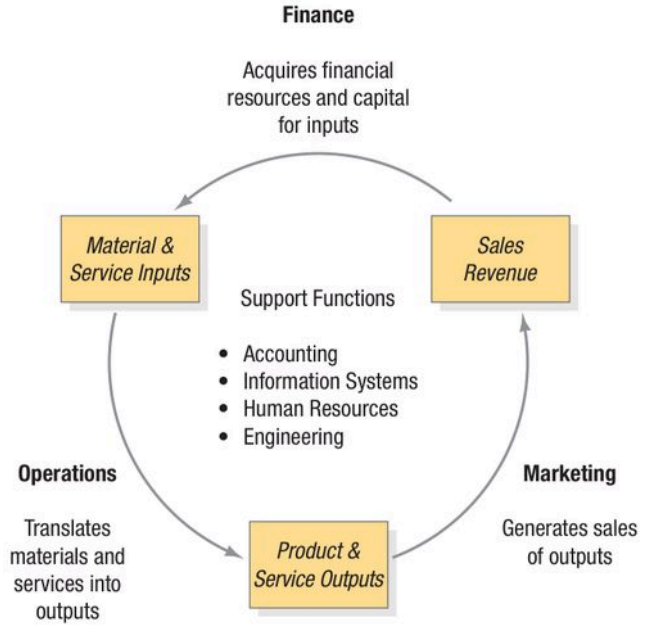



**Figure 1:** Integration between the functional areas of a business (operations, finance and marketing)

Marketing generates demand and customer information, finance provides the capital and funds, and operations converts these resources into services and products. Each function depends on the other two, so the corporate strategy must coordinate them.

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| **28** | Strategies for change include process improvement, process redesign, and process _____.<br>Source: Unit II \| CIE-I, 06/11/2025 | 01 | 1 | 1 |

**Answer:** reengineering

The two complementary strategies for change are process reengineering (radical redesign) and process improvement (systematic, incremental improvement).

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| **29** | Strategic fit refers to the alignment between business strategy and _____ strategy.<br>Source: Unit II \| CIE-I, 06/11/2025 | 01 | 2 | 2 |

**Answer:** operations

Strategic fit means that process decisions fit each other and fit the operations strategy, which in turn implements the corporate (business) strategy. The closer the fit, the more effective the process.

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| **30** | In manufacturing, the four process structures are job process, batch process, line process, and _____ process.<br>Source: Unit II \| CIE-I, 06/11/2025 | 01 | 1 | 1 |

**Answer:** continuous-flow

The four process choices form a continuum: job process, batch process, line process and continuous-flow process.

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| **31** | In a make-to-order process, production begins only after receiving a _____ from the customer.<br>Source: Unit II \| CIE-I, 06/11/2025 | 01 | 1 | 1 |

**Answer:** order (customer order)

In a make-to-order strategy products are made to customer specifications only after the order is received; each order is handled as a single job and nothing is produced ahead of time.

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| **32** | In services, the process structure is primarily influenced by the level of _____ and customization.<br>Source: Unit II \| CIE-II, 07/01/2025 | 01 | 1 | 2 |

**Answer:** customer contact

A good process strategy for a service process depends first on the type and amount of customer contact. The customer-contact matrix combines the degree of customer contact, customization and process characteristics.

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| **33** | In manufacturing, a _____ process is ideal for high-volume, standardized production.<br>Source: Unit II \| CIE-II, 07/01/2025 | 01 | 2 | 2 |

**Answer:** line (at the extreme, continuous-flow)

A line process has high volumes and standardized products, so resources are organized around particular products. A continuous-flow process is the extreme end of high-volume, standardized production.

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| **34** | The first step in a systematic approach to long-term capacity decisions is _____ future demand.<br>Source: Unit II \| CIE-II, 07/01/2025 | 01 | 2 | 2 |

**Answer:** estimating (forecasting) future capacity requirements

The four steps are: (1) estimate future capacity requirements, (2) identify gaps, (3) develop alternative plans and (4) evaluate the alternatives and make a final choice. The estimate is based on forecasts of demand, productivity, competition and technological change.

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| **35** | A systematic approach to capacity planning involves forecasting, identifying alternatives, and evaluating _____.<br>Source: Unit II \| CIE-II, 07/01/2025 | 01 | 1 | 2 |

**Answer:** alternatives

The final step is to evaluate each alternative, both qualitatively and quantitatively, and make a final choice.

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| **36** | _____ ensures that operational capabilities are in line with the company's strategic goals.<br>Source: Unit II \| CIE-II, 07/01/2025 | 01 | 2 | 3 |

**Answer:** Operations strategy

Operations strategy specifies the means by which operations implements corporate strategy. It links long-term and short-term operations decisions to corporate strategy and develops the capabilities the firm needs to be competitive.

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| **37** | A _____ shop deals with customized, low-volume production.<br>Source: Unit II \| Improvement CIE, 12/01/2026 | 01 | 1 | 2 |

**Answer:** job

A job process (job shop) produces a wide variety of products in small quantities; customization is high and volume for any one product is low.

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| **38** | In mass production, the layout used is _____ layout.<br>Source: Unit II \| Improvement CIE, 12/01/2026 | 01 | 1 | 2 |

**Answer:** product (line) layout

Mass production combines a line process with a make-to-stock strategy. Resources are organized around the product, with equipment and workers arranged in the sequence of operations (product layout).

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| **39** | The maximum output a system can produce is called its _____.<br>Source: Unit II \| Improvement CIE, 12/01/2026 | 01 | 1 | 2 |

**Answer:** capacity

Capacity is the maximum rate of output of a process or a system.

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| **40** | The long-term decision of choosing a geographic site for operations is called _____.<br>Source: Unit II \| Improvement CIE, 12/01/2026 | 01 | 2 | 3 |

**Answer:** facility location

Choosing a geographic site is a long-term strategic decision on the design of the supply chain, along with outsourcing and inventory placement.

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| **41** | Continuous processes are suitable for _____ production.<br>Source: Unit II \| Improvement CIE, 12/01/2026 | 01 | 2 | 3 |

**Answer:** high-volume, standardized (commodity) production

Continuous-flow processes have rigid line flows and negligible divergence. Examples are petroleum refining, chemicals, steel, soft drinks and food processing.

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| **42** | What is the objective of a process strategy?<br>Source: Unit II \| SEE Mar/Apr 2023 (18IM54), Q1.3 | 02 | – | – |

**Process strategy** specifies the pattern of decisions made in managing processes so that they achieve their competitive priorities. Its objective is therefore to design and run processes so that the firm can deliver the cost, quality, time and flexibility that its customers value. The decisions are process structure, customer involvement, resource flexibility and capital intensity.

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| **43** | List the four process strategies.<br>Source: Unit II \| SEE Mar/Apr 2023 (18IM54), Q1.4 | 02 | – | – |

The process strategy rests on four basic decisions:

(1) **Process structure:** the process type (job, batch, line or continuous flow in manufacturing; the degree of customer contact in services), including the layout.
(2) **Customer involvement:** the ways in which customers become part of the process and the extent of their participation.
(3) **Resource flexibility:** the ease with which employees and equipment can handle a wide variety of products, output levels, duties and functions.
(4) **Capital intensity:** the mix of equipment and human skills in the process.

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| **44** | Identify the four basic (major) process decisions in operations management.<br>Source: Unit II \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.3; SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.3; SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.3 | 02 | – | – |

The four major process decisions are:

(1) **Process structure:** the process type (job, batch, line or continuous flow in manufacturing; the degree of customer contact in services), including the layout.
(2) **Customer involvement:** the ways in which customers become part of the process and the extent of their participation.
(3) **Resource flexibility:** the ease with which employees and equipment can handle a wide variety of products, output levels, duties and functions.
(4) **Capital intensity:** the mix of equipment and human skills in the process.

They must fit one another and the competitive priorities (strategic fit), and they are linked to the strategies for change (process reengineering and process improvement).

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| **45** | What are the two different but complementary philosophies for process design?<br>Source: Unit II \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.4 | 02 | – | – |

(1) **Process reengineering:** the fundamental rethinking and radical redesign of processes to achieve dramatic improvements in cost, quality, speed and service (a clean-slate approach).
(2) **Process improvement:** the systematic study of the activities and flows of each process to improve it, in small continuing steps, by removing waste.

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| **46** | List the five techniques that are effective for documenting and evaluating processes.<br>Source: Unit II \| SEE Jan/Feb 2021 (18IM54), Q1.4 | 02 | – | – |

(1) **Flowchart:** traces the flow of information, customers, equipment or materials through the steps of a process.
(2) **Service blueprint:** a flowchart of a service process that shows which steps involve high customer contact.
(3) **Process chart:** an organized way of documenting all the activities performed by a person, at a workstation, with a customer or on materials.
(4) **Time study:** a work measurement method that times the work elements and sets a time standard.
(5) **Work sampling:** estimates the proportion of time spent on different activities from random observations.

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| **47** | Differentiate between job shop and flow shop.<br>Source: Unit II \| SEE Feb/Mar 2026 (IM352IA), Q1.3 | 02 | 2 | 3 |

| **Basis** | **Job shop (job process)** | **Flow shop (line process)** |
|---|---|---|
| Product | Low-volume products made to customer order; very high variety | Few major products in higher volume; standardized |
| Flow | Divergent flows; each job follows its own route through the workstations | Connected line flow; resources arranged along the product path |
| Equipment and skills | General-purpose equipment, highly skilled workers; flexible | Specialized equipment, less skill per worker; less flexible |
| Layout and cost | Process layout; higher cost per unit | Product (line) layout; low cost per unit |
| Example | Tool and die shop | Automobile assembly line |

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| **48** | The product-process matrix brings together three elements. List them.<br>Source: Unit II \| SEE Feb/Mar 2025 (IM352IA), Q1.3 | 02 | 1 | 2 |

The product-process matrix brings together three elements:

(1) **Process characteristics (process choice):** job, batch (small and large), line and continuous-flow processes, with their degree of divergence and flow.
(2) **Level of product customization:** from low-volume customized products to high-volume standardized products.
(3) **Volume:** the volume of the product, which increases from job to continuous-flow processes.

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| **49** | State three approaches to inventory that should be coordinated with process choice.<br>Source: Unit II \| SEE Jan/Feb 2021 (18IM54), Q1.3 | 02 | – | – |

Three inventory approaches to be coordinated with the process choice are:

(1) **Make-to-stock (MTS):** items are produced ahead and held in stock for immediate delivery.
(2) **Assemble-to-order (ATO):** standard components and subassemblies are held and the final product is assembled after the customer order is received.
(3) **Make-to-order (MTO):** production starts only after the customer order is received.

(A fourth strategy, engineer-to-order, is used when the product is designed for each customer.)

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| **50** | List any two strategies used to manage production and inventory.<br>Source: Unit II \| SEE Apr/May 2024 (21IM53), Q1.3 | 02 | 1 | 1 |

*Any two of make-to-stock, assemble-to-order and make-to-order (see Part A Q49); engineer-to-order is the fourth strategy.*

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| **51** | List any four aspects of continuous flow processes.<br>Source: Unit II \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.4 | 02 | – | – |

Four aspects of a continuous-flow process:

- Extreme end of high-volume, standardized production with rigid line flows.
- Process divergence is negligible; there is usually one primary material (liquid, gas or powder) that moves without stopping.
- Highly automated and capital intensive, with specialized equipment and little flexibility.
- Very low variety (commodity products), low cost per unit; examples are oil refineries and chemical plants.

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| **52** | List any four of the key differences between manufacturing and service processes.<br>Source: Unit II \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.2 | 02 | – | – |

| **Basis** | **Manufacturing process** | **Service process** |
|---|---|---|
| Output | Tangible, can be stored | Intangible, cannot be stored |
| Customer contact | Low | High |
| Time response | Longer response time is acceptable (delivery from stock) | Short response time; customer is present |
| Demand and capacity | Smoothed by inventory | Capacity must match demand when it occurs; peaks are costly |
| Quality measurement | Easy (physical specifications) | Difficult (customer perception) |

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| **53** | Bring out any four distinctions between job process and batch process.<br>Source: Unit II \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.4 | 02 | – | – |

| **Basis** | **Job process** | **Batch process** |
|---|---|---|
| Volume | Low volume, made to customer order | Higher volume; the same or similar products repeated in batches |
| Variety and customization | Very high; products are unique | Moderate; products or parts are standard within a batch |
| Flow and divergence | Most divergent flows | Less divergence; some flows are repeated |
| Setups and equipment | General-purpose equipment, frequent changeovers | Setups between batches; larger lot sizes reduce setup cost |

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| **54** | Differentiate between long-term and short-term capacity decisions.<br>Source: Unit II \| SEE Feb/Mar 2026 (IM352IA), Q1.4 | 02 | 2 | 3 |

| Basis | Long-term capacity decisions | Short-term capacity decisions |
|---|---|---|
| Concern | Capacity planning: when and how much capacity to add or reduce | Constraint management: making the most of existing capacity |
| Topics | Economies of scale, capacity cushions, timing and sizing strategies, systematic approach | Theory of constraints, bottleneck identification, product mix, line balancing |
| Nature | Strategic, large investments, hard to reverse | Operational, schedule and mix decisions within current resources |

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| **55** | List out any two of the long-term capacity (capacity planning) decisions and any two of the short-term constraint management decisions, as part of overall capacity management.<br>Source: Unit II \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.5 | 02 | – | – |

- **Long-term (capacity planning):** economies and diseconomies of scale; sizing capacity cushions; timing and sizing of expansion; a systematic approach to long-term capacity decisions.
- **Short-term (constraint management):** theory of constraints; identification and management of bottlenecks; product mix decisions using bottlenecks; managing constraints in a line process (line balancing).

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| **56** | What is meant by the concept 'economies of scale' in operations capacity planning?<br>Source: Unit II \| SEE Feb/Mar 2025 (IM352IA), Q1.4 | 02 | 2 | 2 |

**Economies of scale:** the average unit cost of a service or product tends to drop as the facility's size (and volume) increases, because fixed costs are spread over more units, construction costs per unit capacity fall, purchasing costs fall with bulk buying, and processing costs fall through specialization and learning. Beyond a certain size, **diseconomies of scale** appear and the unit cost rises (complexity, loss of focus, congestion).

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| **57** | Highlight any two dimensions of capacity strategy before making capacity decisions.<br>Source: Unit II \| SEE Apr/May 2024 (21IM53), Q1.4 | 02 | 1 | 1 |

Operations managers must examine three dimensions of capacity strategy:

(1) **Sizing capacity cushions:** the reserve capacity kept above average demand to handle surges and temporary losses.
(2) **Timing and sizing expansion:** when to add capacity and in what size increments (expansionist or wait-and-see strategy).
(3) **Linking process capacity and other operating decisions:** capacity must match the other decisions on inventory, quality, workforce and supply chain.

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| **58** | How is capacity planning measured?<br>Source: Unit II \| SEE Mar/Apr 2023 (18IM54), Q1.5 | 02 | – | – |

Capacity is measured in one of two ways. **Output measures** (units produced per time, for example cars per day) suit high-volume, standardized processes. **Input measures** (machine hours, number of workstations or workers, seats, available seat-miles) suit low-volume, flexible processes with high variety. Capacity is then compared with the demand expressed in the same unit, and the **utilization** is monitored.

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| **59** | Define the terms capacity and utilization.<br>Source: Unit II \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.5 | 02 | – | – |

**Capacity:** the maximum rate of output of a process or a system.

**Utilization:** the degree to which equipment, space or workforce is currently being used, measured as the ratio of average output rate to maximum capacity, expressed as a percentage.

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| **60** | List the measures of capacity utilization.<br>Source: Unit II \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.5 | 02 | – | – |

- **Output measures** of capacity (units of output per unit time): suitable for individual, high-volume, standardized processes.
- **Input measures** of capacity (machine hours, workers, workstations, seats): suitable for low-volume, flexible processes.
- **Utilization** = (average output rate / maximum capacity) × 100%, and the capacity cushion = 100% − average utilization.

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| **61** | Write the formula to calculate the utilization of any equipment.<br>Source: Unit II \| SEE Jan/Feb 2021 (18IM54), Q1.5 | 02 | – | – |

$$\text{Utilization} = \frac{\text{Average output rate}}{\text{Maximum capacity}} \times 100\%$$

The output rate and the capacity must be measured in the same units; the capacity cushion is 100% minus the average utilization.

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| **62** | List the steps of a systematic approach to long-term capacity decisions.<br>Source: Unit II \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.6 | 02 | – | – |

(1) **Estimate future capacity requirements** (from demand forecasts, productivity, competition and technology).
(2) **Identify gaps** between the requirements and the current capacity.
(3) **Develop alternatives** for closing the gaps (including the base case of doing nothing).
(4) **Evaluate the alternatives** qualitatively and quantitatively and make a choice.

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| **63** | In time series models, regular fluctuations occurring within a year due to weather or festivals are called _____ variations.<br>Source: Unit III \| CIE-II, 15/12/2025 | 01 | 1 | 1 |

**Answer:** seasonal

A seasonal pattern is a repeatable increase or decrease in demand that depends on the time of day, week, month or season (a period of less than a year).

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| **64** | The _____ component of a forecast captures recurring patterns over specific periods, such as seasons or months.<br>Source: Unit III \| CIE-II, 07/01/2025 | 01 | 2 | 3 |

**Answer:** seasonal

The seasonal component of a time series is the repeatable pattern of increases or decreases in demand within a year, for example by month or quarter.

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| **65** | List any four basic patterns of demand time series.<br>Source: Unit III \| SEE Feb/Mar 2025 (IM352IA), Q1.5 | 02 | 1 | 3 |

Demand time series show these patterns (any four):

(1) **Horizontal:** fluctuation of data around a constant mean.
(2) **Trend:** systematic increase or decrease in the mean of the series over time.
(3) **Seasonal:** repeatable pattern of increases or decreases depending on the time of day, week, month or season.
(4) **Cyclical:** less predictable gradual increases or decreases over longer periods (years or decades).
(5) **Random:** the unforecastable variation in demand.

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| **66** | Identify and list the five basic patterns of most demand time series.<br>Source: Unit III \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.8 | 02 | – | – |

*The five basic patterns are horizontal, trend, seasonal, cyclical and random (see Part A Q65).*

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| **67** | A hotel predicts weekend occupancy using past booking patterns. This uses a _____ time series model.<br>Source: Unit III \| CIE-II, 15/12/2025 | 01 | 2 | 1 |

**Answer:** seasonal

Weekend occupancy repeats every week, so it is a seasonal pattern. Using only the past data of the same period (for example, only weekends) is a time-series approach in which one series is kept for each season.

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| **68** | Qualitative forecasting methods rely heavily on _____ judgment.<br>Source: Unit III \| CIE-II, 15/12/2025 | 01 | 1 | 1 |

**Answer:** human (managerial or expert)

Judgment methods translate the opinions of managers, experts, consumers and the salesforce into quantitative estimates; they are used when there is no adequate history of demand.

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| **69** | A company uses a senior manager's opinion to forecast sales of a new product. This reflects the _____ forecasting method.<br>Source: Unit III \| CIE-II, 15/12/2025 | 01 | 1 | 1 |

**Answer:** executive opinion

In the executive opinion method the opinions, experience and technical knowledge of one or more managers are summarized to arrive at a single forecast. It is also used for technological forecasting.

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| **70** | Mention any two qualitative methods of forecasting.<br>Source: Unit III \| SEE Apr/May 2024 (21IM53), Q1.5 | 02 | 1 | 2 |

Two qualitative (judgment) methods: **executive opinion** (managers' opinions are combined into one forecast) and the **Delphi method** (consensus of anonymous experts through repeated rounds). The others are salesforce estimates and market research.

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| **71** | List out any four of the judgment methods for demand forecasting with a one-line description for each of these.<br>Source: Unit III \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.8 | 02 | – | – |

(1) **Salesforce estimates:** forecasts compiled from the estimates of future demand made by the members of the company's salesforce.
(2) **Executive opinion:** the opinions, experience and technical knowledge of one or more managers are summarized into a single forecast.
(3) **Market research:** a systematic approach to determine external consumer interest in a service or product by testing hypotheses through surveys.
(4) **Delphi method:** a process of gaining consensus from a group of experts while maintaining their anonymity.

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| **72** | What are the causal forecasting methods?<br>Source: Unit III \| SEE Mar/Apr 2023 (18IM54), Q1.7 | 02 | – | – |

**Causal methods** use historical data on independent variables (such as promotional campaigns, economic conditions and competitors' actions) to predict demand. The main causal method is **linear regression**, in which the dependent variable (demand) is related to one or more independent variables by a linear equation, Y = a + bX. With several independent variables it becomes multiple regression. Causal methods are good for predicting turning points and for long-range forecasts.

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| **73** | The difference between the actual value and the forecasted value is known as the _____.<br>Source: Unit III \| CIE-II, 15/12/2025 | 01 | 1 | 2 |

**Answer:** forecast error

Forecast error for period t is $E_t = D_t - F_t$, the actual demand minus the forecast.

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| **74** | What is 'forecast error' in the context of demand forecasting?<br>Source: Unit III \| SEE Feb/Mar 2025 (IM352IA), Q1.6 | 02 | 2 | 3 |

**Forecast error** for a period is the difference found by subtracting the forecast from the actual demand:

$$E_t = D_t - F_t$$

It is the starting point for the measures CFE, MSE, standard deviation, MAD and MAPE. Random error results from unpredictable factors; it can be reduced by choosing a good model but never eliminated.

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| **75** | Highlight the importance of forecast errors.<br>Source: Unit III \| SEE Apr/May 2024 (21IM53), Q1.6 | 02 | 2 | 2 |

- Forecasts almost always contain errors, so accuracy must be **measured** for every technique.
- Error measures **select the best method** and the best parameters (n, weights, alpha).
- The cumulative error (CFE, bias) shows a **systematic** fault, which causes the greatest disruption to planning; the tracking signal alerts the analyst when the method stops tracking demand.
- Errors tell managers how much **safety capacity or stock** to plan for, and they guide improvement of the forecasting process.

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| **76** | The smoothing constant in exponential smoothing ranges between 0 and _____.<br>Source: Unit III \| CIE-II, 07/01/2025 | 01 | 1 | 3 |

**Answer:** 1 (the smoothing parameter $\alpha$ lies between 0 and 1.0)

F(t+1) = $\alpha$ D(t) + (1 − $\alpha$) F(t). A large $\alpha$ gives more weight to recent demand (more responsive); a small $\alpha$ gives more stable forecasts.

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| **77** | What is meant by exponential smoothing?<br>Source: Unit III \| SEE Feb/Mar 2026 (IM352IA), Q1.5 | 02 | 2 | 3 |

**Exponential smoothing** is a sophisticated weighted moving-average method that calculates the average of a time series by implicitly giving recent demands more weight than earlier demands, all the way back to the first period. It needs only the last forecast, the actual demand and the smoothing parameter $\alpha$ (between 0 and 1):

$$F_{t+1} = \alpha D_t + (1 - \alpha) F_t$$

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| **78** | What is the role of forecasting in capacity planning?<br>Source: Unit III \| SEE Feb/Mar 2026 (IM352IA), Q1.6 | 02 | 2 | 3 |

Forecasts are the starting point of capacity planning. Operations needs forecasts to plan output levels and long-term capacities; at the process level, output forecasts are used to design the processes and to identify and deal with in-house bottlenecks. The first step of the systematic approach to long-term capacity decisions, estimating the future capacity requirements, is based on demand forecasts, so forecast errors lead directly to capacity gaps (shortage or excess).

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| **79** | What is the purpose of a tracking signal and how is it calculated?<br>Source: Unit III \| SEE Mar/Apr 2023 (18IM54), Q1.8; SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.8 | 02 | – | – |

A **tracking signal** indicates whether a forecasting method is accurately predicting actual changes in demand. It is the number of MADs represented by the cumulative sum of forecast errors:

$$\text{Tracking signal} = \frac{CFE}{MAD}$$

CFE and MAD are updated every period, and the tracking signal is compared with predetermined limits (plotted on a control chart). While the signal stays within the limits, the CFE is close to zero and the method is tracking demand; when it goes outside the limits, the model is no longer tracking demand and must be reviewed.

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| **80** | A _____ is any resource whose capacity is less than or equal to the demand placed on it.<br>Source: Unit IV \| CIE-III, 27/01/2025 | 01 | 1 | 3 |

**Answer:** bottleneck (constraint)

A bottleneck is a process, step or workstation with the lowest capacity, so its capacity is less than or equal to the demand placed on it.

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| **81** | In product mix decisions, TOC prioritizes products based on _____ per unit of the bottleneck resource.<br>Source: Unit IV \| CIE-III, 27/01/2025 | 01 | 1 | 3 |

**Answer:** contribution margin

In the bottleneck method, products are ranked by the contribution margin per minute of processing time at the bottleneck workstation, and the product with the highest value is scheduled first.

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| **82** | The Theory of Constraints uses the _____ process to repeat improvements once a bottleneck is resolved.<br>Source: Unit IV \| CIE-III, 27/01/2025 | 01 | 1 | 3 |

**Answer:** five-step improvement (focusing) process: identify, exploit, subordinate, elevate and repeat

The last step, "do not let inertia set in", repeats steps 1 to 4 because the constraint may shift after improvements.

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| **83** | In manufacturing, reducing _____ time at a bottleneck can significantly improve efficiency.<br>Source: Unit IV \| CIE-III, 27/01/2025 | 01 | 1 | 3 |

**Answer:** setup (changeover)

Setup time at a bottleneck is unproductive time; reducing it (and processing large lots per setup) preserves bottleneck capacity.

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| **84** | A production line experiencing a delay at one critical machine identifies it as the _____ resource.<br>Source: Unit IV \| CIE-II, 15/12/2025 | 01 | 2 | 4 |

**Answer:** bottleneck

The resource that delays the whole line is the bottleneck; an hour lost there is an hour lost for the whole system.

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| **85** | A bottleneck is a stage that limits the _____ of the system.<br>Source: Unit IV \| Improvement CIE, 12/01/2026 | 01 | 2 | 3 |

**Answer:** throughput (output)

The bottleneck sets the throughput of the entire system, so the system can produce no more than the bottleneck allows.

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| **86** | The Theory of _____ focuses on improving the weakest link in a process.<br>Source: Unit IV \| Improvement CIE, 12/01/2026 | 01 | 2 | 3 |

**Answer:** Constraints

The theory of constraints (TOC) is a systematic management approach that focuses on actively managing the constraints that impede a firm's progress toward its goal.

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| **87** | State the concept of Theory of Constraints (TOC).<br>Source: Unit IV \| SEE Feb/Mar 2026 (IM352IA), Q1.7 | 02 | 2 | 3 |

The **theory of constraints (TOC)** is a systematic management approach, developed by Eli Goldratt, that focuses on actively managing the constraints (bottlenecks) that impede a firm's progress toward its goal of maximizing profits and using its resources effectively. Instead of improving the efficiency of every individual process, it concentrates on the bottlenecks that limit the system as a whole, and aims to make materials flow rapidly through the entire system (higher throughput, lower inventory and operating expense).

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| **88** | What is meant by 'bottleneck' in the context of constraint management?<br>Source: Unit IV \| SEE Feb/Mar 2025 (IM352IA), Q1.7 | 02 | 2 | 3 |

A **bottleneck** is a process, step or workstation with the lowest capacity (the highest total time per unit processed or the highest average utilization and workload). Its capacity is less than or equal to the demand placed on it, so it limits the throughput of the whole system.

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| **89** | Differentiate between constraint and bottleneck.<br>Source: Unit IV \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.7 | 02 | – | – |

| Basis | Constraint | Bottleneck |
|---|---|---|
| Meaning | Anything that limits the system's performance in achieving its goal; it can be an internal resource, market demand, lack of capital or a policy (such as the planning and scheduling system). | A process, step or workstation with the lowest capacity; a physical resource whose capacity is less than or equal to the demand on it. |
| Scope | Broader term; includes non-physical constraints | A special (physical) kind of constraint |
| Example | Market demand below capacity; a scheduling policy | The welding station loaded at 105% of its capacity |

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| **90** | What are the five steps of the Theory of Constraints?<br>Source: Unit IV \| SEE Mar/Apr 2023 (18IM54), Q1.6 | 02 | – | – |

(1) **Identify the system bottleneck(s).**
(2) **Exploit the bottleneck(s):** schedule it to maximize its throughput and pass only good-quality parts to it.
(3) **Subordinate all other decisions** to step 2: non-bottlenecks support the bottleneck schedule and do not produce more than it can handle.
(4) **Elevate the bottleneck(s):** increase its capacity (shift, machine, redesign) if it is still a constraint.
(5) **Do not let inertia set in:** the constraint may shift, so repeat steps 1 to 4.

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| **91** | List any four of the seven key principles of the Theory of Constraints.<br>Source: Unit IV \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.6; SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.6 | 02 | – | – |

The seven key principles of the TOC (any four):

(1) The focus should be on balancing flow, not on balancing capacity.
(2) Maximizing the output and efficiency of every resource may not maximize the throughput of the entire system.
(3) An hour lost at a bottleneck is an hour lost for the whole system; an hour saved at a non-bottleneck is a mirage.
(4) Inventory is needed only in front of the bottlenecks and at assembly and shipping points.
(5) Work should be released into the system only as frequently as the bottlenecks need it.
(6) Activating a non-bottleneck resource is not the same as utilizing a bottleneck resource.
(7) Every capital investment must be viewed from the perspective of its global impact on throughput, inventory and operating expense.

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| **92** | What is bottleneck scheduling, and why is it critical?<br>Source: Unit IV \| CIE-III, 27/01/2025; SEE Feb/Mar 2026 (IM352IA), Q1.10 | 02 | 2 | 4 |

**Bottleneck scheduling** means creating schedules that maximize the throughput of the bottleneck resource while meeting the promised completion dates. In the drum-buffer-rope system the bottleneck schedule is the **drum** (it sets the production rate of the plant), the time **buffer** protects the bottleneck from starving, and the **rope** ties material release to the bottleneck's pace.

It is critical because the bottleneck determines the throughput of the whole system: any idle time at the bottleneck is lost output for the entire system, whereas idle time at non-bottlenecks costs nothing in throughput.

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| **93** | Define line balancing.<br>Source: Unit IV \| SEE Jan/Feb 2021 (18IM54), Q1.8 | 02 | – | – |

**Line balancing** is the assignment of work to stations in a line process so as to achieve the desired output rate with the smallest number of workstations. Work elements are grouped into stations so that the station workloads are as even as possible and no station exceeds the cycle time.

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| **94** | What is meant by load balancing in operations?<br>Source: Unit IV \| SEE Feb/Mar 2026 (IM352IA), Q1.8 | 02 | 2 | 3 |

*In a line process the workload is balanced between the stations so that no station becomes a bottleneck; this is line balancing (see Part A Q93). The aim is workstations with workloads as evenly balanced as possible, so that the utilization of the bottleneck is not much higher than that of the other stations.*

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| **95** | MRP stands for _____ Requirements Planning.<br>Source: Unit IV \| Improvement CIE, 12/01/2026 | 01 | 1 | 3 |

**Answer:** Material

Material requirements planning (MRP) is a computerized information system developed to help manufacturers manage dependent demand inventory and schedule replenishment orders.

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| **96** | ERP stands for _____ Resource Planning.<br>Source: Unit IV \| Improvement CIE, 12/01/2026 | 01 | 1 | 4 |

**Answer:** Enterprise

Enterprise resource planning (ERP) systems are large, integrated information systems that support many enterprise processes and data storage needs.

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| **97** | _____ provides a detailed plan of what needs to be produced and when, serving as an input for MRP.<br>Source: Unit IV \| CIE-II, 07/01/2025 | 01 | 1 | 3 |

**Answer:** Master production schedule (MPS)

The MPS details how many end items will be produced within specified periods of time. It breaks the sales and operations plan into specific product schedules and is a key input to MRP.

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| **98** | The detailed schedule of what is to be produced is called the _____.<br>Source: Unit IV \| Improvement CIE, 12/01/2026 | 01 | 2 | 3 |

**Answer:** master production schedule (MPS)

The MPS gives the timing and size of production quantities for each end item.

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| **99** | The _____ file contains information about the quantities of materials required to make a product.<br>Source: Unit IV \| CIE-II, 07/01/2025 | 01 | 2 | 3 |

**Answer:** bill of materials (BOM)

The BOM is a record of all the components of an item, the parent–component relationships and the usage quantities.

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| **100** | _____ integrates all business processes, including finance, HR and supply chain, into a unified system.<br>Source: Unit IV \| CIE-II, 07/01/2025 | 01 | 2 | 3 |

**Answer:** ERP (enterprise resource planning)

ERP revolves around a single comprehensive database shared across the enterprise, so that all the functional applications (manufacturing, supply chain, finance, human resources, sales, customer service) share the same data.

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| **101** | What is the meaning of MRP explosion?<br>Source: Unit IV \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.10 | 02 | – | – |

**MRP explosion** is the process that converts the requirements of various final products (from the master production schedule and other demand) into a material requirements plan. It specifies the replenishment schedules (timing and quantity) of all the subassemblies, components and raw materials needed to produce the final products, using the bill of materials, the inventory records and the lot-sizing rules and lead times.

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| **102** | List the inputs required for MRP explosion.<br>Source: Unit IV \| SEE Feb/Mar 2025 (IM352IA), Q1.8; SEE Apr/May 2024 (21IM53), Q1.8 (any two inputs) | 02 | 1 | 3 |

The MRP explosion has three key inputs:

(1) **Authorized master production schedule** (together with other sources of demand such as replacement parts).
(2) **Inventory records** (kept up to date by inventory transactions).
(3) **Bills of materials** (from engineering and process designs).

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| **103** | List the inputs for the MPS (master production schedule).<br>Source: Unit IV \| SEE Mar/Apr 2023 (18IM54), Q1.10 | 02 | – | – |

Inputs for developing the master production schedule: the **sales and operations plan** (authorized production plan for the product family), the **forecast** and the **customer orders booked**, the **current on-hand inventory**, the **order policy (lot size) and lead time**, and the **capacity and resource limits** (machine, labor, storage, working capital).

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| **104** | What is the key difference between ERP and MRP?<br>Source: Unit IV \| SEE Mar/Apr 2023 (18IM54), Q1.9 | 02 | – | – |

| Basis | MRP | ERP |
|---|---|---|
| Meaning | A computerized system that helps manufacturers manage dependent demand inventory and schedule replenishment orders. | A large integrated information system that supports many enterprise processes across all functions on a single database. |
| Scope | Materials and production planning of manufacturing | The whole enterprise: manufacturing, supply chain, finance, HR, sales, customer service |
| Relationship | MRP is one module (resource planning) of ERP; MRP II extended MRP and ultimately evolved into ERP | Contains MRP as its manufacturing application |

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| **105** | List any two challenges faced during ERP implementation.<br>Source: Unit IV \| SEE Apr/May 2024 (21IM53), Q1.7 | 02 | 1 | 2 |

- **Cost and complexity:** firms can spend excessive amounts on ERP systems that are complex to use and costly to manage, especially when they are customized extensively.
- **Process redesign and legacy systems:** the major processes must be analyzed, redundant and convoluted information flows may have to be reengineered, and new software must be coordinated with the existing legacy systems.
- **Training and rollout:** users must be trained on the screens; problems in the roll-out can disrupt operations (Cadbury built up excess inventory and took a large profit hit).

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| **106** | Give one example each for dependent and independent demand.<br>Source: Unit IV \| SEE Jan/Feb 2021 (18IM54), Q1.7 | 02 | – | – |

| Basis | Independent demand | Dependent demand |
|---|---|---|
| Meaning | Demand for a final product, influenced only by market conditions; it must be forecast. | Demand for an item that occurs because its quantity depends on the production plans for other items; it is calculated, not forecast. |
| Example | A finished bicycle | The wheel rims or spokes of the bicycle (two rims per bicycle) |

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| **107** | Define the periodic order quantity and lot-for-lot lot-sizing rules to bring out their working mechanics.<br>Source: Unit IV \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.10 | 02 | – | – |

**Periodic order quantity (POQ):** a rule that issues an order at predetermined intervals of P periods, and the order quantity equals the total gross requirements for those P periods less the projected on-hand inventory at the end of the previous week, so that the inventory is zero at the end of the Pth period.

**Lot-for-lot (L4L):** a special case of POQ with P = 1: the order quantity covers exactly the gross requirement of a single week, less the inventory on hand, so inventory is minimized but the number of orders (setups) is maximized.

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| **108** | Give the precise definition of ordering cost and setup cost.<br>Source: Unit IV \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.7 | 02 | – | – |

**Ordering cost:** the cost of preparing a purchase order for a supplier or a production order for manufacturing.

**Setup cost:** the cost involved in changing over a machine or workspace to produce a different item.

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| **109** | Differentiate between push and pull workflow.<br>Source: Unit IV \| SEE Jan/Feb 2021 (18IM54), Q1.10 | 02 | – | – |

| **Basis** | **Push method** | **Pull method** |
|---|---|---|
| Meaning | Production of the item begins in advance of customer needs. | Customer demand activates production of the service or item. |
| Driver | Forecast and schedule | Actual customer demand |
| Inventory | Builds inventory ahead of need | Inventory is minimized |
| Example | Material pushed down the line after the bottleneck toward the customer; make-to-stock | The drum beat pulls material forward before the bottleneck; make-to-order |

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| **110** | What is sequencing?<br>Source: Unit V \| SEE Feb/Mar 2026 (IM352IA), Q1.9 | 02 | 2 | 3 |

**Sequencing** is deciding the order in which the jobs or customers waiting at a workstation are processed. It is done with a **priority sequencing rule**, a rule that specifies the job or customer processing sequence when several jobs are waiting in line at a workstation (for example FCFS, EDD, SPT or critical ratio).

| | | | | |
|---|---|---|---|---|
| **111** | What is shortest processing time sequencing in scheduling?<br>Source: Unit V \| SEE Feb/Mar 2025 (IM352IA), Q1.9 | 02 | 2 | 4 |

In the **shortest processing time (SPT)** rule the job requiring the shortest processing time is the next job to be processed. Arranging the jobs in increasing order of processing time minimizes the average flow time of the jobs at a single workstation, but it can increase the past due time of the long jobs.

| | | | | |
|---|---|---|---|---|
| **112** | A production manager selects the job with the shortest processing time to reduce average waiting time. This applies the _____ rule.<br>Source: Unit V \| CIE-II, 15/12/2025 | 01 | 2 | 3 |

**Answer:** shortest processing time (SPT)

SPT processes the shortest job first, which reduces the average flow time (average time in the system) and so the average waiting time.

| | | | | |
|---|---|---|---|---|
| **113** | List any two priority sequencing rules commonly used in practice.<br>Source: Unit V \| SEE Apr/May 2024 (21IM53), Q1.9 | 02 | 1 | 1 |

Two commonly used priority sequencing rules:

- **First-come, first-served (FCFS):** the job or customer arriving at the workstation first has the highest priority.
- **Earliest due date (EDD):** the job with the earliest due date is processed next.

(Shortest processing time and critical ratio are other priority rules.)

| | | | | |
|---|---|---|---|---|
| **114** | What is flow time in a scheduling problem?<br>Source: Unit V \| SEE Apr/May 2024 (21IM53), Q1.10 | 02 | 2 | 2 |

**Flow time** is the amount of time a job spends in the service or manufacturing system: the waiting time for machines, processing time including setups, moving time and delays. For jobs at a single workstation, all available at time 0:

$$\text{Flow time} = \text{Finish time} + \text{Time since the job arrived at the workstation}$$

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| **115** | Using the Earliest Due Date (EDD) rule, determine the sequence for the following jobs based on their due dates.<br><br>Job \| Processing time (hr) \| Due date (hr)<br>A \| 4 \| 10<br>B \| 3 \| 8<br>C \| 2 \| 6<br>D \| 5 \| 12<br><br>Source: Unit V \| CIE-III, 27/01/2025 | 02 | 2 | 4 |

The EDD rule sequences the jobs in increasing order of due date: C (6), B (8), A (10), D (12).

| Sequence | Job | Processing time (hr) | Start | Finish | Due date | Hours past due |
|---|---|---|---|---|---|---|
| 1 | C | 2 | 0 | 2 | 6 | 0 |
| 2 | B | 3 | 2 | 5 | 8 | 0 |
| 3 | A | 4 | 5 | 9 | 10 | 0 |
| 4 | D | 5 | 9 | 14 | 12 | 2 |

**Answer:** Sequence C – B – A – D. Finish times are 2, 5, 9 and 14 hours; only job D is late, by 2 hours (average flow time 7.5 hours).

| | | | | |
|---|---|---|---|---|
| **116** | Differentiate between backlog and back order.<br>Source: Unit V \| SEE Jan/Feb 2021 (18IM54), Q1.6 | 02 | – | – |

| Basis | Backlog | Back order |
|---|---|---|
| Meaning | An accumulation of customer orders that a manufacturer has promised for delivery at some future date. | A customer order that cannot be filled immediately but is filled as soon as possible. |
| Nature | Planned, normal business practice (make-to-order); a demand option to level demand | Unplanned shortage; a last resort that lowers customer service |
| Effect | Reduces uncertainty of future production and levels demand; a disadvantage if too big | Adds to the next period's demand requirement; customers may go elsewhere |

| | | | | |
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| **117** | What is aggregation? List the three dimensions along which companies perform aggregation.<br>Source: Unit V \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q1.10 | 02 | – | – |

**Aggregation** is the act of clustering several similar services or products so that forecasts and plans can be made for whole families (product families) instead of for each individual item. Companies perform aggregation along three dimensions:

(1) **Services or products:** items with similar demand and common process, labor and materials requirements are grouped into families.
(2) **Workforce:** workers with different skills are treated as one aggregate group when they are flexible.
(3) **Time:** the plan looks at months, quarters or seasons (typically over a year) rather than weeks or days.

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| **118** | Clearly define the sales and operations plan (S&OP) and the resource plan.<br>Source: Unit V \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q1.9 (S&OP); SEE Fast Track Jul 2019 (16IM55/12IM62), Q1.9 (S&OP and resource plan) | 02 | – | – |

**Sales and operations plan (S&OP):** a plan of future aggregate resource levels so that supply is in balance with demand. It states a company's or department's production rates, workforce levels and inventory holdings that are consistent with demand forecasts and capacity constraints, and it is time-phased (projected for several months or quarters into the future).

**Resource plan:** a plan that determines the requirements for materials and other resources on a more detailed level than the sales and operations plan (the master production schedule and material requirements planning in manufacturing, the workforce schedule in services).

| Sl. No. | Questions | M | BT | CO |
|---|---|---|---|---|
| | **PART – B** | | | |
| **1** | Define operations management. Briefly explain the key terms involved in the definition of operations management with a schematic model of a production system.<br>Source: Unit I \| CIE-I, 06/11/2025 and 25/11/2024 (same question) | 10 | 2 | 1 |

### Definition

**Operations management** refers to the systematic design, direction and control of processes that transform inputs into services and products for internal as well as external customers.

### Key terms in the definition

- **Process:** any activity or group of activities that takes one or more inputs, transforms them and provides one or more outputs for its customers.
- **Operation:** a group of resources performing all or part of one or more processes. Processes are clustered together into operations for organizational purposes.
- **Supply chain:** the interrelated series of processes within a firm and across different firms that produce a service or product to the satisfaction of customers.
- **Supply chain management:** the synchronization of a firm's processes with those of its suppliers and customers to match the flow of materials, services and information with customer demand.
- **Inputs:** human resources (workers and managers), capital (equipment and facilities), purchased materials and services, land and energy.
- **Outputs:** services (which may take the form of information) or tangible products.
- **Customers:** external customers are end users or intermediaries (manufacturers, financial institutions, retailers) who buy the firm's finished services or products; internal customers are employees whose inputs are the outputs of earlier processes within the firm.
- **Suppliers:** external suppliers are businesses or individuals who provide resources, services, products and materials; internal suppliers are employees or processes that supply important information or materials.

### Schematic model of a production system

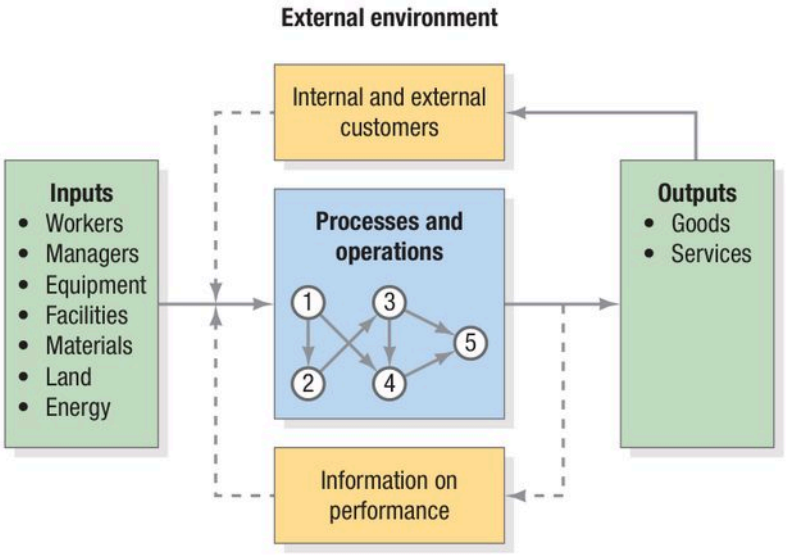



**Figure 2:** Processes and operations: schematic model of a production system

Inputs enter the system and are transformed by processes performed at operations (the numbered circles) into outputs of goods and services. The arrows represent flows; they can cross because one job or customer can have different requirements, and hence a different flow pattern, than the next. The dashed lines represent two special inputs: participation by customers and information on performance from internal and external sources. The whole system operates within the external environment.

| | | | | |
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| **2** | With suitable example, distinguish between manufacturing and service processes.<br>Source: Unit I \| CIE-I, 06/11/2025 and 25/11/2024 (same question) | 10 | 2 | 2 |

Processes are classified as manufacturing or service processes. The two key differences between them are (1) the nature of their output and (2) the degree of customer contact.

### Manufacturing process

A manufacturing process converts materials into goods that have a physical form (products). The transformation changes the materials on one or more of the following five dimensions:

(1) Physical properties
(2) Shape
(3) Size (for example, length, breadth and height of a block of wood)
(4) Surface finish
(5) Joining of parts and materials

Its outputs can be produced, stored and transported in anticipation of future demand, and it usually has less customer contact. **Examples:** an assembly line producing a car; a tailor producing an outfit; washing machines produced to meet retail forecasts.

### Service process

A process that does not change the properties of materials on at least one of the five dimensions is a service (non-manufacturing) process. Service processes produce intangible, perishable outputs that usually cannot be held in finished-goods inventory, and they have a higher degree of customer contact; customers may take an active part in the process. **Examples:** a bank's auto-loan process (output: a car loan), postal delivery of a letter, shopping in a supermarket, treatment at a medical clinic.

### Distinction between manufacturing and service processes

| Basis | Manufacturing process | Service process |
|---|---|---|
| Nature of output | Physical, durable output | Intangible, perishable output |
| Inventory | Output can be inventoried | Output cannot be inventoried |
| Customer contact | Low customer contact | High customer contact |
| Response time | Long response time | Short response time |
| Resources | Capital intensive | Labour intensive |
| Quality | Quality easily measured | Quality not easily measured |
| Examples | Car assembly, steel making, garment stitching | Banking, hospitals, airlines, postal service |

Most organizations lie on a continuum between the two. **Example:** at McDonald's, cooking a hamburger is a manufacturing process because it changes the physical properties of the material (dimension 1), and assembling the burger with the bun is also manufacturing (dimension 5); most of the other processes visible to customers, such as order taking and serving, are service processes.

| | | | | |
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| **3** | A retailer must decide whether to build a small or a large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6, respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = \$223,000) or to expand (payoff = \$270,000). If a small facility is built and demand is low, there is no reason to expand and the payoff is \$200,000. If a large facility is built and demand proves to be low, the choice is to do nothing (\$40,000) or to stimulate demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7, respectively. If it is modest, the payoff is estimated to be only \$20,000; the payoff grows to \$220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is \$800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative – building a small facility or building a large facility – has the higher expected payoff?<br>Source: Unit I \| CIE-I, 06/11/2025 | 16 | 4 | 3 |

### Decision tree

A decision tree is a schematic model of the alternatives available to the decision maker and their possible consequences. Square nodes are decision points, and the branches leaving them are the alternatives. Circular (chance) nodes are events, and the probability of each event is shown on its branch; the probabilities leaving a chance node sum to 1.0. The tree is read from left to right.

The first decision (node 1) is whether to build a small or a large facility. Decision node 2 (expand or not) is reached only if a small facility is built and demand is high. Decision node 3 (advertise or not) is reached only if a large facility is built and demand is low. Payoffs are in thousands of dollars.

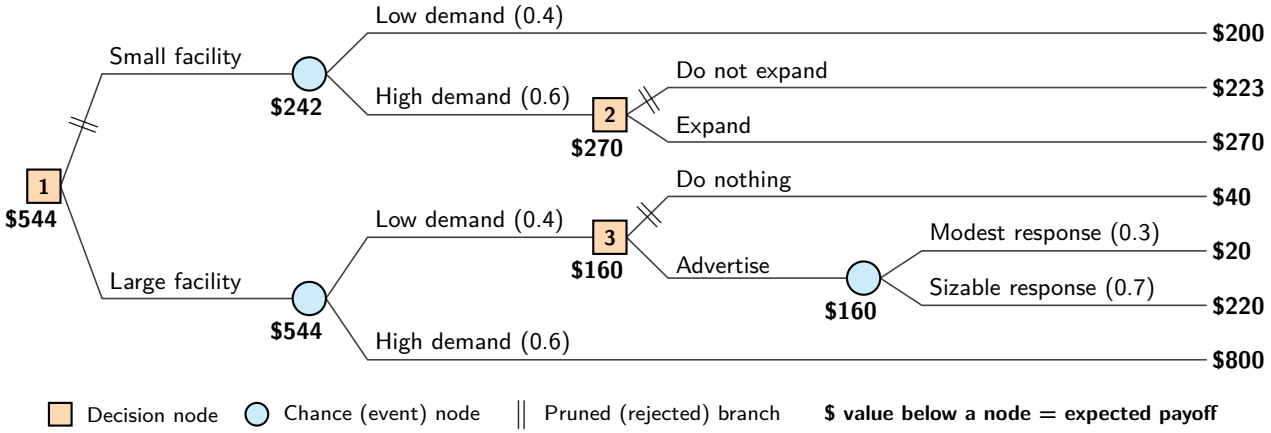



**Figure 3:** Decision tree for the retailer (payoffs in \$000)

### Analysis (rolling back from right to left)

For an event node, multiply the payoff of each branch by its probability and add the products. For a decision node, choose the alternative with the best expected payoff and prune the others.

Event node (advertising): $0.3(20) + 0.7(220) = 6 + 154 = 160$

Decision node 3: Advertise (160) > Do nothing (40) $\Rightarrow$ 160 (prune Do nothing)

Decision node 2: Expand (270) > Do not expand (223) $\Rightarrow$ 270 (prune Do not expand)

Event node (small facility): $0.4(200) + 0.6(270) = 80 + 162 = 242$

Event node (large facility): $0.4(160) + 0.6(800) = 64 + 480 = 544$

Decision node 1: Large (544) > Small (242) $\Rightarrow$ 544 (prune Small facility)

**Answer:** Build the large facility: expected payoff \$544,000 against \$242,000 for the small facility. Only this first decision is made now; later decisions are taken after demand is known.

| | | | | |
|---|---|---|---|---|
| **4** | A manager is deciding whether to build a small or a large facility. Much depends on the future demand that the facility must serve, and demand may be small or large. The manager knows with certainty the payoffs that will result under each alternative, shown in the payoff table below. What is the best alternative for each decision rule: Maximin, Maximax, Laplace, Minimax Regret? | 14 | 4 | 3 |

| Alternative | Low demand | High demand |
|---|---|---|
| Small facility | 200 | 270 |
| Large facility | 160 | 800 |
| Do nothing | 0 | 0 |

Source: Unit I | CIE-I, 06/11/2025 (14 M); CIE-I, 25/11/2024 (10 M)

The "do nothing" alternative is **dominated**: its payoff is no better than that of another alternative for every event, so it is not considered further.

### Decision rules

- **Maximin:** choose the alternative that is the "best of the worst". This rule is for the pessimist, who anticipates the worst case for each alternative.
- **Maximax:** choose the alternative that is the "best of the best". This rule is for the optimist, who has high expectations and prefers to "go for broke".
- **Laplace:** choose the alternative with the best weighted payoff, giving equal probability (1/n) to each of the n events. This rule is for the realist.
- **Minimax regret:** choose the alternative with the best "worst regret". A regret (opportunity loss) is the difference between a given payoff and the best payoff in the same column (event).

### Solution (payoffs in $000)

**Maximin and maximax:** the worst and best payoffs in each row are

| Alternative | Worst payoff | Best payoff |
|---|---|---|
| Small facility | 200 | 270 |
| Large facility | 160 | 800 |

The best of the worst payoffs is $200,000, so the pessimist (maximin) builds a small facility. The best of the best payoffs is $800,000, so the optimist (maximax) builds a large facility.

**Laplace:** with two events, each is given a probability of 0.5.

| Alternative | Weighted payoff |
|---|---|
| Small facility | 0.5(200) + 0.5(270) = 235 |
| Large facility | 0.5(160) + 0.5(800) = 480 |

The best weighted payoff is $480,000, so the realist builds a large facility.

**Minimax regret:** the best payoff is 200 for low demand and 800 for high demand.

| Alternative | Low demand | High demand | Maximum regret |
|---|---|---|---|
| Small facility | 200 − 200 = 0 | 800 − 270 = 530 | 530 |
| Large facility | 200 − 160 = 40 | 800 − 800 = 0 | 40 |

The smallest maximum regret (40) belongs to the large facility. The biggest regret arises from building only a small facility when demand turns out to be high.

### Summary

| Decision rule | Best alternative |
|---|---|
| Maximin | Small facility |
| Maximax | Large facility |
| Laplace | Large facility |
| Minimax regret | Large facility |

**Answer:** The pessimist (maximin) chooses the small facility; the optimist (maximax), the realist (Laplace) and the minimax-regret decision maker choose the large facility.

| | | | | |
|---|---|---|---|---|
| **5** | White Valley Ski Resort is planning the ski lift operation for its new ski resort. Management is trying to determine whether one or two lifts will be necessary; each lift can accommodate 250 people per day. Skiing normally occurs in the 14-week period from December to April, during which the lift will operate 7 days per week. The first lift will operate at 90 percent capacity if economic conditions are bad, the probability of which is believed to be about 0.3. During normal times the first lift will be utilized at 100 percent capacity, and the excess crowd will provide 50 percent utilization of the second lift. The probability of normal times is 0.5. Finally, if times are really good, the probability of which is 0.2, the utilization of the second lift will increase to 90 percent. The equivalent annual cost of installing a new lift, recognizing the time value of money and the lift's economic life, is \$50,000. The annual cost of installing two lifts is only \$90,000 if both are purchased at the same time. If used at all, each lift costs \$200,000 to operate, no matter how low or high its utilization rate. Lift tickets cost \$20 per customer per day. Should the resort purchase one lift or two?<br>Source: Unit I \| CIE-I, 25/11/2024 | 20 | 4 | 3 |

### Payoffs

Operating season = 14 weeks × 7 days = 98 days. Revenue from one lift at 100 percent capacity = 250 customers × 98 days × \$20 = \$490,000. Payoff = revenue − (annual installation cost + operating cost). Values are in \$000.

| Alternative | Economic condition | Payoff calculation (revenue − cost) |
|---|---|---|
| One lift | Bad times (0.3) | 0.9(490) − (50 + 200) = 191 |
| | Normal times (0.5) | 1.0(490) − (50 + 200) = 240 |
| | Good times (0.2) | 1.0(490) − (50 + 200) = 240 |
| Two lifts | Bad times (0.3) | 0.9(490) − (90 + 200) = 151 |
| | Normal times (0.5) | 1.5(490) − (90 + 400) = 245 |
| | Good times (0.2) | 1.9(490) − (90 + 400) = 441 |

In bad times only the first lift is used, so the second lift incurs no operating cost.

### Decision tree




**Figure 4:** Decision tree for the ski lifts (payoffs in \$000)

### Expected payoffs

$$\text{One lift: } 0.3(191) + 0.5(240) + 0.2(240) = 57.3 + 120 + 48 = 225.3$$

$$\text{Two lifts: } 0.3(151) + 0.5(245) + 0.2(441) = 45.3 + 122.5 + 88.2 = 256.0$$

At the decision node the alternative with the higher expected payoff is chosen, and the one-lift branch is pruned.

**Answer:** Purchase two lifts: expected payoff \$256,000 against \$225,300 for one lift.

| | | | | |
|---|---|---|---|---|
| **6** | Consider amazon.com, whose Website enjoys millions of "hits" each day and puts customers in touch with millions of services and products. What are amazon.com's competitive priorities and what should its operations strategy focus on?<br>Source: Unit I \| SEE Jan/Feb 2021 (18IM54), Q2a | 06 | – | – |

**Competitive priorities of amazon.com**

- **Variety (primary priority):** handling a wide assortment of services or products efficiently. Amazon.com uses information technology and streamlined customer relationship and order fulfillment processes to reliably deliver a vast variety of items.
- **Delivery speed and on-time delivery:** quickly filling each order and meeting the promised delivery date (one-day and two-day delivery).
- **Low-cost operations:** competitive prices made possible by high volumes, automation and efficient fulfillment centers.
- **Consistent quality:** accurate orders, a reliable website and dependable service on every transaction.
- **Volume flexibility:** handling large demand peaks, for example during festive and sale seasons.

**What its operations strategy should focus on**

Operations strategy specifies how operations implements corporate strategy; any gap between a competitive priority and the capability to deliver it must be closed. For amazon.com the strategy should focus on:

(1) **Information technology:** a fast, reliable website, recommendation and search systems, and real-time inventory information linked to suppliers.
(2) **Order fulfillment process:** a network of automated fulfillment centers, inventory placement close to customers, and efficient picking, packing and shipping.
(3) **Supply chain and supplier relationships:** coordination with suppliers and third-party sellers, and reliable logistics partners and own delivery fleet.
(4) **Capacity planning:** enough warehouse, server and delivery capacity (with a cushion) to handle peak demand.
(5) **Customer relationship process:** easy ordering, tracking, returns and customer service to retain millions of customers.

| | | | | |
|---|---|---|---|---|
| **7** | Identify nine competitive priorities used in operations strategy and explain how a consistent pattern of decisions can develop organizational capabilities.<br>Source: Unit I \| SEE Feb/Mar 2025 (IM352IA), Q2a | 10 | 2 | 1 |

**Competitive priorities** are the critical operational dimensions that a process or supply chain must possess to satisfy internal or external customers, now and in the future. The nine broad priorities fall into four capability groups: cost, quality, time and flexibility.

| Competitive priority | Definition | Example |
|---|---|---|
| **Cost** | | |
| 1. Low-cost operations | Delivering a service or product at the lowest possible cost to the satisfaction of customers | Costco designs all processes for efficiency, stacks products on pallets in warehouse-type stores and negotiates aggressively with suppliers. |
| **Quality** | | |
| 2. Top quality | Delivering an outstanding service or product | Rolex is known globally for precision timepieces. |
| 3. Consistent quality | Producing services or products that meet design specifications on a consistent basis | McDonald's standardizes work methods, staff training and procurement to give the same quality in every store. |
| **Time** | | |
| 4. Delivery speed | Quickly filling a customer's order | Dell's integrated, agile supply chain delivers computers with short lead times. |
| 5. On-time delivery | Meeting delivery-time promises | UPS delivers a very large volume of shipments on time across the globe. |

| | | |
|---|---|---|
| 6. Development speed | Quickly introducing a new service or product | Zara brings fashionable designs from the runway to the market quickly. |
| **Flexibility** | | |
| 7. Customization | Satisfying the unique needs of each customer by changing service or product designs | Ritz-Carlton customizes services to individual guest preferences. |
| 8. Variety | Handling a wide assortment of services or products efficiently | Amazon.com reliably delivers a vast variety of items. |
| 9. Volume flexibility | Accelerating or decelerating the rate of production quickly to handle large fluctuations in demand | The postal service handles severe demand peaks with flexibly designed sorting and dispatch processes. |

**How a consistent pattern of decisions develops capabilities**

- The current operations strategy of any firm is really the **pattern of decisions** made for its processes and supply chains, even if it is not formally stated.
- After the competitive priorities of a process are set, its **competitive capabilities** are assessed. Any gap between a priority and the capability to achieve it must be closed by the operations strategy; developing capabilities and closing gaps is the thrust of operations strategy.
- Good, preferably quantitative, **measures** are identified for each priority; data are gathered to determine current capabilities; gaps are found by comparing capabilities with management's targets; and unacceptable gaps are closed by appropriate actions.
- When decisions on process design, capacity, inventory, quality and the supply chain are made **consistently** in support of the chosen priorities, they reinforce one another and build distinctive capabilities over time. Priorities must evolve with business conditions and customer preferences, so the cycle is repeated.

| | | | | |
|---|---|---|---|---|
| **8** | Describe the process view of operations in typical organizations with the help of a block diagram.<br>Source: Unit I \| SEE Feb/Mar 2025 (IM352IA), Q2b | 06 | 2 | 1 |

A **process** is any activity or group of activities that takes one or more inputs, transforms them and provides one or more outputs for its customers. The process view of the firm gives a more relevant picture of how firms actually work than the departmental (functional) view, because a process has its own objectives, involves a work flow that cuts across departmental boundaries and requires resources from several departments.

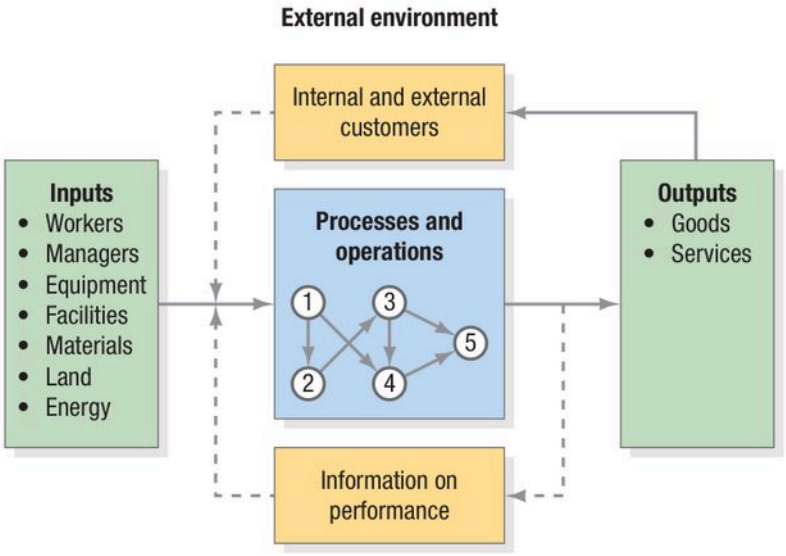



**Figure 5:** Process view of operations: processes and operations

- **Inputs:** human resources (workers and managers), capital (equipment and facilities), purchased materials and services, land and energy.
- **Processes and operations:** the numbered circles are operations through which services, products or customers pass and where processes are performed; the arrows are flows, which can cross because different jobs or customers have

different requirements.

- **Outputs:** goods and services delivered to internal and external customers.
- **Special inputs (dashed lines):** participation by customers and information on performance from internal and external sources.

The figure can represent a whole firm, a department, a small group or even a single individual: each has inputs and uses processes at various operations to provide outputs. An organization is only as effective as its processes, so understanding how processes work is the key to success.

| | | | | |
|---|---|---|---|---|
| **9** | Differentiate between competitive priorities and competitive capabilities with suitable examples.<br>Source: Unit I \| SEE Apr/May 2024 (21IM53), Q2a | 08 | 2 | 1 |

**Competitive priorities:** the critical operational dimensions a process or supply chain must possess to satisfy internal or external customers, now and in the future. They are planned for processes and the supply chain and must be present to maintain or build market share. Not all priorities are critical for a given process; management selects the most important ones.

**Competitive capabilities:** the cost, quality, time and flexibility dimensions that a process or supply chain actually possesses and is able to deliver. When a capability falls short of the priority attached to it, management must close the gap or revise the priority.

| Basis | Competitive priorities | Competitive capabilities |
|---|---|---|
| Meaning | What the process must be good at (planned, desired) | What the process is actually able to deliver (achieved) |
| Nature | Target or goal set by management from market analysis | Measured current performance |
| Use | Guides the design of processes and the operations strategy | Compared with priorities to identify performance gaps |
| Example | Low-cost operations: Costco designs all processes for efficiency and negotiates aggressively with suppliers | Cost per billing statement of $0.0813 against a target of $0.06 shows a gap in low-cost capability |

**Example: billing and payment process of a credit card division**

| Competitive priority | Measure | Capability / gap | Action |
|---|---|---|---|
| Low-cost operations | Cost per billing statement | $0.0813; target is $0.06 | Eliminate microfilming and storage of billing statements |
| Consistent quality | Percent errors in bill information | 0.90%; acceptable | No action |
| Volume flexibility | Utilization | 98%; too high to support a rapid increase in volume | Acquire temporary employees |

The process shows a significant gap in low-cost capability, so management redesigns it to reduce cost without impairing the other priorities.

| | | | | |
|---|---|---|---|---|
| **10** | IKEA shows how important operations management is for its own success and the success of any type of organization. In this view identify some of the activities that IKEA's operations managers are involved in through highlighting some eight important aspects.<br>Source: Unit I \| SEE Apr/May 2024 (21IM53), Q2b | 08 | 3 | 2 |

IKEA competes on low cost, good design and wide variety, and its success depends on well-managed operations. Operations managers are involved in both strategic and tactical decisions. Eight important aspects are:

(1) **Developing and maintaining capabilities:** building and sustaining the capabilities needed to serve customers, such as IKEA's ability to offer stylish furniture at low prices.
(2) **Designing new processes:** for example, the flat-pack design, self-service warehouse areas in stores and customer self-assembly, which cut handling and transport cost.
(3) **Developing and organizing the supply chain:** selecting and coordinating suppliers worldwide and linking them with distribution centers and stores.
(4) **Managing key performance measures:** controlling cost and quality, the measures through which IKEA gains its competitive edge.
(5) **Process improvement and performance measurement:** continuously improving store layouts, packaging and logistics to reduce waste and cost.
(6) **Managing and planning projects:** opening new stores and distribution centers, launching new product ranges and implementing new systems.
(7) **Production and staffing plans (capacity):** planning output with suppliers and staffing stores to meet seasonal demand.
(8) **Managing inventories and scheduling resources:** keeping the right stock in stores and warehouses and scheduling deliveries, staff and replenishment.

These decisions must reflect corporate strategy; at IKEA every operations decision supports the strategy of low cost with good design.

| | | | | |
|---|---|---|---|---|
| **11** | Explain the term operations management and identify the three major functional areas of business organizations and briefly describe how they are interrelated.<br>Source: Unit I \| SEE Mar/Apr 2023 (18IM54), Q2a | 08 | – | – |

**Operations management** refers to the systematic design, direction and control of processes that transform inputs into services and products for internal as well as external customers.

### Three major functional areas

Operations is one of the key functions of an organization. The three mainline functions of any business are (1) **operations**, (2) **marketing** and (3) **finance**. Each has its own knowledge and skill areas, responsibilities, processes and decision domains.

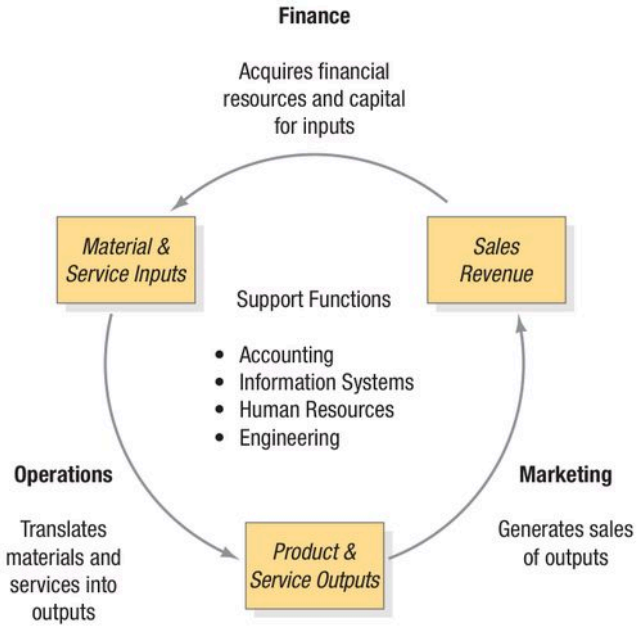



**Figure 6:** Integration between the functional areas of a business

### How they are interrelated

- **Finance** generates resources, capital and funds from investors and from sales of goods and services.
- Based on business strategy, **finance and operations** decide how to invest these resources and convert them into physical assets and material inputs.
- **Operations** transforms these material and service inputs into product and service outputs, which must match what can be sold in the selected markets.
- **Marketing** generates sales revenue from the outputs, which becomes returns to investors and capital for supporting operations, completing the cycle.
- Support functions such as accounting, information systems, human resources and engineering provide essential information, services and managerial support.

How well these functions work together determines the effectiveness of the organization; they should be integrated and pursue a common strategy.

| | | | | |
|---|---|---|---|---|
| **12** | Describe the new trends and challenges in operations management.<br>Source: Unit I \| SEE Mar/Apr 2023 (18IM54), Q2b | 08 | – | – |

Several trends have a great impact on operations management: productivity improvement; global competition; and ethical, workforce-diversity and environmental issues. In addition, accelerating change in information technology, e-commerce, robotics and the Internet is changing the design of services and products and of sales, order fulfillment and purchasing processes.

### 1. Productivity improvement

All firms face pressure to improve their processes and supply chains to compete with domestic and foreign competitors. Productivity is the value of outputs produced divided by the value of input resources:

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}}$$

### 2. Global competition

To prosper, businesses must view customers, suppliers, facility locations and competitors in global terms. Five developments spurred the need for sound global strategies: (1) improved transportation and communication technologies, (2) loosened regulation of financial institutions, (3) increased demand for imported services and goods, (4) reduced import quotas and trade barriers through regional trading blocks such as the EU and NAFTA, and (5) comparative cost advantages. The challenges include the risk of losing proprietary technology to offshore suppliers and political risks.

**3. Ethical, workforce diversity and environmental issues**

Businesses face more ethical dilemmas than ever, intensified by global presence and rapid technological change. Managing a diverse workforce fairly is essential. Environmental problems such as toxic waste, polluted water, poor air quality and climate change are now seen as survival issues rather than only quality-of-life issues. Issues of ethics, workforce diversity and the environment are becoming part of every manager's job.

**4. Technology and digitalization**

Information technology, e-commerce, automation and robotics change how services and products are designed, sold and delivered, and require new skills and new process designs.

| | | | | |
|---|---|---|---|---|
| **13** | A retailer must decide whether to build a small or large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6 respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If the small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate maximum demand through local advertising. The response to advertising may be either modest or sizable, with probabilities estimated to be 0.3 and 0.7 respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative – building a small facility or building a large facility – has the higher expected payoff?<br>Source: Unit I \| SEE Jan/Feb 2021 (18IM54), Q2b | 10 | – | – |

*The problem is identical to Part B Q3; the decision tree, analysis and answer are the same.*

| | | | | |
|---|---|---|---|---|
| **14** | Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. Identify and discuss the factors connecting corporate strategy and key operations management decisions with block diagram.<br>Source: Unit I \| SEE Jan/Feb 2021 (16IM55); Nov/Dec 2019 (16IM55), Q2a (identical) | 08 | – | – |

Developing a customer-driven operations strategy begins with **corporate strategy**, which coordinates the firm's overall goals with its core processes. It determines the markets the firm will serve and its responses to changes in the environment, provides resources to develop core competencies and core processes, and identifies the strategy for international markets. A **market analysis** then categorizes customers, identifies their needs and assesses competitors' strengths; this is used to develop **competitive priorities**, which guide the design of services, products and processes.

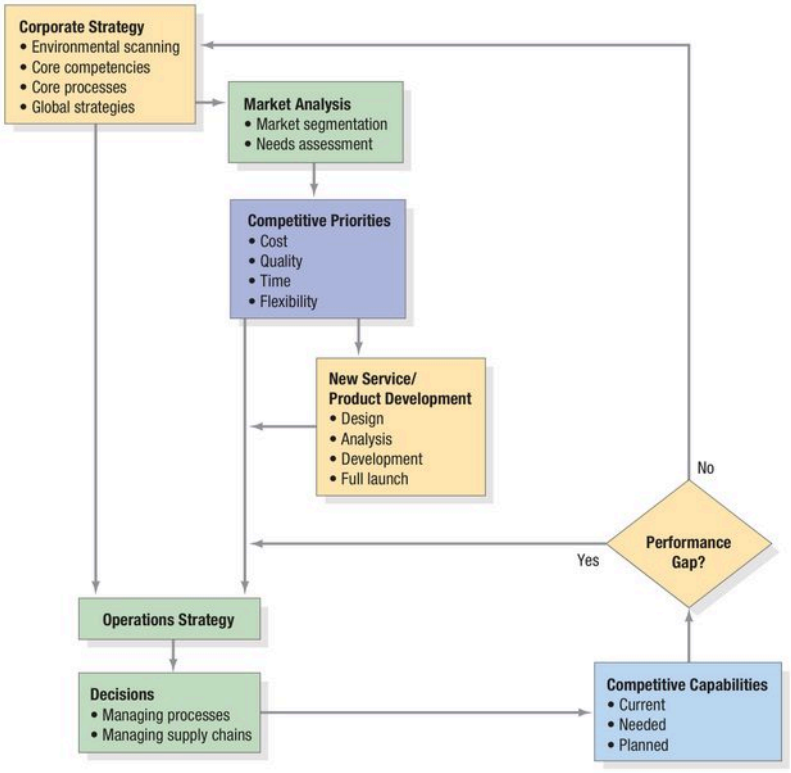



**Figure 7:** Connection between corporate strategy and key operations management decisions

### Factors connecting corporate strategy and operations decisions

(1) **Corporate strategy:** involves four considerations: (a) environmental scanning, monitoring trends in the industry, marketplace and society for opportunities or threats; (b) identifying and developing core competencies, the unique resources and strengths of the organization; (c) developing core processes (customer relationship, new service/product development, order fulfillment and supplier relationship); and (d) developing global strategies.

(2) **Market analysis:** divides customers into market segments (market segmentation) and identifies the needs of each segment (needs assessment).

(3) **Competitive priorities:** cost, quality, time and flexibility, set for each process from the market analysis.

(4) **New service/product development:** design, analysis, development and full launch of the services and products the priorities call for.

(5) **Operations strategy:** specifies how operations will achieve the competitive priorities.

(6) **Decisions:** managing processes and managing supply chains.

(7) **Competitive capabilities:** current, needed and planned capabilities are compared with the priorities. If there is a performance gap, the operations strategy is revised; developing the operations strategy is therefore a continuous process.

| | | | | |
|---|---|---|---|---|
| **15** | Calculate the productivity for the following operations: (i) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. (ii) A team of workers makes 400 units of a product, which is sold in the market for \$10 each. The accounting department reports that for this job the actual costs are \$400 for labour, \$1,000 for materials, and \$300 for overhead.<br>Source: Unit I \| SEE Jan/Feb 2021 (16IM55); Nov/Dec 2019 (16IM55), Q2b (identical) | 08 | – | – |

**Productivity** is the value of outputs produced divided by the value of input resources. **Labour productivity** is an index of output per person or per hour worked; **multifactor productivity** is the value of output divided by the sum of labour, materials and overhead costs.

**(i) Labour productivity**

$$\text{Labour productivity} = \frac{\text{Policies processed}}{\text{Employee hours}} = \frac{600 \text{ policies}}{(3 \text{ employees})(40 \text{ hours/employee})} = 5 \text{ policies/hour}$$

**(ii) Multifactor productivity**

$$\text{Multifactor productivity} = \frac{\text{Value of output}}{\text{Labour} + \text{Materials} + \text{Overhead}} = \frac{(400 \text{ units})(\$10/\text{unit})}{\$400 + \$1{,}000 + \$300} = \frac{\$4{,}000}{\$1{,}700} = 2.35$$

**Answer:** (i) 5 policies per hour; (ii) multifactor productivity = 2.35.

Multifactor productivity should be as high as possible. These measures are compared with performance in earlier periods and with future goals; if they fall short, the process is investigated for improvement.

| | | | | |
|---|---|---|---|---|
| **16** | Define operations management. Explain with the help of a block diagram the process view of operations management and write how processes work.<br>Source: Unit I \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q2a | 06 | – | – |

**Operations management** refers to the systematic design, direction and control of processes that transform inputs into services and products for internal as well as external customers.

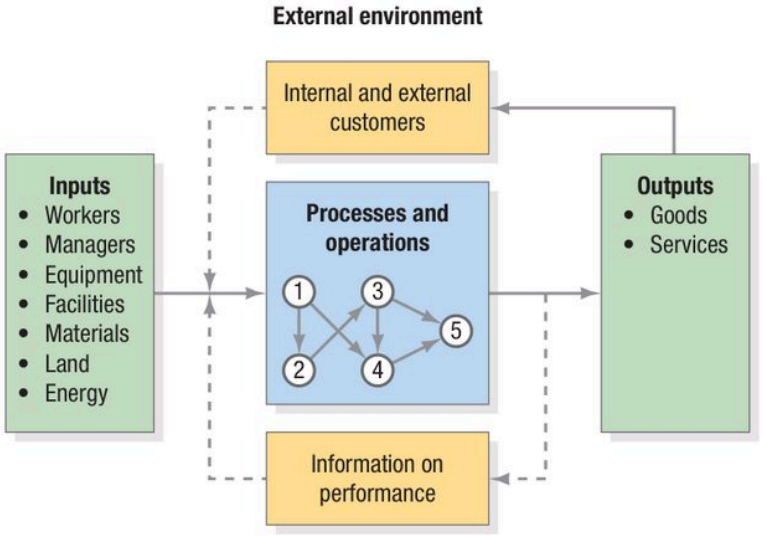



**Figure 8:** Process view of operations management

**How processes work**

- A **process** takes one or more inputs, transforms them and provides one or more outputs for its customers; processes are grouped into **operations**, each a group of resources performing all or part of one or more processes.
- **Inputs** (workers, managers, equipment, facilities, materials, land and energy) flow into the processes performed at the operations (numbered circles); the arrows show the flows, which can cross because different jobs or customers need different routes.
- The processes produce **outputs** of goods and services for internal and external customers.

- Customers participate in the process, and **information on performance** from internal and external sources is fed back to control and improve the processes (dashed lines).
- Processes are linked together to form a **supply chain** that cuts across departmental and company boundaries.

*The key terms of the definition are explained in detail in Part B Q1, and the process view in Part B Q8.*

| | | | | |
|---|---|---|---|---|
| **17** | A manager is trying to decide whether to build a small, medium or large facility. Demand can be low, average or high with estimated probabilities being 0.25, 0.40 and 0.35 respectively. A small facility is expected to earn an after tax net present value of just \$18,000 if demand is low. If demand is average, the small facility is expected to earn \$75,000; it can be increased to medium size to earn a net present value of \$60,000. If demand is high, the small facility is expected to earn \$75,000 and can be expanded to medium size to earn \$60,000 or to large size to earn \$125,000; a medium sized facility is expected to lose an estimated \$25,000 if demand is low and earn \$140,000 if demand is average. If the demand is high the medium sized facility is expected to earn a net present value of \$150,000; it can be expanded to a large size for a net pay off of \$145,000. If a large facility is built and demand is high, earnings are expected to be \$220,000. If demand is average for the large facility the present value is expected to be \$125,000; if demand is low, the facility is expected to lose \$60,000. (i) Draw a decision tree for this problem. (ii) What should the management do to achieve the highest expected pay off?<br>Source: Unit I \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q2b | 10 | – | – |

In a decision tree, square nodes are decision points and circular nodes are chance events with their probabilities. The tree is solved from right to left: at a chance node the expected payoff is the sum of payoff × probability, and at a decision node the alternative with the best expected payoff is chosen and the others are pruned.

**(i) Decision tree (payoffs in \$000)**

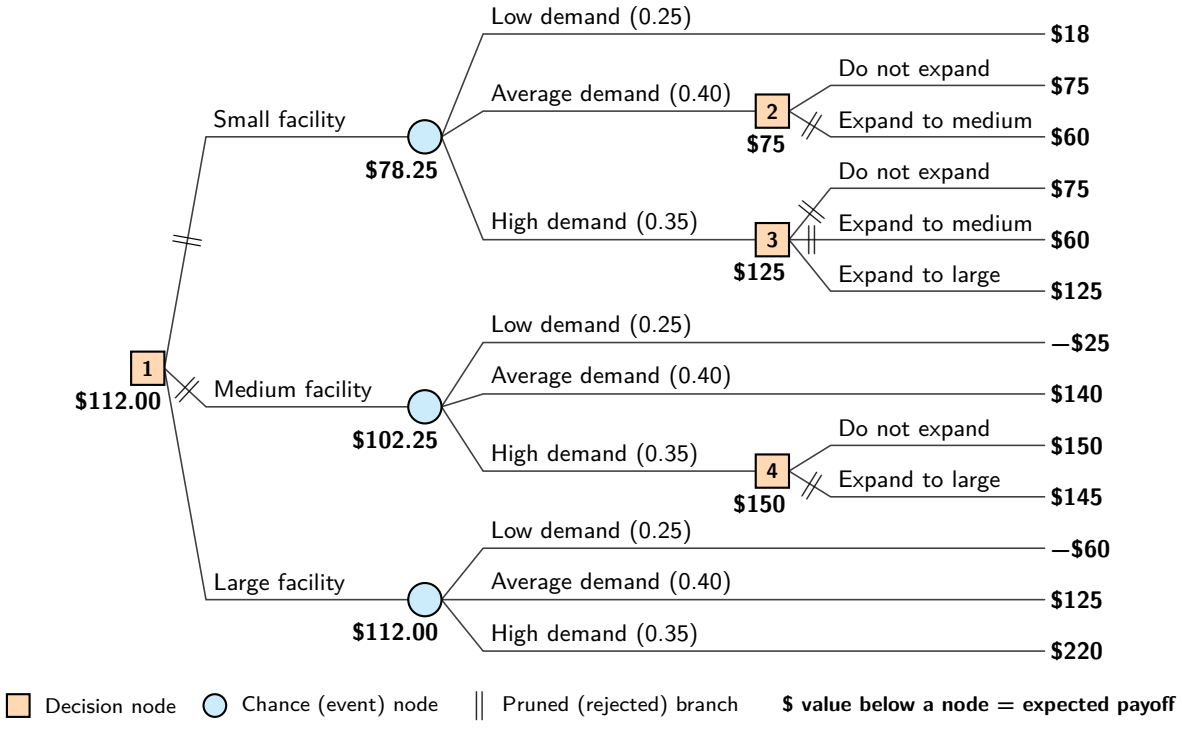



**Figure 9:** Decision tree for the facility-size decision (payoffs in \$000)

**(ii) Analysis, working from right to left**

Decision node 2 (small, average demand): $\max(75,\ 60) = 75$

Decision node 3 (small, high demand): $\max(75,\ 60,\ 125) = 125$

Decision node 4 (medium, high demand): $\max(150,\ 145) = 150$

Small facility: $0.25(18) + 0.40(75) + 0.35(125) = 4.50 + 30.00 + 43.75 = 78.25$

Medium facility: $0.25(-25) + 0.40(140) + 0.35(150) = -6.25 + 56.00 + 52.50 = 102.25$

Large facility: $0.25(-60) + 0.40(125) + 0.35(220) = -15.00 + 50.00 + 77.00 = 112.00$

Decision node 1: $\max(78.25,\ 102.25,\ 112.00) = 112.00$

**Answer:** Management should build the large facility: expected payoff \$112,000 (medium \$102,250; small \$78,250).

| | | | | |
|---|---|---|---|---|
| **18** | How does the process view and the supply chain view of operations management help us understand the need and importance of operations management as a strategic function in the organization? Elaborate on these two perspectives of operations management.<br>Source: Unit I \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q2a | 08 | – | – |

### Process view

A process view of the firm gives a more relevant picture of how firms actually work than the departmental view. Departments have their own objectives, resources and managers, but a process has its own objectives, involves a work flow that cuts across departmental boundaries and requires resources from several departments. Since every department has processes that must be managed effectively to gain competitive advantage, operations management is relevant to all functions, not only to the operations department.

### Supply chain view

Most services and products are produced through a series of interrelated business activities, and each activity should add value to the preceding ones; waste and unnecessary cost are eliminated. The supply chain view adds the strategic insight that processes must add value for customers throughout the supply chain, linking the firm's internal processes with those of its external customers and suppliers.

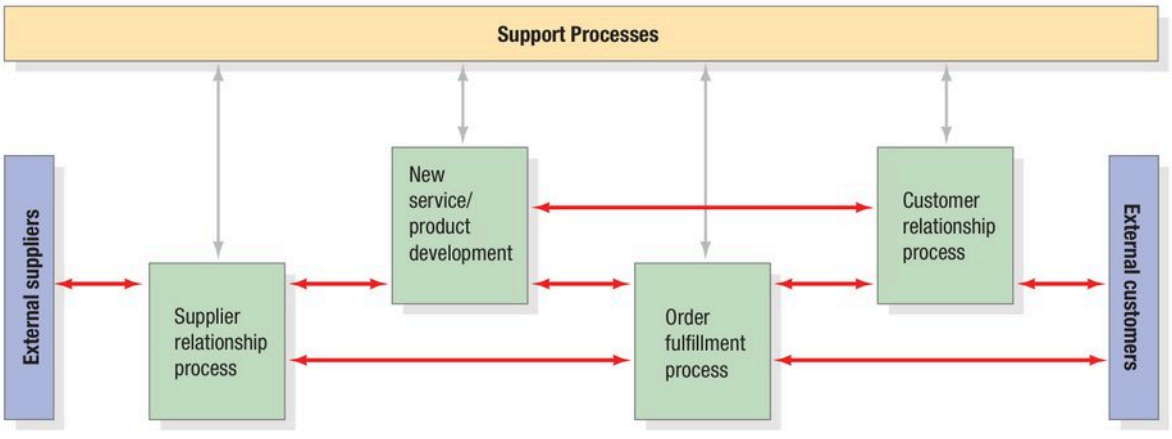



**Figure 10:** Supply chain linkages showing work and information flows

- **Core processes** deliver value to external customers: the **supplier relationship process** selects suppliers and ensures the timely flow of materials, services and information into the firm; the **new service/product development process** designs and develops new services and products; the **order fulfillment process** produces and delivers the service or product; and the **customer relationship process** identifies, attracts and builds relationships with external customers and facilitates their orders.
- **Support processes** provide vital resources and inputs to the core processes and are essential to the management of the business.

**Operations as a strategic function**

Operations strategy is the linchpin that brings these processes together to form supply chains extending beyond the walls of the firm to suppliers and customers. Operations managers are responsible for key performance measures such as cost and quality, which determine how the firm gains a competitive edge. Together, the process and supply chain views show why operations management is a strategic function for the whole organization.

| | | | | |
|---|---|---|---|---|
| **19** | The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before commercializing the device, the owner wants reasonable assurance of success. Variable costs are \$7 per unit produced and sold. Fixed costs are about \$500,000 per year. (i) If the price is reduced to \$15 and forecasted sales for the first year are 10,000 units, what would be the product's total contribution to profits in the first year? (ii) If the selling price is set at \$25, how many units must be produced and sold to break even? Use both algebraic and graphical approaches.<br>Source: Unit I \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q2b | 08 | – | – |

The **break-even quantity** is the volume at which total revenue equals total cost. With fixed cost F, variable cost per unit c and price p:

$$\text{Total cost} = F + cQ, \qquad \text{Total revenue} = pQ, \qquad Q = \frac{F}{p-c}$$

**(i) Contribution to profit at a price of \$15 and 10,000 units**

$$\text{Total profit contribution} = pQ - (F + cQ)$$

$$= 15(10{,}000) - [500{,}000 + 7(10{,}000)] = 150{,}000 - 570{,}000 = -\$420{,}000$$

**Answer:** The product would make a loss of \$420,000 in the first year.

**(ii) Break-even quantity at a price of \$25**

**Algebraic approach:**

$$Q = \frac{F}{p-c} = \frac{500{,}000}{25-7} = \frac{500{,}000}{18} = 27{,}777.8 \approx 27{,}778 \text{ units}$$

**Graphic approach:** draw the total revenue line (25Q) through (0, 0) and (40,000 units, \$1,000,000), and the total cost line (500,000 + 7Q) through (0, \$500,000) and (40,000 units, \$780,000). The lines intersect at the break-even point.

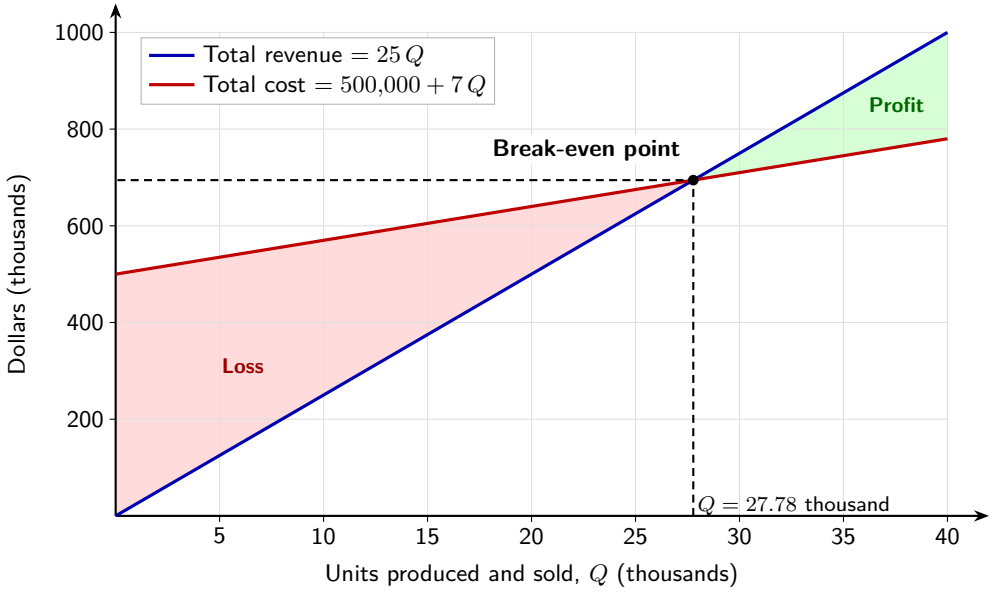



**Figure 11:** Break-even chart (F = \$500,000, c = \$7, p = \$25)

**Answer:** Break-even quantity = 27,778 units (total revenue = total cost = \$694,444).

| | | | | |
|---|---|---|---|---|
| **20** | The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Variable costs are estimated at \$7 per unit produced and sold. Fixed costs are about \$56,000 per year. (i) If the selling price is set at \$25, how many units must be produced and sold to break even? (ii) Forecasted sales for the first year are 10,000 units if the price is reduced to \$15. With this pricing strategy, what would be the product's total contribution to profits in the first year?<br>Source: Unit I \| SEE Feb/Mar 2026 (IM352IA), Q2a | 08 | 3 | 1 |

The **break-even quantity** is the volume at which total revenue equals total cost: total cost = F + cQ, total revenue = pQ, so Q = F/(p − c).

**(i) Break-even quantity at a price of \$25**

**Algebraic approach:**

$$Q = \frac{F}{p-c} = \frac{56{,}000}{25-7} = \frac{56{,}000}{18} = 3{,}111 \text{ units}$$

**Graphic approach:** total revenue = 25Q passes through (0, 0) and (8,000 units, \$200,000); total cost = 56,000 + 7Q passes through (0, \$56,000) and (8,000 units, \$112,000). The lines intersect at Q = 3,111 units.

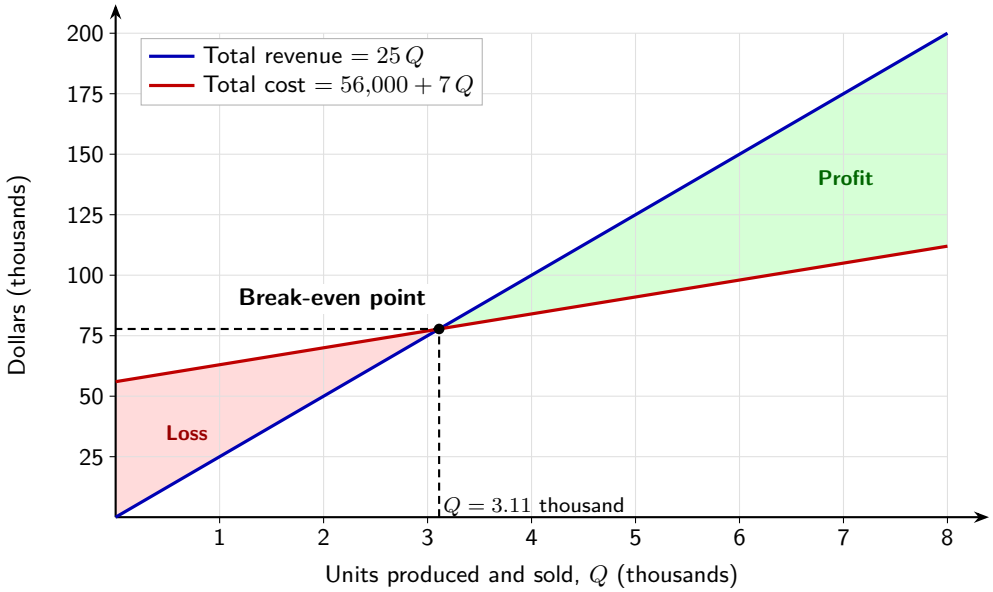



**Figure 12:** Break-even chart (F = \$56,000, c = \$7, p = \$25)

**Answer:** Break-even quantity = 3,111 units (total revenue = total cost ≈ \$77,778).

**(ii) Contribution to profit at a price of \$15 and 10,000 units**

$$\text{Total profit contribution} = pQ - (F + cQ) = 15(10{,}000) - [56{,}000 + 7(10{,}000)]$$

$$= 150{,}000 - 126{,}000 = \$24{,}000$$

**Answer:** Contribution to profit in the first year = \$24,000.

| | | | | |
|---|---|---|---|---|
| **21** | With suitable example, distinguish between manufacturing processes and service processes.<br>Source: Unit I \| SEE Feb/Mar 2026 (IM352IA), Q2b | 08 | 2 | 1 |

*The answer is the same as for Part B Q2.*

| | | | | |
|---|---|---|---|---|
| **22** | Explain the major process strategy decisions and their implications for operations.<br>Source: Unit II \| CIE-II, 07/01/2025 | 10 | 2 | 2 |

Process strategy specifies the pattern of decisions made in managing processes so that they achieve their competitive priorities. There are four basic process decisions: (1) process structure (including layout), (2) customer involvement, (3) resource flexibility and (4) capital intensity. They apply to both service and manufacturing processes.

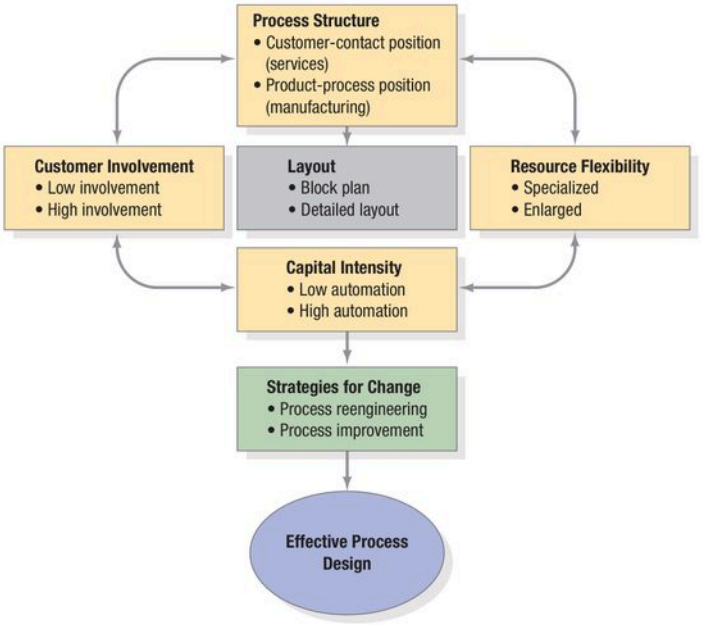



**Figure 13:** Major decisions for effective processes

### The four decisions

**Process structure:** determines the process type relative to the kinds of resources needed, how resources are partitioned between them and their key characteristics. A layout, the physical arrangement of operations, puts these decisions into tangible form.

**Customer involvement:** reflects the ways in which customers become part of the process and the extent of their participation.

**Resource flexibility:** is the ease with which employees and equipment can handle a wide variety of products, output levels, duties and functions.

**Capital intensity:** is the mix of equipment and human skills in a process; the greater the relative cost of equipment, the greater the capital intensity.

### Implications for operations

The choices must fit the situation and make sense together; one process should not be optimized at the expense of others.

**High customer contact (front-office service process):**

(1) **Process structure:** the customer is present, actively involved and receives personal attention, giving high divergence and flexible process flows.
(2) **Customer involvement:** customers are more likely to become part of the process, and the service created for each customer is unique.
(3) **Resource flexibility:** high divergence and flexible flows need a flexible workforce, facilities and equipment.
(4) **Capital intensity:** automation and capital intensity become more likely as volume increases.

**High volume per part type (manufacturing process):**

(1) **Process structure:** high volume and a standard product make a line flow possible (the opposite of a job process producing to specific customer orders).
(2) **Customer involvement:** not a factor in most manufacturing processes, except for choices on product variety and customization.
(3) **Resource flexibility:** with high volume and low divergence, flexibility is not needed; specialization gives more efficient processes.
(4) **Capital intensity:** high volumes justify the large fixed costs of an efficient, automated operation.

| | | | | |
|---|---|---|---|---|
| **23** | Richard Garber is the head designer for Matthews and Novak Design Company. Garber has been called in to design the layout for a newly constructed office building. From statistical samplings over the past three months, Garber developed the following closeness matrix for daily trips between the department's offices.<br><br>a. If other factors are equal, which two offices should be located closest together?<br>b. A figure shows an alternative layout for the department: top row C, F, A; bottom row B, E, D. What is the total weighted-distance score for this plan based on rectilinear distance, assuming that offices A and B are three units of distance apart?<br>Source: Unit II \| CIE-II, 07/01/2025 | 10 | 3 | 2 |

| Department | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| A | – | 25 | 90 | | | 185 |
| B | | – | | | 105 | |
| C | | | – | | 125 | 125 |
| D | | | | – | 25 | |
| E | | | | | – | 105 |
| F | | | | | | – |

### (a) Offices that should be closest together

The closeness factor measures the need for proximity. The largest entry in the matrix is 185 trips per day, between offices A and F.

**Answer:** Offices A and F should be located closest together (185 trips per day).

### (b) Weighted-distance score of the proposed plan

The weighted-distance (wd) score is the sum of (closeness factor × distance) over all pairs of offices; the lower the score, the better the layout. Rectilinear distance is measured along a series of 90-degree turns:

$$d_{AB} = |x_A - x_B| + |y_A - y_B|$$

| | | |
|---|---|---|
| **C** | **F** | **A** |
| **B** | **E** | **D** |

**Figure 14:** Proposed block plan (top row C, F, A; bottom row B, E, D)

Adjacent offices are one unit apart, so A (top right) and B (bottom left) are 2 + 1 = 3 units apart, as stated in the question.

| Pair | Trips (w) | Rectilinear distance (d) | w × d |
|---|---|---|---|
| A–B | 25 | 3 | 75 |
| A–C | 90 | 2 | 180 |
| A–F | 185 | 1 | 185 |
| B–E | 105 | 1 | 105 |
| C–E | 125 | 2 | 250 |
| C–F | 125 | 1 | 125 |
| D–E | 25 | 1 | 25 |
| E–F | 105 | 1 | 105 |
| | | **Total** | **1,050** |

**Answer:** Total weighted-distance score of the plan = 1,050.

| | | | | |
|---|---|---|---|---|
| **24** | Explain the concepts of economies and diseconomies of scale with appropriate examples.<br>Source: Unit II \| Improvement CIE, 12/01/2026; SEE Fast Track Jul 2019 (16IM55/12IM62), Q6a | 10 | 2 | 2 |

### Economies of scale

**Economies of scale** means that the average unit cost of a service or good can be reduced by increasing its output rate. Four principal reasons explain why costs fall as output increases:

(1) **Fixed costs are spread over more units.** Heating, debt service, managers' salaries and depreciation do not vary with output in the short term; as the output (utilization) rate rises, the average unit cost falls.

(2) **Construction costs are reduced.** Building permits, architects' fees and rental of building equipment are needed for small and large facilities alike; doubling the size of a facility usually does not double its construction cost.

(3) **Costs of purchased materials are cut.** Higher volumes give a better bargaining position and quantity discounts. Walmart gains large economies of scale because its stores buy and sell huge volumes of each item.

(4) **Process advantages are found.** High volume shifts the process toward a line process with dedicated resources and justifies more efficient, specialized technology. Benefits include a faster learning effect, lower inventory, better process and job design and fewer changeovers.

### Diseconomies of scale

Beyond a certain point a facility becomes so large that **diseconomies of scale** set in: the average cost per unit increases as the facility size increases. Excessive size brings complexity, loss of focus and inefficiencies. Too many layers of employees and bureaucracy cause management to lose touch with employees and customers, and the less agile organization loses the flexibility needed to respond to changing demand.

### Example: hospitals

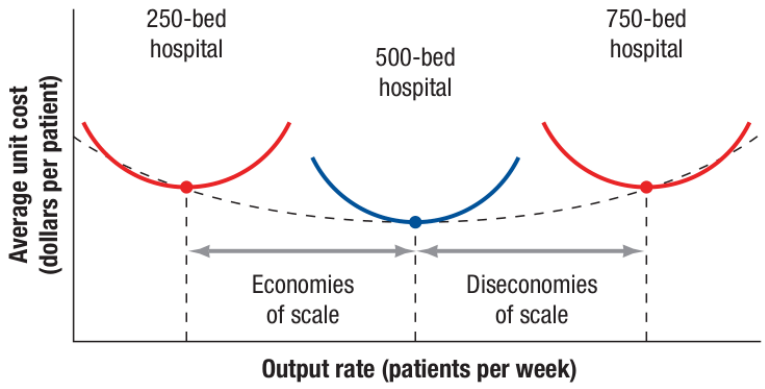



**Figure 15:** Economies and diseconomies of scale (hospital example)

The 500-bed hospital shows economies of scale because its average unit cost at its best operating level is lower than that of the 250-bed hospital. Further expansion to 750 beds raises the average unit cost, showing diseconomies of scale. The

optimal size depends on the number of patients to be served: a small community may be served more cheaply by a 250-bed hospital, and a large community more efficiently by two 500-bed hospitals than by one 1,000-bed hospital.

| | | | | |
|---|---|---|---|---|
| **25** | Explain process reengineering and process improvement, highlighting their differences with suitable examples.<br>Source: Unit II \| Improvement CIE, 12/01/2026 | 10 | 4 | 3 |

Process reengineering and process improvement are two different but complementary strategies for changing processes.

### Process reengineering

**Reengineering** is the fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service and speed. It is about reinvention rather than incremental improvement. It is strong medicine: layoffs and large investments in information technology usually accompany such massive change, but the payoffs can be large. It is carried out by a cross-functional team, uses information technology as the primary enabler, and takes a clean-slate view starting from what the customer wants.

### Process improvement

**Process improvement** is the systematic study of the activities and flows of each process to improve it. Its purpose is to "learn the numbers", understand the process and dig out the details; once a process is really understood it can be improved. Tasks are streamlined, unnecessary steps are eliminated, expensive materials or services are cut, the work environment is improved and jobs are made safer, so that costs and delays fall and customer satisfaction rises. It goes on continuously, whether or not a process is reengineered.

### Differences

| Basis | Process reengineering | Process improvement |
|---|---|---|
| Nature of change | Radical, fundamental rethinking; reinvention of the process | Incremental, continuous improvement of the existing process |
| Scope | Core business processes, usually cross-functional | Any process; one activity or flow at a time |
| Cost and risk | High investment (IT), layoffs, high risk of failure | Low cost and low risk |
| Who carries it out | Cross-functional team led by top management | Individuals or teams who perform the work each day |
| Time frame | One-time, large project | Ongoing, never-ending |
| Results | Dramatic gains in cost, quality, service and speed | Steady, cumulative gains |

### Examples

- **Reengineering:** most reengineering projects redesign processes around information flows, such as customer order fulfillment. A bank that replaces a hand-off between several departments for loan approval with a single case worker supported by an integrated IT system cuts approval time from weeks to hours.
- **Process improvement:** the people who perform the work each day usually understand it best. Workers on an assembly line who study changeover steps and eliminate unnecessary movements reduce setup time and delays (for example, through kaizen or quality circles).

| | | | | |
|---|---|---|---|---|
| **26** | Describe a systematic approach to process analysis with the help of a block diagram.<br>Source: Unit II \| Improvement CIE, 12/01/2026 | 10 | 2 | 2 |

**Process analysis** is the documentation and detailed understanding of how work is performed and how it can be redesigned. A systematic approach to process analysis has six steps.

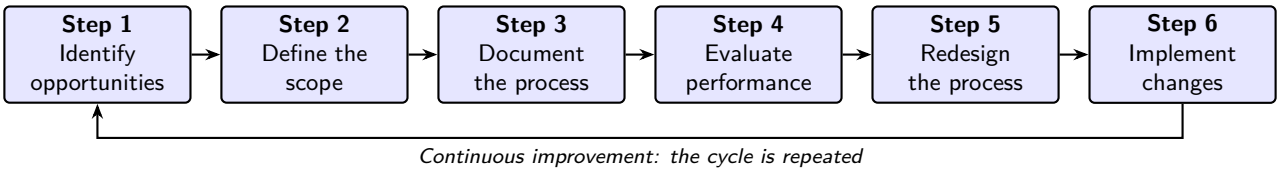



**Figure 16:** Systematic approach to process analysis

(1) **Identify opportunities:** pay attention to the core processes (customer relationship, new service/product development, order fulfillment, supplier relationship) and support processes; look for problems in performance, customer complaints, employee suggestions and benchmarking gaps.

(2) **Define the scope:** establish the boundaries of the process to be analyzed (a broad process or a narrow sub-process), the resources available and the team that will carry out the analysis.

(3) **Document the process:** list the inputs, suppliers, outputs and customers, and document the steps using tools such as flowcharts, swim-lane flowcharts, service blueprints and process charts.

(4) **Evaluate performance:** establish metrics (performance measures) for the process and its steps, collect data and identify disconnects and bottlenecks, using tools such as checklists, histograms, Pareto charts, scatter diagrams and cause-and-effect diagrams.

(5) **Redesign the process:** generate ideas for improvement through questioning and brainstorming, compare against benchmarks and prepare a redesigned process with better flows and fewer non-value-adding steps.

(6) **Implement changes:** win acceptance from the people affected, carry out the change with an implementation plan and monitor the results. The cycle is repeated for continuous improvement.

| | | | | |
|---|---|---|---|---|
| **27** | A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center. | 10 | 3 | 3 |

| Item | Client X | Client Y |
|---|---|---|
| Annual demand forecast (copies) | 2,000 | 6,000 |
| Standard processing time (hours/copy) | 0.5 | 0.7 |
| Average lot size (copies per report) | 20 | 30 |
| Standard setup time (hours) | 0.25 | 0.40 |

Source: Unit II | Improvement CIE, 12/01/2026; SEE Feb/Mar 2026 (IM352IA), Q7b; SEE Jan/Feb 2021 (18IM54), Q5b; SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q5b; SEE Fast Track Oct 2020 (16IM55/12IM62), Q5b

**Setup time** is the time required to change a process from making one service or product to making another. When several products share the machines, the number of machines required is

$$M = \frac{[Dp + (D/Q)s]_{\text{product 1}} + [Dp + (D/Q)s]_{\text{product 2}} + \cdots + [Dp + (D/Q)s]_{\text{product } n}}{N\,[1 - (C/100)]}$$

where D = annual demand, p = processing time per unit, Q = lot size, s = setup time per lot, N = hours available per year from one machine and C = desired capacity cushion (percent).

**Solution**

Client X: $2{,}000(0.5) + (2{,}000/20)(0.25) = 1{,}000 + 25 = 1{,}025$ hours

Client Y: $6{,}000(0.7) + (6{,}000/30)(0.40) = 4{,}200 + 80 = 4{,}280$ hours

Total required $= 1{,}025 + 4{,}280 = 5{,}305$ hours

$N[1 - (C/100)] = (250 \times 1 \times 8)(1 - 0.15) = 2{,}000 \times 0.85 = 1{,}700$ hours per machine

$$M = \frac{5{,}305}{1{,}700} = 3.12 \text{ machines}$$

A fractional machine cannot be bought, so M is rounded up to four machines, unless it is cheaper to cover the shortfall with short-term options such as overtime or stockouts. With the existing three machines the center no longer has the desired 15 percent capacity cushion; overtime can be used as a short-term measure, and a fourth machine should be acquired if demand continues at this level or grows.

**Answer:** Four machines are required (M = 3.12, rounded up); the center currently has three.

| | | | | |
|---|---|---|---|---|
| **28** | Compare process structures in services and manufacturing.<br>Source: Unit II \| SEE Feb/Mar 2026 (IM352IA), Q3a | 08 | 4 | 3 |

**Process structure in services: customer-contact matrix**

The customer-contact matrix brings together (1) the degree of customer contact, (2) customization and (3) process characteristics. It is the starting point for evaluating and improving a service process. The three process structures form a continuum:

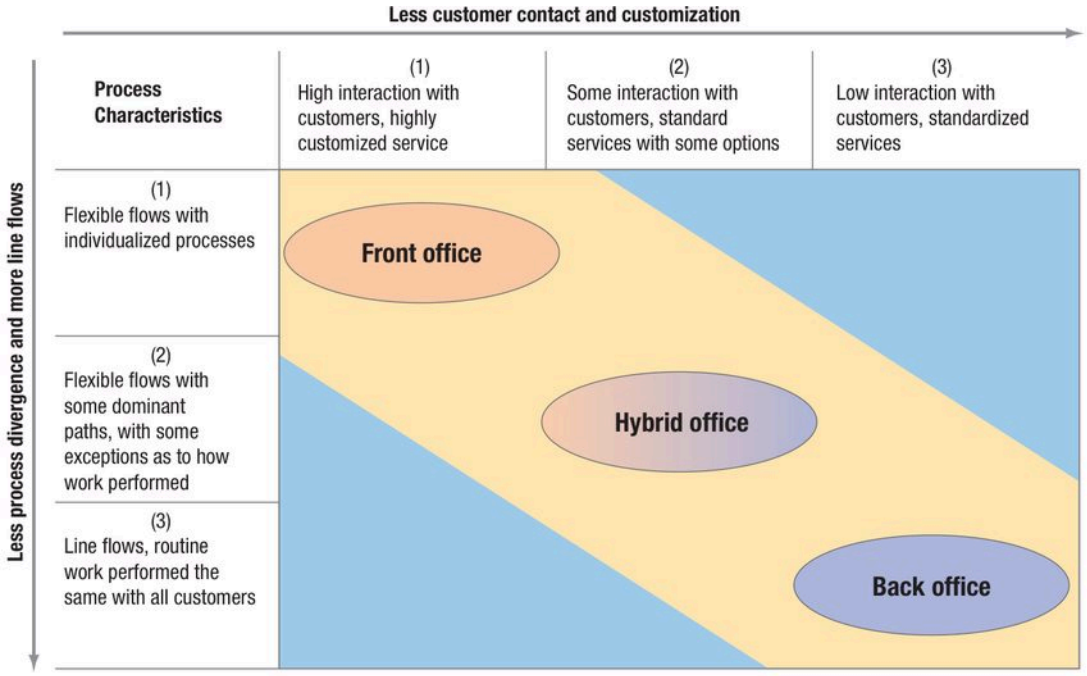



**Figure 17:** Customer-contact matrix for service processes

- **Front office:** high customer contact; the service provider interacts directly with the internal or external customer; highly customized service with flexible flows.
- **Hybrid office:** moderate customer contact and standard services with some options; flexible flows with some dominant paths.
- **Back office:** low customer contact and little customization; standardized, routine work with line flows.

### Process structure in manufacturing: product-process matrix

The product-process matrix brings together (1) volume, (2) product customization and (3) process characteristics; it synchronizes the product with the manufacturing process. The four process choices form a continuum: job, batch, line and continuous-flow processes.

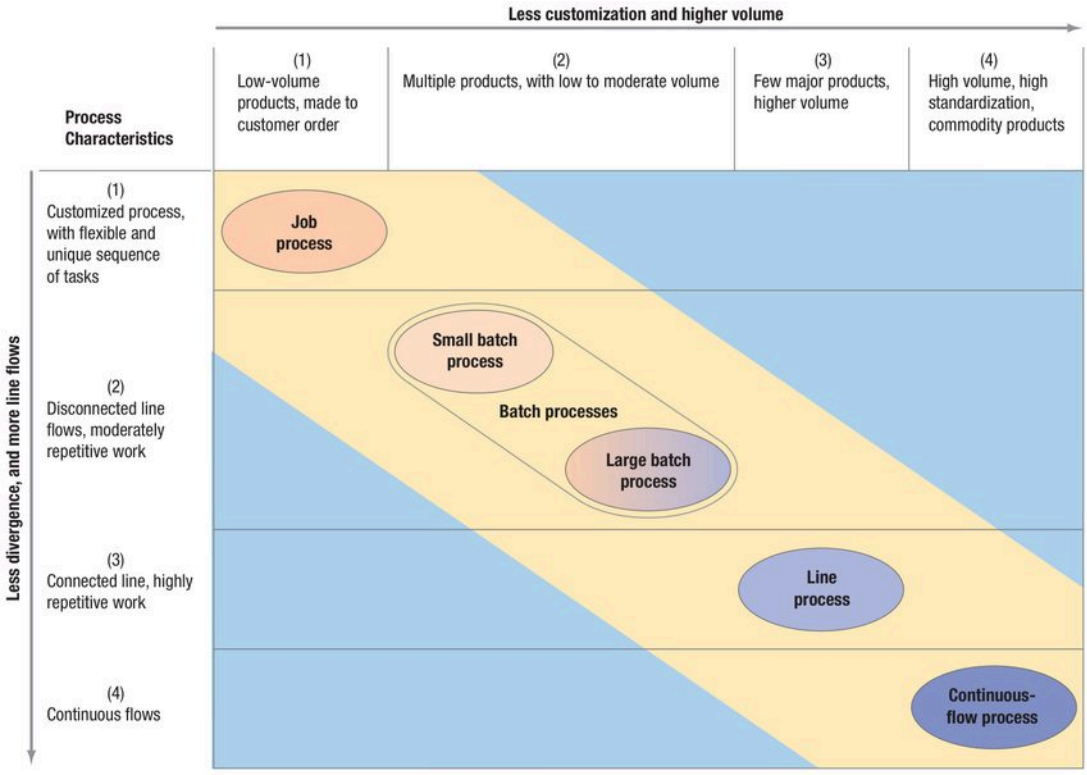



**Figure 18:** Product-process matrix for manufacturing processes

**Comparison**

| Basis | Services | Manufacturing |
|---|---|---|
| Tool used | Customer-contact matrix | Product-process matrix |
| Elements | Degree of customer contact, customization, process characteristics | Volume, product customization, process characteristics |
| Process structures | Three: front office, hybrid office, back office | Four: job, batch, line, continuous-flow |
| Customer contact | Primary feature; customer is present and involved | Normally not a consideration |
| Inventory | Output cannot be stored; capacity must match demand | Inventories can be used (make-to-order, assemble-to-order, make-to-stock) |
| Best position | Along the diagonal of the matrix; processes far off it perform poorly | Along the diagonal of the matrix; processes far off it perform poorly |

| | | | | |
|---|---|---|---|---|
| **29** | Explain long-term capacity planning. Describe capacity timing strategies and capacity sizing strategies with suitable explanations.<br>Source: Unit II \| SEE Feb/Mar 2026 (IM352IA), Q3b | 08 | 4 | 3 |

**Capacity** is the maximum rate of output of a process or system. **Long-term capacity plans** deal with investments in new facilities and equipment at the organizational level; they need top-management approval because they are not easily reversed, and they cover at least two years into the future. Three dimensions of capacity strategy must be examined: (1) sizing capacity cushions, (2) timing and sizing expansion and (3) linking capacity with other operating decisions.

**Capacity sizing: capacity cushion**

The **capacity cushion** is the reserve capacity a process uses to handle sudden increases in demand or temporary losses of capacity; it is the amount by which average utilization falls below 100 percent.

$$\text{Capacity cushion} = 100\% - \text{Average utilization rate}\ (\%)$$

Large cushions are used when demand is uncertain or variable and when customer service is critical; small cushions are used in capital-intensive industries.

**Capacity timing and sizing strategies**

This dimension decides when to adjust capacity and by how much. There are two extreme strategies:

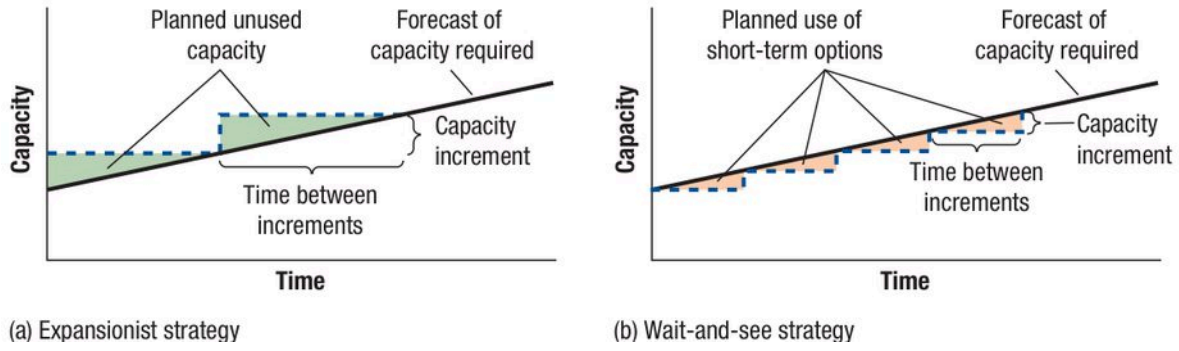



**Figure 19:** Two capacity strategies: (a) expansionist, (b) wait-and-see

- **Expansionist strategy:** large, infrequent jumps in capacity that stay ahead of demand. It minimizes the chance of sales lost to insufficient capacity, can give economies of scale and faster learning, and can pre-empt competitors' expansion.
- **Wait-and-see strategy:** smaller, more frequent jumps that lag behind demand, for example renovating existing facilities rather than building new ones. Shortfalls are met with short-term options such as overtime, temporary workers, subcontractors, stockouts and postponed maintenance. It reduces the risk of over-expansion based on optimistic forecasts or obsolete technology, but risks being pre-empted by a competitor or being unable to respond to high demand.
- **Follow-the-leader:** an intermediate strategy of expanding when others do.

Timing and sizing are related: if demand is increasing and the time between increments increases, the size of the increments must also increase.

### Linking capacity and other decisions

Capacity decisions must be closely linked to processes and supply chains throughout the organization, including competitive priorities, quality, process design, inventory and facility location.

| | | | | |
|---|---|---|---|---|
| **30** | Explain timing and sizing strategies in capacity planning with diagrams.<br>Source: Unit II \| SEE Feb/Mar 2026 (IM352IA), Q4a | 08 | 4 | 3 |

Timing and sizing of capacity decide **when** to adjust capacity levels and **by how much**. The two extreme strategies are shown below; management may choose either or any strategy between them.

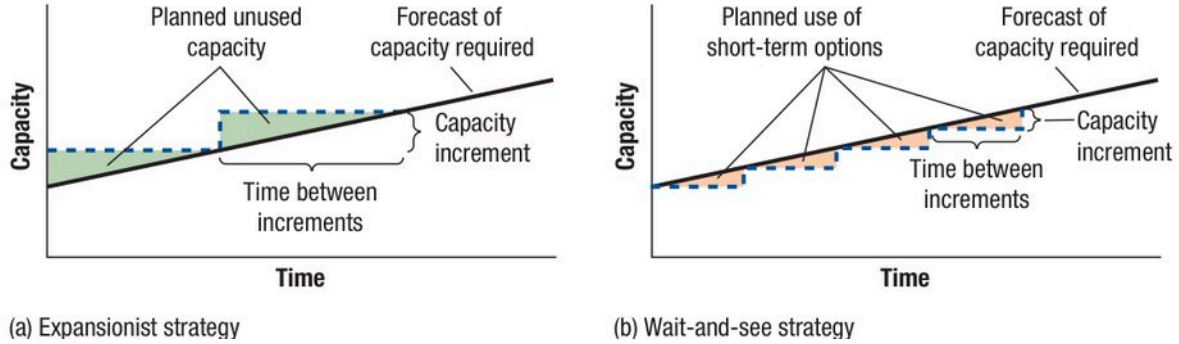



**Figure 20:** Capacity timing and sizing strategies

### (a) Expansionist strategy

- Large, infrequent jumps in capacity; capacity stays **ahead of demand**, giving planned unused capacity.
- Minimizes the chance of sales lost to insufficient capacity.
- Gives economies of scale and a faster rate of learning, helping the firm reduce costs and compete on price.
- A large expansion, or announcing one, can pre-empt the expansion of competitors.
- Risk: over-expansion if demand forecasts are too optimistic.

### (b) Wait-and-see strategy

- Smaller, more frequent jumps; capacity **lags behind demand**.
- Shortfalls are met with short-term options: overtime, temporary workers, subcontractors, stockouts and postponed preventive maintenance.
- Reduces the risk of over-expansion based on optimistic forecasts, obsolete technology or wrong assumptions about competitors.
- Risks: being pre-empted by a competitor or being unable to respond if demand is unexpectedly high.

### Relationship between timing and sizing

Timing and sizing are related: if demand is increasing and the time between increments increases, the size of the increments must also increase. An intermediate strategy is to follow the leader, expanding when others do.

| | | | | |
|---|---|---|---|---|
| **31** | How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer introduced variability?<br>Source: Unit II \| SEE Feb/Mar 2026 (IM352IA), Q4b | 08 | 2 | 2 |

Customer-introduced variability arises because customers differ in when they want service (arrival variability), what they request (request variability), how capable they are (capability variability), how much effort they put in (effort variability) and what they expect (subjective preference variability). Process strategies either **accommodate** this variability or **reduce** it.

### eBay: accommodating variability

As an online auction house, eBay has very high volume and high request variability: customers want service at any time and buy and sell an endless number of items, and their Internet experience varies widely. Such variability would make workforce scheduling very difficult if employees performed the processes. eBay therefore lets customers perform virtually all of the selling and buying processes themselves through its website (high customer involvement through self-service technology). When the customer does much of the work, the right labour is provided at the right moment, so a relatively small workforce can serve a very large customer base.

### McDonald's: reducing variability

McDonald's reduces variability so that it can deliver consistent quality at low cost. It offers a limited, standardized menu; standardizes work methods, staff training and procurement of raw materials; uses line flows and specialized equipment in the kitchen; produces popular items ahead of demand; and limits customer involvement to ordering, collecting and clearing (self-service). The result is the same consistent product and process quality from one store to the next.

### Comparison

| Basis | eBay | McDonald's |
|---|---|---|
| Approach to variability | Accommodates customer-introduced variability | Reduces customer-introduced variability |
| Customer involvement | Very high: customers perform buying and selling processes themselves | Limited: customers order, collect and serve themselves |
| Variety | Endless variety of items and requests | Limited, standardized menu |
| Process structure | Flexible, technology-enabled self-service | Standardized line flows, specialized equipment |
| Competitive priority | Variety and convenience | Consistent quality, low cost, delivery speed |
| Workforce | Small workforce relative to volume | Trained crew following standard procedures |

| | | | | |
|---|---|---|---|---|
| **32** | Explain the four inventory strategies that should be coordinated with process choice in a manufacturing environment. Provide example for each of the strategy.<br>Source: Unit II \| SEE Feb/Mar 2025 (IM352IA), Q3a | 10 | 3 | 2 |

Production and inventory strategies must be coordinated with the process choice. The four strategies are make-to-order, assemble-to-order, make-to-stock and mass customization.

### 1. Make-to-order

Products are made to customer specifications in low volumes after the order is received. The strategy gives a high degree of customization, uses job or small-batch processes with high divergence, and matches the priorities of top quality and customization. **Examples:** specialized medical equipment, castings, expensive custom-built homes.

### 2. Assemble-to-order

A wide variety of products is produced from relatively few standard subassemblies and components after customer orders are received. Stocking finished products would be too costly because the many options make forecasting inaccurate, so the principle of **postponement** is applied: the final activities are delayed until the order is received. It usually combines a batch process for fabrication with a line process for assembly; typical priorities are variety and fast delivery. **Examples:** personal computers configured to order, automobiles with optional features, restaurant meals assembled from prepared ingredients.

### 3. Make-to-stock

Items are held in stock for immediate delivery, minimizing customer delivery time. It suits standardized products with

high volumes and reasonably accurate forecasts, and is the strategy of choice for line and continuous-flow processes. Combining a line process with make-to-stock is called **mass production**. **Examples:** garden tools, electronic components, soft drinks, chemicals.

#### 4. Mass customization

Highly divergent processes generate a wide variety of customized products at reasonably low cost, combining the low cost of mass production with the flexibility of customization. It is closely linked to assemble-to-order and postponement and relies on flexible automation and modular design. **Examples:** made-to-measure clothing ordered online, customized sports shoes, build-to-order laptops.

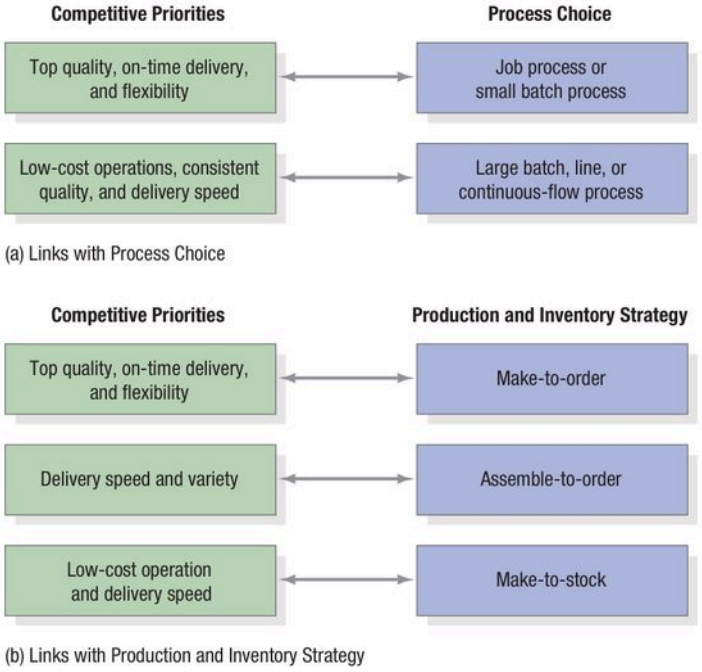



**Figure 21:** Links of competitive priorities with manufacturing strategy

| | | | | |
|---|---|---|---|---|
| **33** | Provide a note on the three elements in a customer-contract matrix in service processes.<br>Source: Unit II \| SEE Feb/Mar 2025 (IM352IA), Q3b | 06 | 2 | 2 |

The customer-contact matrix (the question's "customer-contract matrix") brings together three elements: (1) the degree of customer contact, (2) customization and (3) process characteristics. The matrix is the starting point for evaluating and improving a service process.

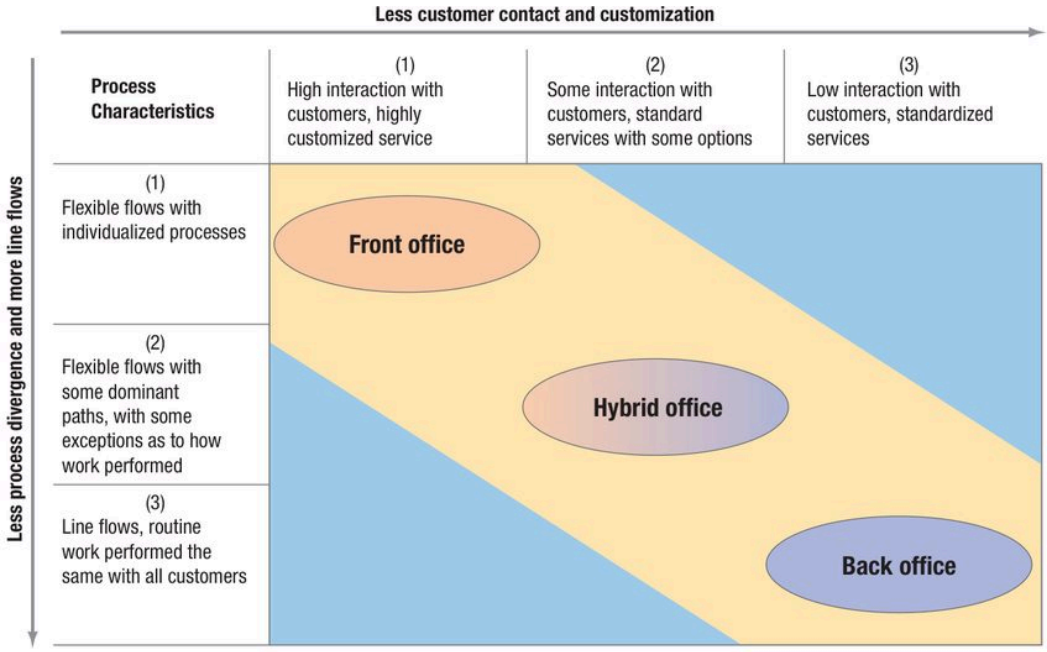



**Figure 22:** Customer-contact matrix for service processes

### (1) Degree of customer contact

Customer contact is the extent to which the customer is present, is actively involved and receives personal attention during the service process. Contact is high when the customer is physically present, people (rather than possessions or information) are processed, involvement is active and visible, attention is personal and delivery is face-to-face.

### (2) Customization

The horizontal dimension of the matrix represents the service provided in terms of customer contact and competitive priorities. A key priority is how much customization is needed. Positions on the left represent high customer contact and highly customized services; positions on the right represent low contact, standardized services and processes out of the customer's sight.

### (3) Process characteristics

The vertical dimension deals with two characteristics of the process: **process divergence**, the extent to which the process is customized with latitude in how its tasks are performed, and **flow**, which may be flexible (customers, materials or information move in diverse ways) or line (they move linearly in a fixed sequence). The resulting structures are the front office, hybrid office and back office along the diagonal.

| | | | | |
|---|---|---|---|---|
| **34** | Operations managers must examine three dimensions of capacity strategy before making capacity decisions. What are they? Explain each with suitable examples.<br>Source: Unit II \| SEE Feb/Mar 2025 (IM352IA), Q4a; SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q5a | 10 | 3 | 2 |

The three dimensions of capacity strategy are (1) sizing capacity cushions, (2) timing and sizing expansion and (3) linking process capacity and other operating decisions.

### (1) Sizing capacity cushions

The **capacity cushion** is the reserve capacity a process uses to handle sudden increases in demand or temporary losses of capacity; capacity cushion = 100% − average utilization rate (%). The appropriate size varies by industry:

- In the capital-intensive paper industry, where machines cost hundreds of millions of dollars, cushions well under 10 percent are preferred.
- The less capital-intensive hotel industry breaks even at 60 to 70 percent utilization (a 30 to 40 percent cushion) and suffers customer-service problems when the cushion drops to 20 percent; cruise ships prefer cushions as small as 5 percent.
- Supermarkets keep a cushion large enough to handle peak demand so that customer service is prompt.

### (2) Timing and sizing expansion

This dimension decides when to adjust capacity and by how much: the **expansionist** strategy (large, infrequent jumps ahead of demand) or the **wait-and-see** strategy (small, frequent jumps behind demand). **Examples:** General Motors increased production capacity of the Chevrolet Volt from 30,000 to 45,000 units in 2012 because of strong public interest; ethanol producers followed an expansionist strategy to stay ahead of rising demand.

### (3) Linking capacity and other decisions

Capacity decisions must be closely linked to processes and supply chains. Decisions on process design, resource flexibility, inventory and facility location all affect capacity cushions. **Example:** capacity cushions can be lowered if less emphasis is placed on fast delivery, if yield losses fall (better quality), or if investment in capital-intensive equipment or worker flexibility increases.

| | | | | |
|---|---|---|---|---|
| **35** | Enumerate the four steps of systematic approach to long term capacity decisions.<br>Source: Unit II \| SEE Feb/Mar 2025 (IM352IA), Q4b; SEE Fast Track Oct 2020 (16IM55/12IM62), Q5a; SEE Fast Track Jul 2019 (16IM55/12IM62), Q5a | 06 | 2 | 2 |

A four-step procedure helps managers make sound long-term capacity decisions:

### Step 1: Estimate future capacity requirements

The capacity requirement is what a process's capacity should be for a future period to meet the demand of its customers, given the desired capacity cushion. The estimate is based on forecasts of demand, productivity, competition and technological change, expressed either as an output measure or as an input measure (for example, number of machines M).

### Step 2: Identify gaps

A **capacity gap** is any difference (positive or negative) between projected capacity requirements and current capacity. Gaps are identified by comparing the requirements with available capacity.

### Step 3: Develop alternatives

Alternative plans are developed to cope with the projected gaps. One alternative, the **base case**, is to do nothing and lose orders from demand that exceeds capacity (or carry the cost of excess capacity). Other alternatives vary the timing and size of capacity additions (expansionist or wait-and-see), and include expanding at another location or using short-term options such as overtime, temporary workers and subcontracting.

### Step 4: Evaluate the alternatives

Each alternative is evaluated both qualitatively (fit with strategy, uncertainty about demand, competitive reaction, technological change) and quantitatively (cash flows, net present value, break-even analysis, decision trees), and a final choice is made.

| | | | | |
|---|---|---|---|---|
| **36** | List and explain operations process characteristics. Also briefly discuss the implication of four characteristics of operations processes.<br>Source: Unit II \| SEE Apr/May 2024 (21IM53), Q3a | 08 | 3 | 2 |

### Process characteristics

Processes are characterized on two dimensions:

- **Process divergence:** the extent to which the process is highly customized, with considerable latitude in how its tasks are performed. A highly divergent process changes with each customer (consulting, law, architecture); a low-divergence process is repetitive and standardized.
- **Flow:** a **flexible flow** means customers, materials or information move in diverse ways, with the path of one job often crisscrossing the path of the next; a **line flow** means they move linearly from one operation to the next in a fixed sequence.

### The four major process decisions (characteristics of operations processes)

**Process structure:** determines the process type relative to the kinds of resources needed, how resources are partitioned between them and their key characteristics. A layout, the physical arrangement of operations, puts these decisions into tangible form.

**Customer involvement:** reflects the ways in which customers become part of the process and the extent of their participation.

**Resource flexibility:** is the ease with which employees and equipment can handle a wide variety of products, output levels, duties and functions.

**Capital intensity:** is the mix of equipment and human skills in a process; the greater the relative cost of equipment, the greater the capital intensity.

### Implications

The choices must fit the situation and each other, and must not work at cross-purposes.

| Decision | High customer contact (front-office service) | High volume (manufacturing) |
|---|---|---|
| Process structure | Customer present, actively involved, receives personal attention; high divergence and flexible flows | High volume and a standard product make a line flow possible |
| Customer involvement | Customers are more likely to become part of the process; each service is unique | Not a factor, except for choices on variety and customization |
| Resource flexibility | High divergence and flexible flows need flexible workers, facilities and equipment | Flexibility not needed; specialization gives more efficient processes |
| Capital intensity | Automation and capital intensity more likely when volume is higher | High volumes justify the large fixed costs of an efficient operation |

| | | | | |
|---|---|---|---|---|
| **37** | How does the degree of customer contact, customization and process characteristics help in structuring processes in a service organization?<br>Source: Unit II \| SEE Apr/May 2024 (21IM53), Q3b | 08 | 3 | 2 |

A good process strategy for a service process depends first and foremost on the type and amount of customer contact. **Customer contact** is the extent to which the customer is present, is actively involved and receives personal attention during the service process.

| Dimension | High contact | Low contact |
|---|---|---|
| Physical presence | Present | Absent |
| What is processed | People | Possessions or information |
| Contact intensity | Active, visible | Passive, out of sight |
| Personal attention | Personal | Impersonal |
| Method of delivery | Face-to-face | Regular mail or e-mail |

Table: Dimensions of customer contact in service processes

The customer-contact matrix brings together the degree of customer contact, customization and process characteristics:

- **Customer contact and customization (horizontal axis):** positions on the left represent high contact and highly customized services, where the customer is present and active, the process is visible and attention is personal. The right side represents low contact, passive involvement, impersonal attention and a process out of sight.
- **Process characteristics (vertical axis):** process divergence (how customized the tasks are) and flow (flexible or line).

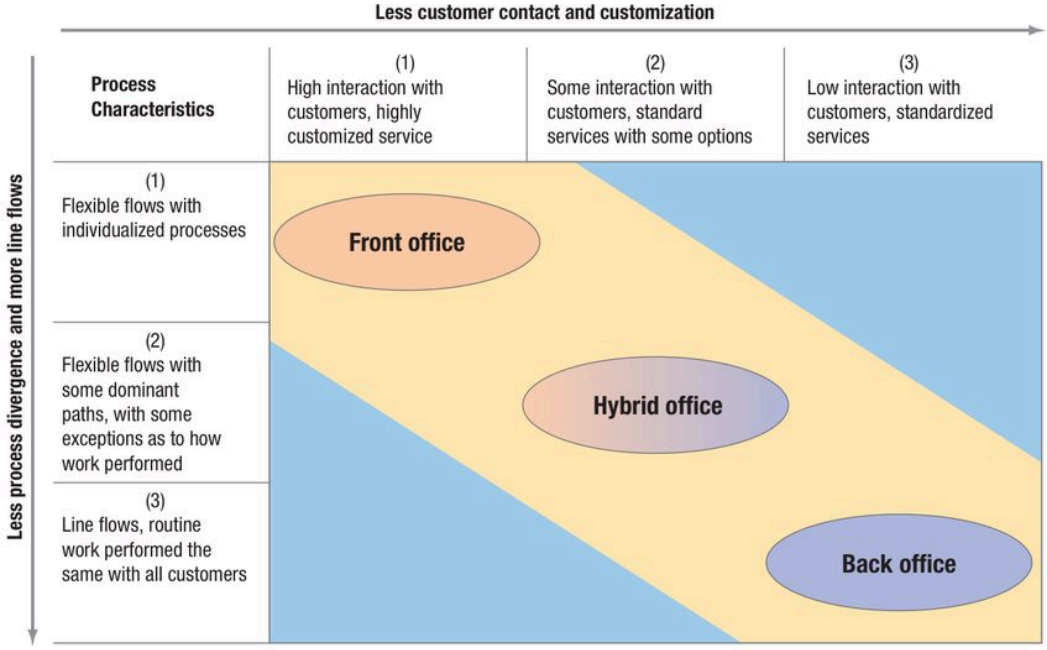



**Figure 23:** Customer-contact matrix for service processes

### Resulting process structures

- **Front office:** high customer contact with direct interaction; customized services with many options, considerable divergence and flexible work flows that vary from one customer to the next.
- **Hybrid office:** moderate contact and standard services with some options; work flows from one workstation to the next with some dominant paths.
- **Back office:** low contact and little customization; standardized, routine work with line flows from one provider to the next.

| | | | | |
|---|---|---|---|---|
| **38** | Explain the concept of economies of scale when planning capacity for long term implementation.<br>Source: Unit II \| SEE Apr/May 2024 (21IM53), Q4a | 08 | 2 | 2 |

Deciding the best level of long-term capacity requires considering the efficiency of operations. **Economies of scale** means that the average unit cost of a service or good can be reduced by increasing its output rate. Four principal reasons explain why costs fall as output and facility size increase:

(1) **Fixed costs are spread over more units.** Heating, debt service, managers' salaries and depreciation do not vary with output in the short term; as the output (utilization) rate rises, the average unit cost falls.

(2) **Construction costs are reduced.** Building permits, architects' fees and rental of building equipment are needed for small and large facilities alike; doubling the size of a facility usually does not double its construction cost.

(3) **Costs of purchased materials are cut.** Higher volumes give a better bargaining position and quantity discounts. Walmart gains large economies of scale because its stores buy and sell huge volumes of each item.

(4) **Process advantages are found.** High volume shifts the process toward a line process with dedicated resources and justifies more efficient, specialized technology. Benefits include a faster learning effect, lower inventory, better process and job design and fewer changeovers.

Beyond a certain size, **diseconomies of scale** set in: average unit cost rises because excessive size brings complexity, loss of focus, bureaucracy and inflexibility.

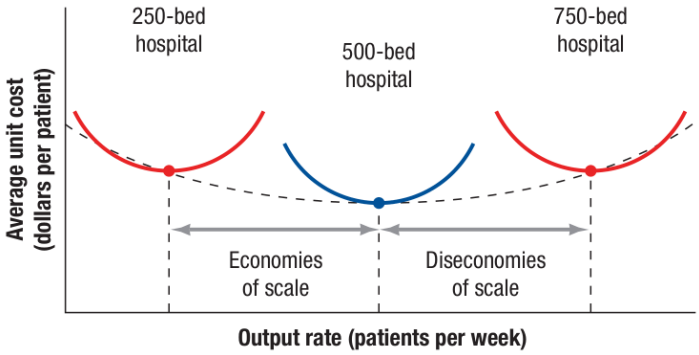



**Figure 24:** Economies and diseconomies of scale

In the hospital example, the 500-bed hospital has a lower average cost per patient than the 250-bed and 750-bed hospitals. This does not mean 500 beds is optimal for all hospitals: the optimal size depends on the number of patients to be served, so capacity planning must match facility size to long-term demand.

| | | | | |
|---|---|---|---|---|
| **39** | The best choice for a manufacturing process depends on the volume and degree of customization required of the process. Justify the above statement with the help of a suitable matrix.<br>Source: Unit II \| SEE Apr/May 2024 (21IM53), Q4b | 08 | 3 | 3 |

The **product-process matrix** brings together (1) volume, (2) product customization and (3) process characteristics. It synchronizes the product to be manufactured with the manufacturing process.

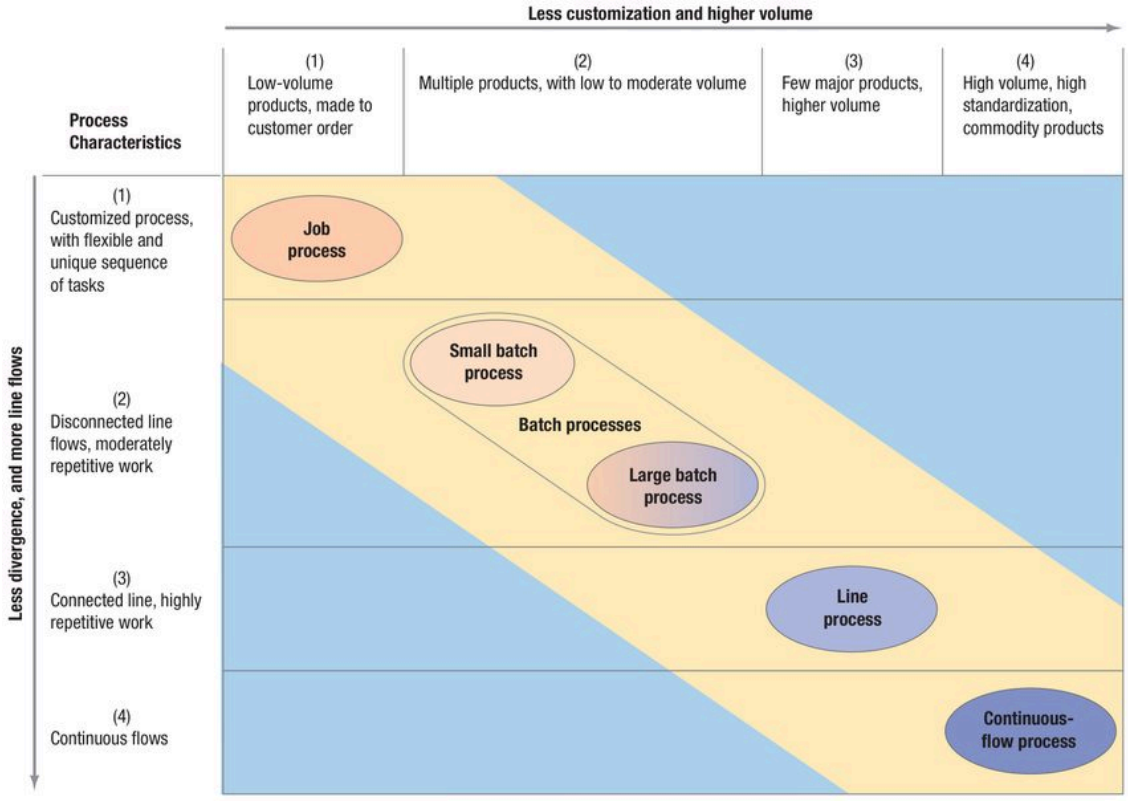



**Figure 25:** Product-process matrix for manufacturing processes

The matrix shows that the best choice of manufacturing process depends on the volume and the degree of customization required. Moving from top-left to bottom-right, volume increases and customization decreases, so the process changes from job to batch to line to continuous flow. A process positioned too far from the diagonal is unlikely to be a top performer: it either has too much flexibility (high cost) or too little (poor customization).

**Justification: the four process choices**

**Job process:** creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable

divergence in the steps performed. Customization is high and the volume of any one product is low; the workforce and equipment are flexible. Firms often bid for work, make products to order and handle each order as a single unit (a job). **Examples:** machining a metal casting for a customized order, producing customized cabinets.

**Batch process:** differs from the job process in volume, variety and quantity: volumes are higher because the same or similar products or parts are produced repeatedly in batches. Volumes are moderate, but divergence is still too great to dedicate a separate process to each product. The flow is flexible with some dominant paths, and some segments have a line flow. **Examples:** standard components that feed an assembly line, some processes that make capital equipment.

**Line process:** lies between the batch and continuous processes; volumes are high and products are standardized, so resources are organized around particular products. Divergence is minimal, little inventory is held between steps, each step performs the same process over and over, and equipment is specialized. **Examples:** assembly of computers, automobiles, appliances and toys.

**Continuous-flow process:** is the extreme end of high-volume, standardized production with rigid line flows and negligible divergence. Usually one primary material (a liquid, gas or powder) moves without stopping through the process until the whole batch is finished, often for several shifts or months. **Examples:** petroleum refining, chemicals, steel, soft drinks and food.

| 40 | Describe the four basic process strategies.<br>Source: Unit II \| SEE Mar/Apr 2023 (18IM54), Q3a | 08 | – | – |
|---|---|---|---|---|

Process strategy rests on four basic process decisions: (1) process structure (including layout), (2) customer involvement, (3) resource flexibility and (4) capital intensity. They are linked to each other and to the strategies for change.

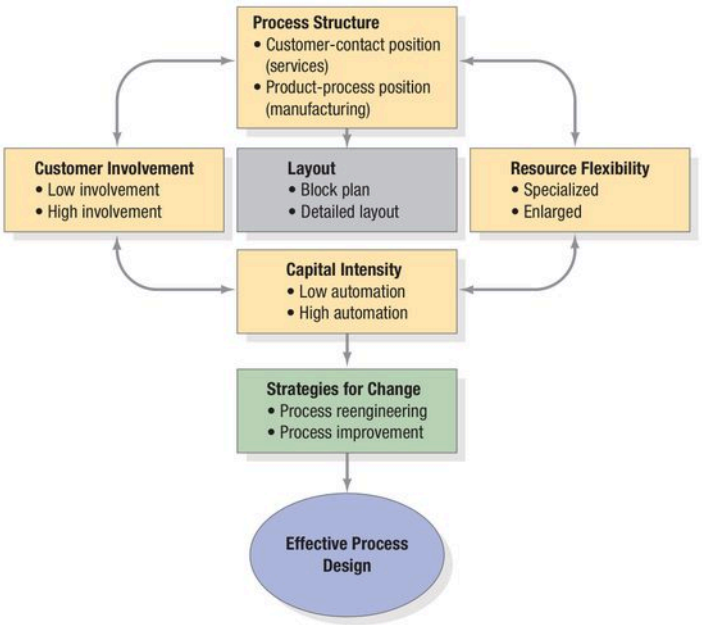



**Figure 26:** Major process strategy decisions for effective processes

**Process structure:** determines the process type relative to the kinds of resources needed, how resources are partitioned between them and their key characteristics. A layout, the physical arrangement of operations, puts these decisions into tangible form.

**Customer involvement:** reflects the ways in which customers become part of the process and the extent of their participation.

**Resource flexibility:** is the ease with which employees and equipment can handle a wide variety of products, output levels, duties and functions.

**Capital intensity:** is the mix of equipment and human skills in a process; the greater the relative cost of equipment, the greater the capital intensity.

**Strategies for change:** two basic strategies are used to analyze and modify processes: (1) process reengineering (radical redesign) and (2) process improvement (systematic, incremental improvement). Together the decisions lead to effective process design.

| 41 | List the different tools for process analysis and design and explain them.<br>Source: Unit II \| SEE Mar/Apr 2023 (18IM54), Q3b | 08 | – | – |
|---|---|---|---|---|

Process analysis and design use tools for documenting processes, evaluating their performance and generating improvements.

| Tool | Explanation |
|---|---|
| Flowchart | A diagram that traces the flow of information, customers, equipment or materials through the various steps of a process. |
| Swim-lane flowchart | A flowchart that groups the functional areas responsible for different sub-processes into lanes. |
| Service blueprint | A special flowchart of a service process that shows which steps have high customer contact. |
| Process chart | An organized way of documenting all the activities performed by a person or group, at a workstation, with a customer or on materials. |
| Time study | A work measurement method in which a trained analyst selects the work elements, times them, determines the sample size and sets a time standard. |
| Work sampling | A method that estimates the proportion of time spent by people or machines on different activities, based on observations randomized over time. |
| Metrics | Performance measures established for a process and the steps within it. |
| Checklist | A form for recording how often certain process characteristics or failures occur. |
| Histogram | A summary of data measured on a continuous scale showing the frequency distribution of a process failure (central tendency and dispersion). |
| Pareto chart | A bar chart on which factors are plotted in decreasing order of frequency, separating the vital few causes from the trivial many. |
| Scatter diagram | A plot of two variables showing whether they are related. |
| Cause-and-effect diagram | A (fishbone) diagram that relates a key performance problem to its potential causes. |
| Brainstorming | Letting a group of people knowledgeable about the process propose ideas for change by saying whatever comes to mind. |
| Benchmarking | A systematic procedure that measures a firm's processes, services and products against those of industry leaders. |

| 42 | Define strategic fit. Describe the elements of strategic fit.<br>Source: Unit II \| SEE Mar/Apr 2023 (18IM54), Q4a | 08 | – | – |
|---|---|---|---|---|

### Definition

**Strategic fit** means that the four major process decisions fit the situation and each other, and support the competitive priorities and operations strategy. When the fit is more strategic, the process is more effective; a more effective process matches key process characteristics and has a close strategic fit. Managers must understand how the decisions tie together in order to spot ways of improving poorly designed processes.

### Elements of strategic fit

**Process structure:** determines the process type relative to the kinds of resources needed, how resources are partitioned between them and their key characteristics. A layout, the physical arrangement of operations, puts these decisions into tangible form.

**Customer involvement:** reflects the ways in which customers become part of the process and the extent of their participation.

**Resource flexibility:** is the ease with which employees and equipment can handle a wide variety of products, output levels, duties and functions.

**Capital intensity:** is the mix of equipment and human skills in a process; the greater the relative cost of equipment, the greater the capital intensity.

**Competitive priorities:** the process should reflect its desired priorities. Front offices generally emphasize top quality and customization, whereas back offices emphasize low-cost operation, consistent quality and on-time delivery. The process

structure selected then points the way to the appropriate choices on customer involvement, resource flexibility and capital intensity.

*Decision patterns that test strategic fit for service and manufacturing processes are given in Part B Q50.*

| | | | | |
|---|---|---|---|---|
| **43** | What is the Six Sigma model of quality improvement? Describe the five steps of the Six Sigma improvement model.<br>Source: Unit II \| SEE Mar/Apr 2023 (18IM54), Q4b | 08 | – | – |

**Six Sigma** is a comprehensive and flexible system for achieving, sustaining and maximizing business success by minimizing defects and variability in processes. A process operating at six-sigma quality produces no more than 3.4 defects per million opportunities. It relies on statistical methods, a strong customer focus and trained specialists: **Black Belts** (employees with the highest level of training, who lead and teach Six Sigma project teams full time), Green Belts and Master Black Belts.

**Six Sigma improvement model (DMAIC)**

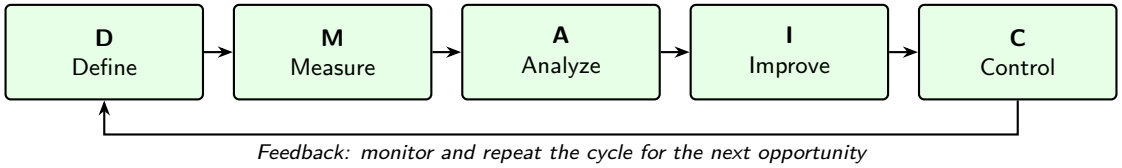



**Figure 27:** Six Sigma improvement model

(1) **Define:** determine the characteristics of the process output that are critical to customer satisfaction and identify any gaps between these characteristics and the process capabilities; define the project scope, goals and team.
(2) **Measure:** quantify the work the process does that affects the gap; select what to measure, identify data sources and prepare a data-collection plan.
(3) **Analyze:** use the data to find the root causes of defects and variation (Pareto charts, cause-and-effect diagrams, scatter diagrams, statistical analysis) and identify improvement opportunities.
(4) **Improve:** modify or redesign existing methods to meet the new performance objectives; test the solutions on a small scale and implement them.
(5) **Control:** monitor the process (for example with control charts and standard procedures) to make sure that high performance levels are maintained.

| | | | | |
|---|---|---|---|---|
| **44** | How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer-introduced variability?<br>Source: Unit II \| SEE Jan/Feb 2021 (18IM54), Q3a | 06 | – | – |

*The question is identical to Part B Q31; the answer is the same.*

| | | | | |
|---|---|---|---|---|
| **45** | A work sampling study was conducted to determine the proportion of the time a worker is idle. The following information was gathered on a random basis. | 10 | – | – |

| Day | Number of times worker idle | Total number of observations |
|---|---|---|
| Monday | 17 | 44 |
| Tuesday | 18 | 56 |
| Wednesday | 14 | 48 |
| Thursday | 16 | 60 |

(i) Based on these preliminary results, what percent of the time is the worker working? (ii) If the idle time is judged to be excessive, what additional categories might you add to a follow-up work sampling study to identify the root causes?

Source: Unit II | SEE Jan/Feb 2021 (18IM54), Q3b

**Work sampling** estimates the proportion of time spent by people or machines on different activities, based on observations randomized over time.

### (i) Percentage of time the worker is working

$$\text{Idle observations} = 17 + 18 + 14 + 16 = 65$$

$$\text{Total observations} = 44 + 56 + 48 + 60 = 208$$

$$\text{Proportion idle} = \frac{65}{208} = 0.3125 = 31.25\%$$

$$\text{Proportion working} = 1 - 0.3125 = 0.6875 = 68.75\%$$

**Answer:** The worker is working 68.75% of the time (idle 31.25% of the time).

### (ii) Additional categories for a follow-up study

To find the root causes of the idle time, the single "idle" category should be split into specific activities, for example:

- Waiting for materials, parts or tools
- Waiting for instructions, inspection or approval from the supervisor
- Machine breakdown or repair (waiting for maintenance)
- Machine setup or changeover
- Material handling and movement between workstations
- Rework of defective items
- Waiting because of an upstream bottleneck (no work available)
- Personal needs, fatigue and rest allowances
- Cleaning and housekeeping, and administrative work (paperwork, meetings)

Recording observations in these categories shows which causes contribute most to idle time so that corrective action can be targeted.

| | | | | |
|---|---|---|---|---|
| **46** | Performance criteria important in creating a layout can go well beyond communication and materials handling. Identify the types of layout performance criteria that might be most important in the following settings: (i) Airport (ii) Bank (iii) Classroom (iv) Product designers' office.<br>Source: Unit II \| SEE Jan/Feb 2021 (18IM54), Q4a | 08 | – | – |

Layout performance criteria go beyond communication and materials handling. They include the flow and convenience of customers, safety and security, flexibility, capital and operating cost, use of space, the work environment and image presented to customers. The most important criteria in each setting are:

| Setting | Important layout performance criteria |
|---|---|
| (i) Airport | Smooth, one-way passenger flow from check-in through security to boarding gates; minimum walking distance; capacity for peak traffic and queues; safety and security screening; efficient baggage handling; clear signage and wayfinding; access to shops and amenities; flexibility for future expansion. |
| (ii) Bank | Customer convenience and short waiting time; privacy for customers at counters and in cabins; security of cash, vaults and records; visibility and supervision of the banking hall; a pleasant, professional image; accessibility (ATMs, ramps); efficient flow of documents between front office and back office. |
| (iii) Classroom | Clear sight lines to the board and screen from every seat; good acoustics, lighting and ventilation; easy interaction between teacher and students; flexibility to rearrange seating for group work; space for audio-visual equipment; safe movement and emergency exits. |
| (iv) Product designers' office | Communication and collaboration among designers (closeness of team members); flexibility to reconfigure team spaces; space for prototypes, models and display boards; quiet zones for concentration; natural light and a creative work environment; easy access to computing, printing and testing facilities. |

When relative locations are the main concern (communication, material handling), the weighted-distance method is used to compare block plans; criteria that depend on the absolute location of a single department (for example, a reception area at the entrance) are considered separately.

| | | | | |
|---|---|---|---|---|
| **47** | Vera Johnson and Merris Williams manufacture vanishing cream. Their packaging process has four steps: (1) mix, (2) fill, (3) cap, and (4) label. They have had the reported process failures analyzed, which shows the following (60 failures in total): lumps of unmixed product – 7; over or under filled jars – 18; jar lids did not seal – 6; labels rumpled or missing – 29. (No separate instruction is printed in the paper.)<br>Source: Unit II \| SEE Jan/Feb 2021 (18IM54), Q4b | 08 | – | – |

A **process failure** is any performance shortfall, such as an error, delay, waste or rework. A **Pareto chart** plots the failure categories in decreasing order of frequency so that the vital few causes can be separated from the trivial many (the 80-20 rule).

**Pareto analysis of the 60 process failures**

| Process failure | Process step | Count | % of total | Cumulative % |
|---|---|---|---|---|
| Labels rumpled or missing | (4) Label | 29 | 48.3 | 48.3 |
| Over- or under-filled jars | (2) Fill | 18 | 30.0 | 78.3 |
| Lumps of unmixed product | (1) Mix | 7 | 11.7 | 90.0 |
| Jar lids did not seal | (3) Cap | 6 | 10.0 | 100.0 |
| **Total** | | **60** | **100.0** | |

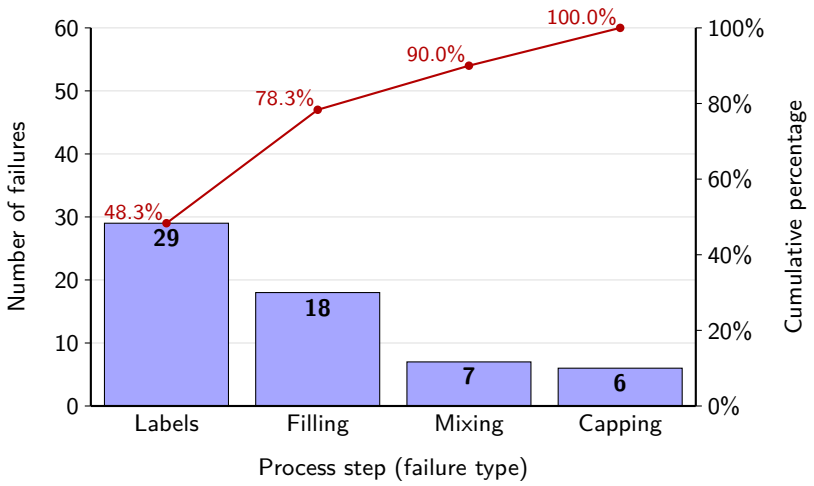



**Figure 28:** Pareto chart of process failures in vanishing-cream packaging

**Answer:** The labelling step causes 48.3% of all failures, and labelling and filling together cause 78.3%. Improvement efforts should focus first on the labelling step and then on the filling step.

A cause-and-effect diagram for the labelling step (machine, materials, methods, manpower) can then be used to find the root causes of rumpled or missing labels.

| | | | | |
|---|---|---|---|---|
| **48** | Identify and discuss with block diagram the important four decisions that are essential for an effective process design.<br>Source: Unit II \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q3a (identical) | 08 | – | – |

*The four decisions for effective process design and their block diagram are given in Part B Q40.*

| | | | | |
|---|---|---|---|---|
| **49** | Identify and discuss the strategies used for structuring the process in manufacturing.<br>Source: Unit II \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q3b (identical) | 08 | – | – |

**Process choice** is the way of structuring the process by organizing resources around the process or around the product. Organizing around the process means, for example, grouping all milling machines together to process all products needing that operation. Organizing around the product means bringing together all the people and equipment needed for a specific product and dedicating them to it. The four process choices form a continuum:

**Job process:** creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed. Customization is high and the volume of any one product is low; the workforce and equipment are flexible. Firms often bid for work, make products to order and handle each order as a single unit (a job). **Examples:** machining a metal casting for a customized order, producing customized cabinets.

**Batch process:** differs from the job process in volume, variety and quantity: volumes are higher because the same or similar products or parts are produced repeatedly in batches. Volumes are moderate, but divergence is still too great to dedicate a separate process to each product. The flow is flexible with some dominant paths, and some segments have a line flow. **Examples:** standard components that feed an assembly line, some processes that make capital equipment.

**Line process:** lies between the batch and continuous processes; volumes are high and products are standardized, so resources are organized around particular products. Divergence is minimal, little inventory is held between steps, each step performs the same process over and over, and equipment is specialized. **Examples:** assembly of computers, automobiles, appliances and toys.

**Continuous-flow process:** is the extreme end of high-volume, standardized production with rigid line flows and negligible divergence. Usually one primary material (a liquid, gas or powder) moves without stopping through the process until the whole batch is finished, often for several shifts or months. **Examples:** petroleum refining, chemicals, steel, soft drinks and food.

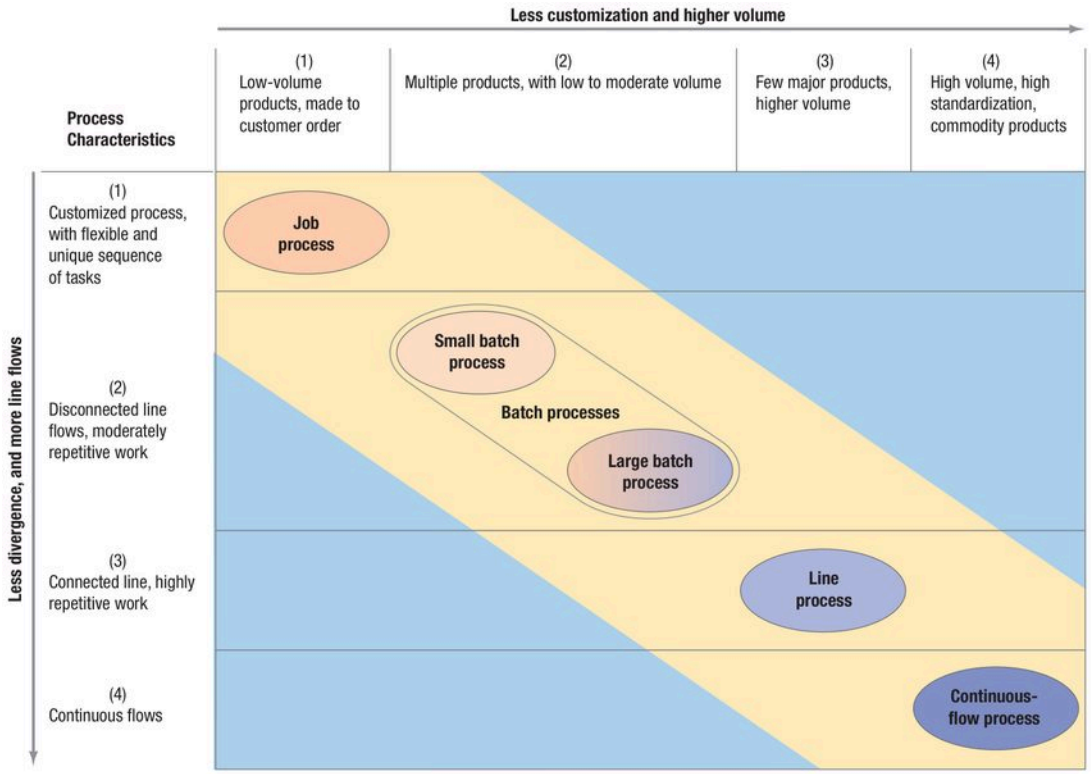



**Figure 29:** Product-process matrix for manufacturing processes

The process choice is coordinated with the production and inventory strategy: job and small-batch processes with make-to-order, batch and line processes with assemble-to-order, and line and continuous-flow processes with make-to-stock.

| | | | | |
|---|---|---|---|---|
| **50** | The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective. As an industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit.<br>Source: Unit II \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q4a (identical) | 08 | – | – |

The four major process decisions must fit the situation and each other; the closer the fit, the more effective the process. Strategic fit is tested separately for services and for manufacturing.

### Services

The process should reflect its desired competitive priorities. Front offices generally emphasize top quality and customization, whereas back offices emphasize low-cost operation, consistent quality and on-time delivery. The process structure then points the way to the choices on customer involvement, resource flexibility and capital intensity.

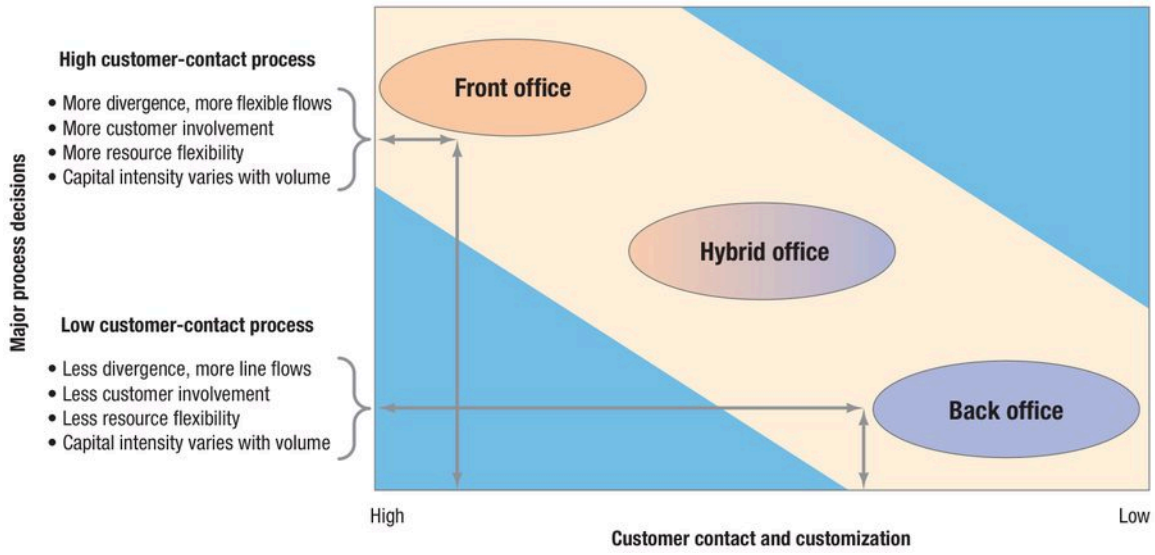



**Figure 30:** Decision patterns for service processes

High customer contact at a front-office process means more divergence and more flexible flows, more customer involvement and more resource flexibility, while capital intensity varies with volume. A low-contact back office shows the opposite pattern. These are general tendencies rather than rigid rules, but they show how service process decisions can be linked coherently.

### Manufacturing

Job and small-batch processes fit when top quality, on-time delivery and flexibility (customization, variety, volume flexibility) are emphasized. Large-batch, line and continuous-flow processes fit low-cost operations, consistent quality and delivery speed. For production and inventory strategies, make-to-order matches flexibility and top quality, assemble-to-order matches delivery speed and variety, and make-to-stock matches low cost and delivery speed.

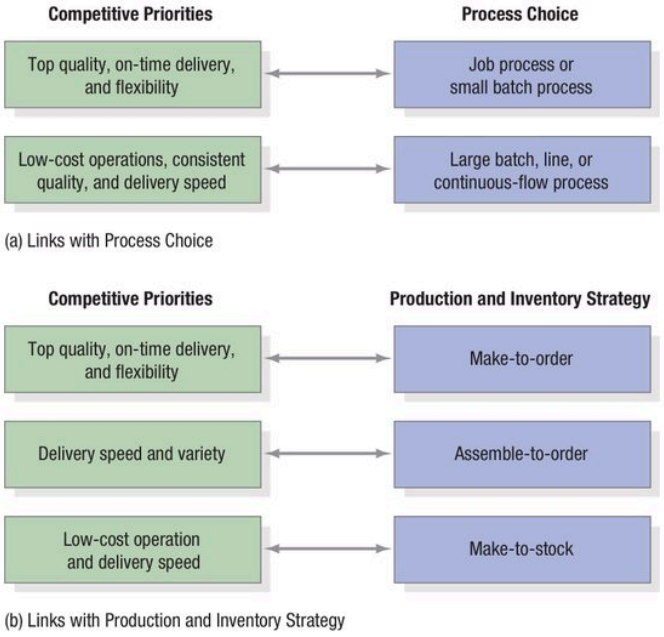



**Figure 31:** Links of competitive priorities with manufacturing strategy

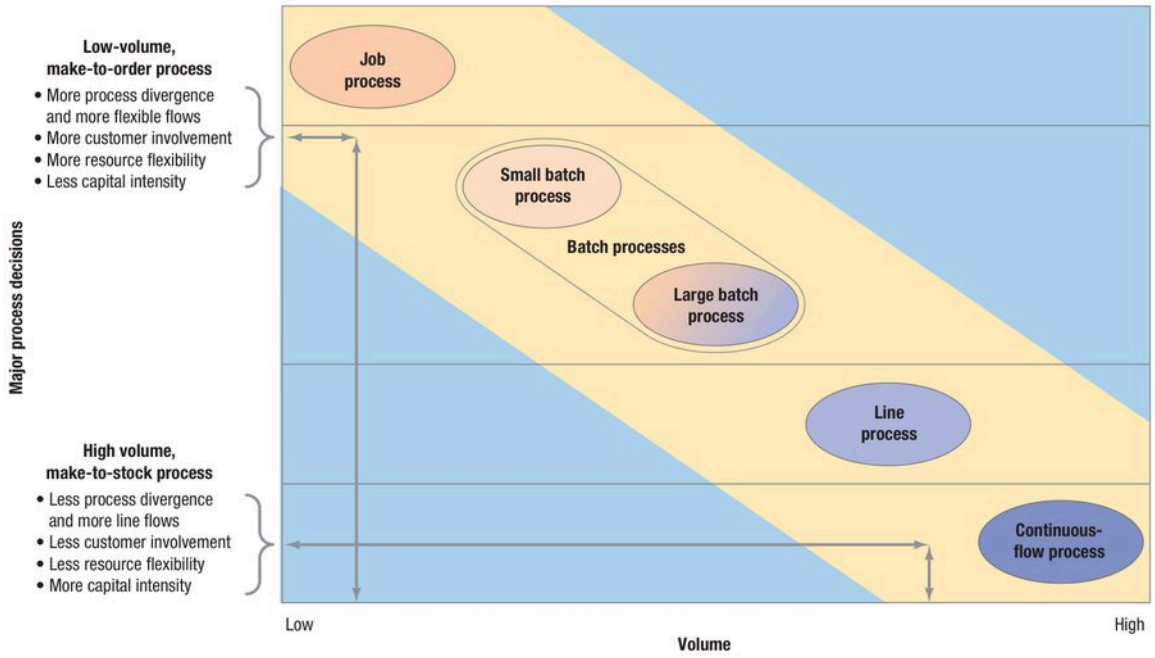



**Figure 32:** Decision patterns for manufacturing processes

High volume per part type means a line flow, little customer involvement, low resource flexibility (specialization) and high capital intensity; low-volume, make-to-order processes show the opposite pattern.

### Gaining focus

Fit is improved by **focus**: narrowing the range of demands on a facility so that management can concentrate on fewer tasks and lead the workforce toward a single goal. **Plants within plants (PWPs)** are separate operations within one facility, each with its own competitive priorities, processes and workforce.

| 51 | Identify and discuss the key elements of process reengineering.<br>Source: Unit II \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q4b (identical) | 08 | – | – |
|---|---|---|---|---|

**Reengineering** is the fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service and speed. Its key elements are:

| Element | Description |
|---|---|
| Critical processes | The emphasis of reengineering is on core business processes; normal process-improvement activities continue with the other processes. |
| Strong leadership | Senior executives must provide strong leadership; otherwise cynicism, resistance ("we tried that before") and boundaries between departments can block radical change. |
| Cross-functional teams | A team with members from each functional area affected by the change carries out the project. Self-managing teams and employee empowerment are the rule rather than the exception. |
| Information technology | Information technology is the primary enabler; most reengineering projects design processes around information flows, such as customer order fulfillment. |
| Clean-slate philosophy | Reengineering starts with the way the customer wants to deal with the company; teams begin with internal and external customer objectives for the process. |
| Process analysis | Despite the clean slate, the team must understand the current process: what it does, how well it performs and what factors affect it, looking at every procedure involved throughout the organization. |

Table: Key elements of reengineering

Reengineering has led to many successes, but it is not simple or easily done, nor is it appropriate for all processes or all organizations.

| | | | | |
|---|---|---|---|---|
| **52** | Explain why processes exist everywhere in all organizations. Take the following supply chain activities and give a brief description for each: (i) Warehousing (ii) Customer service (iii) Logistics.<br>Source: Unit II \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q3a | 08 | – | – |

### Why processes exist everywhere

- Processes are the basic unit of work, and every activity that transforms inputs into outputs for a customer is a process.
- **Supply chains have processes:** supply chain processes are business processes that have external customers or suppliers, such as purchasing, warehousing and logistics.
- **Processes are found throughout the whole organization**, not just in operations: in accounting, finance, human resources, management information systems and marketing (for example, billing, recruiting, order taking).
- Managers of all these processes must ensure that they add as much customer value as possible, and must understand that many processes cut across organizational lines, whether the firm is organized by function, product, region or process.

### Supply chain activities

| Activity | Description |
|---|---|
| (i) Warehousing | Receiving shipments from suppliers, verifying quality, placing them in inventory and reporting receipt for inventory records. |
| (ii) Customer service | Providing information to answer questions or resolve problems, using automated information services as well as voice-to-voice contact with customers. |
| (iii) Logistics | Selecting the transportation mode (train, ship, truck, airplane or pipeline), scheduling inbound and outbound shipments and providing intermediate inventory storage. |

| | | | | |
|---|---|---|---|---|
| **53** | Briefly discuss on the process strategy decisions. Use a block diagrammatic representation to bring out the connections among major process strategy decisions for effective processes.<br>Source: Unit II \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q3b | 08 | – | – |

*The process strategy decisions and the block diagram of the major process decisions are given in Part B Q40.*

The diagram brings out the connections among the decisions: process structure (customer-contact position for services, product-process position for manufacturing) is linked to layout, customer involvement, resource flexibility and capital intensity, and all of them lead, through the strategies for change, to effective process design. The key to successful process decisions is to make choices that fit the situation and make sense together.

| | | | | |
|---|---|---|---|---|
| **54** | Define the following terms clearly: (i) Customer increment (ii) Resource flexibility (iii) Capital intensity (iv) Economies of scope.<br>Source: Unit II \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q4a | 08 | – | – |

### (i) Customer involvement ("customer increment" in the question)

Customer involvement reflects the ways in which customers become part of the process and the extent of their participation, for example self-service, choosing product options, the timing of the service and co-creation of the product.

### (ii) Resource flexibility

Resource flexibility is the ease with which employees and equipment can handle a wide variety of products, output levels, duties and functions (a flexible workforce and general-purpose equipment).

### (iii) Capital intensity

Capital intensity is the mix of equipment and human skills in a process; the greater the relative cost of equipment, the greater the capital intensity.

### (iv) Economies of scope

Economies of scope reflect the ability to produce multiple products more cheaply in combination than separately. Normally, high capital intensity means low resource flexibility; programmable (flexible) automation breaks this inverse relationship and makes both high capital intensity and high flexibility possible, creating economies of scope.

| | | | | |
|---|---|---|---|---|
| **55** | Position a process on the customer-contact matrix or product-process matrix. Bring out the characteristics of the following types of process choices: (i) Job processes (ii) Batch processes (iii) Line processes (iv) Continuous flow processes.<br>Source: Unit II \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q4b | 08 | – | – |

A process is positioned on the customer-contact matrix (services) or the product-process matrix (manufacturing) according to its volume, customization and process characteristics. The desirable positions, called process choices, lie along the diagonal and effectively connect the product with the process.

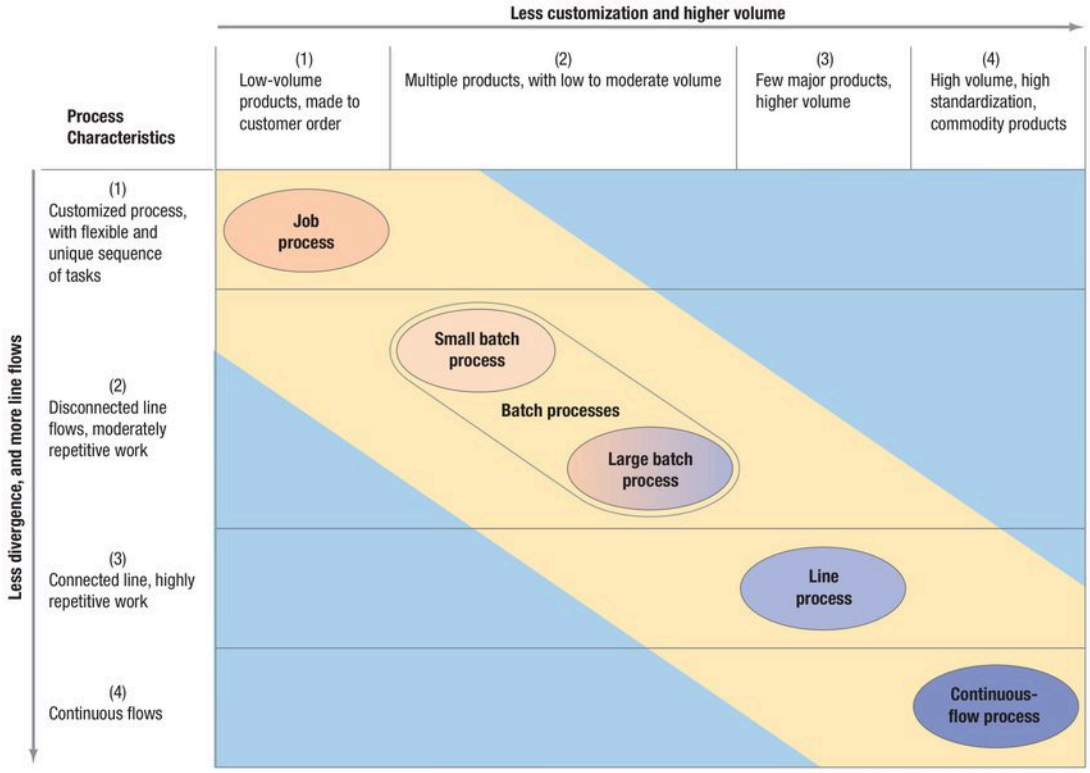



**Figure 33:** Product-process matrix showing the four process choices

The best choice of manufacturing process depends on the volume and degree of customization required.

### (i) Job process

**Job process:** creates the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed. Customization is high and the volume of any one product is low; the workforce and equipment are flexible. Firms often bid for work, make products to order and handle each order as a single unit (a job). **Examples:** machining a metal casting for a customized order, producing customized cabinets.

### (ii) Batch process

**Batch process:** differs from the job process in volume, variety and quantity: volumes are higher because the same or similar products or parts are produced repeatedly in batches. Volumes are moderate, but divergence is still too great to dedicate a separate process to each product. The flow is flexible with some dominant paths, and some segments have a line flow. **Examples:** standard components that feed an assembly line, some processes that make capital equipment.

### (iii) Line process

**Line process:** lies between the batch and continuous processes; volumes are high and products are standardized, so resources are organized around particular products. Divergence is minimal, little inventory is held between steps, each step performs the same process over and over, and equipment is specialized. **Examples:** assembly of computers, automobiles, appliances and toys.

### (iv) Continuous-flow process

**Continuous-flow process:** is the extreme end of high-volume, standardized production with rigid line flows and negligible divergence. Usually one primary material (a liquid, gas or powder) moves without stopping through the process until the whole batch is finished, often for several shifts or months. **Examples:** petroleum refining, chemicals, steel, soft drinks and food.

| | | | | |
|---|---|---|---|---|
| **56** | Construct the product-process matrix and show in this matrix the process characteristic categories, level of product customization and the elements of volume. Elaborate on the type of manufacturing process structure that exist in various types of organization briefly.<br>Source: Unit II \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q3a | 08 | – | – |

The product-process matrix brings together three elements: (1) **volume** (horizontal axis: from low-volume products made to customer order, through multiple products with low to moderate volume and a few major products with higher volume, to high-volume commodity products), (2) **level of product customization** (decreasing from left to right) and (3) **process characteristics** (vertical axis: from customized processes with flexible flows, through disconnected line flows, to connected lines and continuous flows).

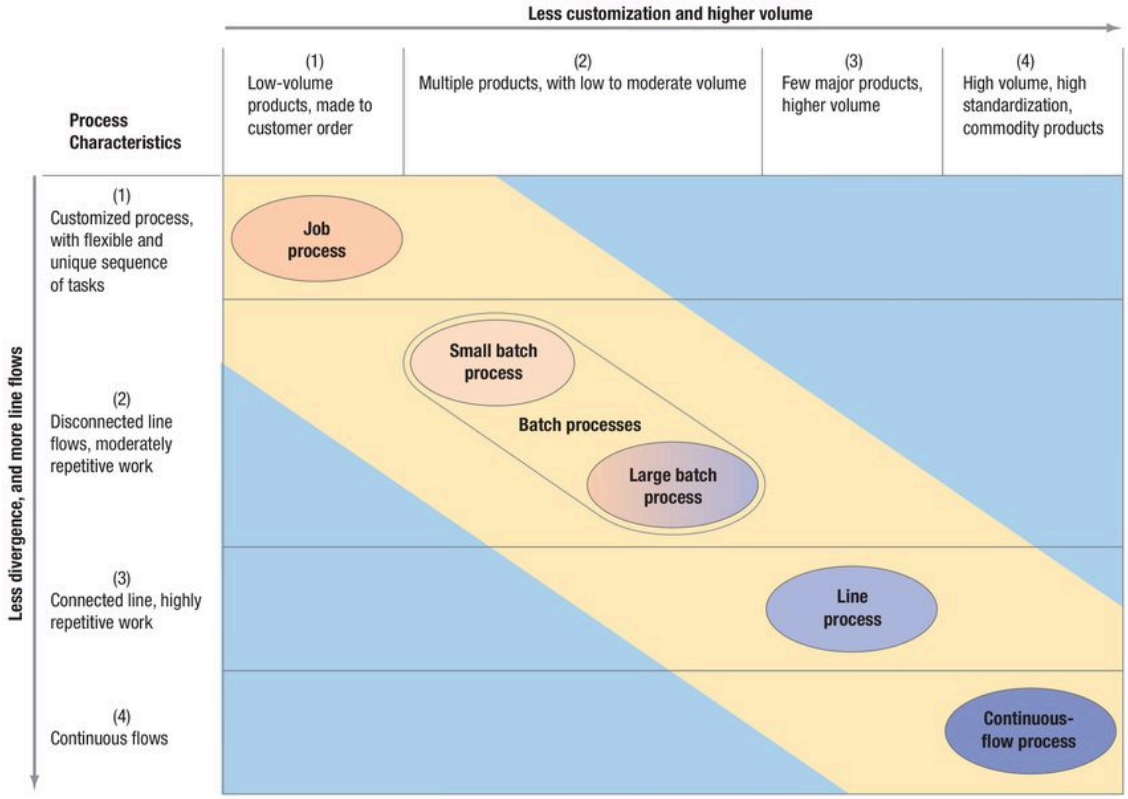



**Figure 34:** Product-process matrix: process characteristics, product customization and volume

**Process structures found in various types of organization**

| Process structure | Characteristics | Organizations that use it |
|---|---|---|
| Job process | Low volume, high customization, flexible flows | Tool rooms, machine shops making customized castings, custom cabinet makers |
| Batch process | Moderate volume, some repetition, dominant paths | Component makers feeding assembly lines, capital equipment manufacturers, bakeries |
| Line process | High volume, standardized products, line flows | Assembly of computers, automobiles, appliances and toys |
| Continuous-flow process | Very high volume, commodity products, continuous flow | Petroleum refineries, chemical plants, steel mills, soft-drink and food plants |

*The full description of each process choice is given in Part B Q55.*

| | | | | |
|---|---|---|---|---|
| **57** | "After a process is documented, metrics data collected and disconnects identified, the process analyst or design team puts together a set of changes that will make the process better". Compare this with a doctor diagnosing the patient and recommending treatments based on the diagnosis. Explain how questioning and brainstorming will assist the process designer in analyzing the gaps in the process. Take an example to support your answer.<br>Source: Unit II \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q3b | 08 | – | – |

### Process analysis compared with medical diagnosis

Process analysis is the documentation and detailed understanding of how work is performed and how it can be redesigned. It closely resembles a doctor treating a patient:

| Doctor and patient | Process analyst and process |
|---|---|
| Records the symptoms and the patient's history | Documents the process (flowcharts, service blueprints, process charts) |
| Takes measurements and runs tests | Collects metrics data on cost, time, quality and customer satisfaction |
| Diagnoses the illness and its causes | Identifies disconnects, bottlenecks and root causes of poor performance |
| Prescribes a treatment | Proposes a set of changes (redesigned process) |
| Follows up the patient's recovery | Implements the changes and monitors the results |

Just as the treatment is only as good as the diagnosis, the redesign is only as good as the analysis that precedes it.

### Role of questioning

The analyst systematically questions every step of the documented process: **what** is being done, **why** it is done (is it necessary and does it add value?), **where** and **when** it is done, **who** does it and **how** it is done and how well. Questioning exposes unnecessary steps, duplication, delays, unnecessary movement and hand-offs, and suggests whether a step can be eliminated, combined, re-sequenced or simplified.

### Role of brainstorming

**Brainstorming** lets a group of people knowledgeable about the process propose ideas for change by saying whatever comes to mind. Criticism is not allowed while ideas are generated, quantity is encouraged and ideas are built on one another; the ideas are then evaluated and the best are selected for the redesign. It draws on the experience of the employees who perform the work every day.

### Example

A hospital finds that patients wait more than three hours to be discharged. The team documents the discharge process and measures the time at each step (the diagnosis shows that most delay occurs while waiting for the pharmacy and for billing clearance). Questioning asks why billing waits until the doctor signs the discharge summary and why medicines are ordered only after discharge is approved. Brainstorming produces ideas such as preparing bills and ordering medicines in parallel the evening before, and using a discharge checklist. After implementation, discharge time falls to under one hour, which is the "treatment" prescribed by the analysis.

| | | | | |
|---|---|---|---|---|
| **58** | Develop the customer-contact matrix and indicate the elements of degree of customer contact, customization and process characteristic. Elaborate on the process divergence and flow based on the customer-contact matrix. How does this matrix provide the starting point for evaluating and improving a process?<br>Source: Unit II \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q4a | 08 | – | – |

The customer-contact matrix brings together (1) the degree of customer contact, (2) customization and (3) process characteristics. It is the starting point for evaluating and improving a service process.

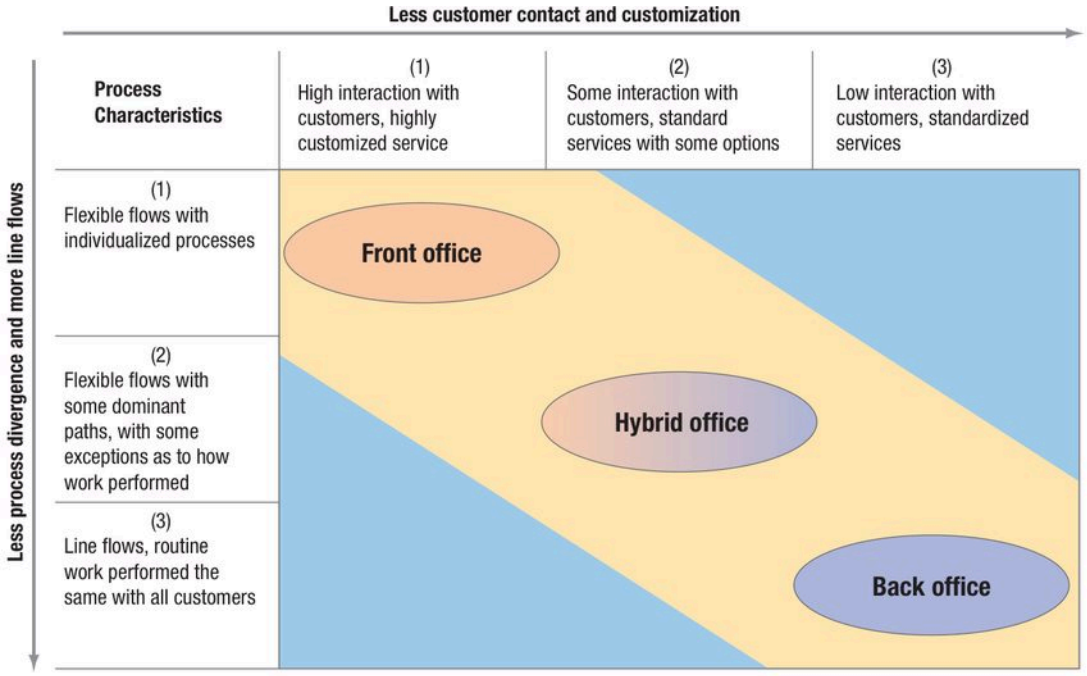



**Figure 35:** Customer-contact matrix for service processes

### Degree of customer contact and customization

The horizontal dimension represents the service in terms of customer contact and competitive priorities. A key priority is how much customization is needed: positions on the left represent high contact and highly customized service (high interaction with customers), and positions on the right represent low contact and standardized services.

### Process divergence and flow

The vertical dimension deals with two process characteristics. **Process divergence** is the extent to which the process is highly customized, with considerable latitude in how its tasks are performed; if the process changes with each customer, every performance is unique, as in consulting, law and architecture. A low-divergence service is repetitive and standardized and is performed the same way for all customers.

**Flow** may be flexible, where customers, materials or information move in diverse ways and paths crisscross, or line, where they move linearly from one operation to the next in a fixed sequence. Front offices have flexible flows with individualized processes, hybrid offices have flexible flows with some dominant paths, and back offices have line flows.

### Starting point for evaluating and improving a process

After analyzing a process and plotting its position on the matrix, it may be found to be improperly positioned: too far to the left or right, or too far to the top or bottom. Opportunities for improvement then become apparent: perhaps more customization and customer contact is needed, or perhaps the process is too divergent, with unnecessarily flexible flows, so that reducing divergence would cut costs and improve productivity. Processes along the diagonal have the best strategic fit.

| | | | | |
|---|---|---|---|---|
| **59** | Capital intensity is the mix of equipment and human skills in the process. Carefully examine the advantages and disadvantages of automating a manufacturing process. Distinguish between fixed automation and flexible automation. Give two examples for fixed and flexible automation respectively. Justify automation in the context of capital intensity.<br>Source: Unit II \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q4b | 08 | – | – |

**Capital intensity** is the mix of equipment and human skills in a process. **Automation** is a system, process or piece of equipment that is self-acting and self-regulating. Automation is often thought necessary for competitive advantage, but it has both advantages and disadvantages and must be examined carefully.

### Advantages of automation

- Substituting labour-saving equipment and technology for labour improves productivity.
- Gives consistent quality and reduces human error.
- Fixed automation yields the lowest variable cost per unit when volumes are high.
- Improves safety by removing workers from dangerous or monotonous tasks.

### Disadvantages of automation

- Prohibitive investment cost for low-volume operations; capital-intensive operations need high utilization to be justified.
- High fixed costs and, for fixed automation, inflexibility when products or volumes change.
- Automation does not always align with competitive priorities: a firm offering a unique product or high-quality personal service may need hand labour and individual attention.

### Fixed automation and flexible automation

| Basis | Fixed automation | Flexible (programmable) automation |
|---|---|---|
| Meaning | Produces one type of part or product in a fixed sequence of simple operations | Can be changed (reprogrammed) easily to handle various products |
| Suitable for | Line and continuous-flow processes; high, stable volumes, stable designs, long life cycles | Low- and high-customization processes; variety of products in moderate volumes |
| Investment and flexibility | Large initial investment; relatively inflexible | High investment, but flexible; creates economies of scope |
| Unit cost | Lowest variable cost per unit at high volume | Higher unit cost than fixed automation at very high volume |
| Examples | Gillette razor-cartridge production lines (1.2 billion cartridges a year); dedicated automotive transfer lines and bottling lines | Industrial robots (welding, painting); CNC machines; robots at Just Born candy company reprogrammed for each marshmallow configuration |

### Automation in the context of capital intensity

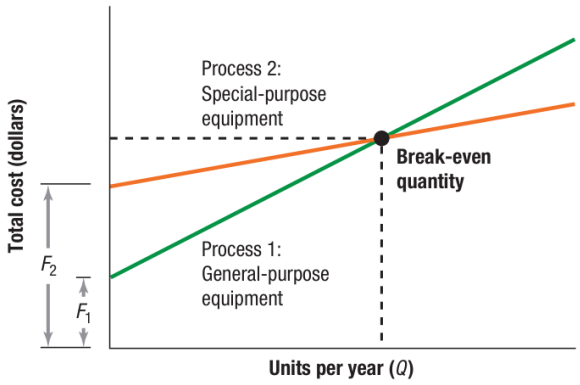



**Figure 36:** Relationship between process costs and product volume

Process 1 uses flexible, general-purpose equipment (low fixed cost F1, high variable cost) and is best at low volumes and high customization. Process 2 uses special-purpose, automated equipment (high fixed cost F2, low variable cost, flatter total-cost line) and is best when volumes are high and customization is low. Beyond the break-even quantity, spreading the high fixed cost over more units makes the automated process cheaper. Automation is therefore justified when volume is high enough to use the capital-intensive equipment fully. Generally, when capital intensity is high, resource flexibility is low, unless flexible automation is used.

**60** You have been asked to put together a capacity plan for a critical operation at the Surefoot Sandal company. Your capacity measure is the number of machines. Three products (men's, women's and children's sandals) are manufactured. The time standards (processing and setup), lot sizes and demand forecasts are given below. The firm operates 8-hour shifts, 5 days per week, 50 weeks per year. Experience shows that a capacity cushion of 5 percent is sufficient. | 08 | – | –

| Product | Processing (h/pair) | Setup (h/lot) | Lot size (pairs/lot) | Demand forecast (pairs/year) |
|---|---|---|---|---|
| Men's sandal | 0.05 | 0.5 | 240 | 80,000 |
| Women's sandal | 0.10 | 2.2 | 180 | 60,000 |
| Children's sandal | 0.02 | 3.8 | 360 | 120,000 |

(i) If the operation currently has two machines, what is the capacity gap? (ii) How many machines are needed?

Source: Unit II | SEE Fast Track Jul 2019 (16IM55/12IM62), Q5b

Capacity is measured by an input measure (number of machines). When several products with setups are made, the capacity requirement is

$$M = \frac{[Dp + (D/Q)s]_1 + [Dp + (D/Q)s]_2 + \cdots + [Dp + (D/Q)s]_n}{N\,[1 - (C/100)]}$$

where D = annual demand, p = processing time per unit, Q = lot size, s = setup time per lot, N = hours per year from one machine and C = capacity cushion (%).

$$N = 8 \text{ h/day} \times 5 \text{ days/week} \times 50 \text{ weeks} = 2,000 \text{ hours per machine per year}$$

$$N[1 - (C/100)] = 2,000(1 - 0.05) = 1,900 \text{ hours}$$

| Product | D | p (h) | Q | s (h) | D·p (h) | (D/Q)·s (h) | Total hours |
|---|---|---|---|---|---|---|---|
| Men's | 80,000 | 0.05 | 240 | 0.5 | 4,000 | 166.67 | 4,166.67 |
| Women's | 60,000 | 0.10 | 180 | 2.2 | 6,000 | 733.33 | 6,733.33 |
| Children's | 120,000 | 0.02 | 360 | 3.8 | 2,400 | 1,266.67 | 3,666.67 |
| **Total** | | | | | | | **14,566.67** |

$$M = \frac{14,566.67}{1,900} = 7.67 \text{ machines}$$

### (i) Capacity gap

A capacity gap is the positive or negative difference between the projected capacity requirement and the current capacity.

$$\text{Gap} = M - \text{current capacity} = 7.67 - 2 = 5.67 \text{ machines}$$

### (ii) Machines needed

A fractional machine cannot be bought, so the fraction is rounded up (unless overtime or other short-term options are cost-effective): 7.67 rounds up to 8 machines.

**Answer:** Capacity gap = 5.67 machines (a shortage); the operation needs 8 machines, that is 6 more than the two it has now.

| 61 | Explain the different forecasting methods used in operations management.<br>Source: Unit III \| SEE Feb/Mar 2026 (IM352IA), Q5a | 08 | 3 | 3 |
|---|---|---|---|---|

Forecasting methods fall into two general types: **judgment methods** and **quantitative methods**. Quantitative methods include **causal methods** and **time-series analysis** (and trend projection with regression, a hybrid of the two). The choice depends on the demand pattern, whether a history file exists, the forecast horizon and the item being forecast.

**Demand patterns the methods must handle**

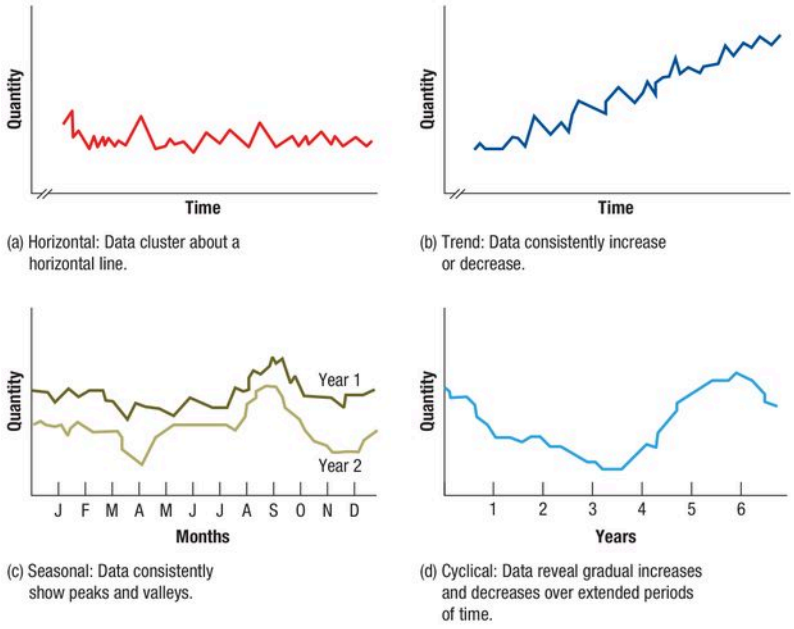



**Figure 37:** Patterns of demand: horizontal, trend, seasonal and cyclical (all contain random variation)

| Group | Methods and when they are used |
|---|---|
| **Judgment methods** | Used when there is no adequate history (new product, new technology) or to adjust quantitative forecasts. (1) **Salesforce estimates:** compiled from the estimates of the salespeople; (2) **executive opinion:** managers' opinions and experience summarized into one forecast (also technological forecasting); (3) **market research:** surveys to test hypotheses about consumer interest; (4) **Delphi method:** consensus of anonymous experts after repeated rounds. |
| **Causal methods** | Use historical data on independent variables (advertising, economic conditions, competitors' actions) to predict demand; good for turning points and long-range forecasts. The main method is **linear regression**, $Y = a + bX$ (dependent variable related linearly to one or more independent variables). |
| **Time-series methods** | Use only the past pattern of the dependent variable. **Naive forecast:** $F(t+1) = D(t)$. **Simple moving average:** average of the n most recent demands. **Weighted moving average:** each of the n demands has its own weight (weights sum to 1). **Exponential smoothing:** $F(t+1) = \alpha D(t) + (1 - \alpha)F(t)$. They estimate the average of a horizontal series. |
| **Trend projection with regression** | A hybrid of time-series and causal methods: $F(t) = a + bt$, with the time period t as the independent variable; handles a trend and can forecast several periods ahead. |
| **Seasonal methods** | Multiplicative seasonal method (average demand per season × seasonal index) and additive seasonal method (average demand ± a seasonal constant). |

Two further ideas improve accuracy: **combination forecasts** (averaging forecasts from several methods or sources) and **focus forecasting** (choosing each period the method that has produced the least error).

| | | | | |
|---|---|---|---|---|
| **62** | Explain the Delphi method. Describe its main advantage and limitations.<br>Source: Unit III \| SEE Mar/Apr 2023 (18IM54), Q7b | 08 | – | – |

**Delphi method:** a process of gaining consensus from a group of experts while maintaining their anonymity. A coordinator sends questions to each member of the group of outside experts, who may not even know who else is participating. The coordinator prepares a statistical summary of the responses along with a summary of the arguments for particular responses. The report is sent back to the same group for another round, and the participants may modify their previous answers. The rounds continue until consensus is obtained.

### Main advantage

- It is useful when **no historical data** are available from which to develop statistical models, and when managers inside the firm have no experience on which to base informed projections (new products, new technology).
- Because the experts are **anonymous**, no individual dominates and each member can revise a view without losing face, so the final forecast reflects the group's knowledge.

### Limitations

- It is **time-consuming**: several rounds of questions, summaries and replies are needed, and a coordinator is required to run them.
- The result is still a **judgment**; it depends on the quality of the experts chosen and may be hedged or biased, so it can only be validated against actual demand later (forecast error).
- It does not use historical demand data, so quantitative methods are preferred whenever an adequate history file exists.

| | | | | |
|---|---|---|---|---|
| **63** | With the help of suitable examples, explain any two qualitative methods of forecasting.<br>Source: Unit III \| SEE Apr/May 2024 (21IM53), Q5b | 06 | 3 | 3 |

Judgment (qualitative) methods translate the opinions of managers, experts, consumers and the salesforce into quantitative estimates. Two of them are described below.

### 1. Executive opinion

The opinions, experience and technical knowledge of one or more managers or customers are summarized to arrive at a single forecast. All the factors that go into judgmental forecasts fall in this category. **Example:** a senior management team of a phone maker estimates the sales of a new model, based on their experience with earlier models and the planned promotions. Executive opinion is also used for technological forecasting, to keep abreast of the latest advances in technology.

### 2. Delphi method

A process of gaining consensus from a group of experts while maintaining their anonymity. A coordinator sends questions to the experts, summarizes the answers statistically with the supporting arguments, and sends the summary back for another round; the rounds continue until consensus is obtained. **Example:** a firm asks outside technology experts, without letting them know each other, to forecast the market for a new battery technology for which no sales history exists.

(The other two judgment methods are salesforce estimates and market research.)

| | | | | |
|---|---|---|---|---|
| **64** | Given the data below, what is the simple linear regression model that can be used to predict sales in future weeks?<br>Source: Unit III \| SEE Mar/Apr 2023 (18IM54), Q7a | 08 | – | – |

| Week | Sales |
|---|---|
| 1 | 150 |
| 2 | 157 |
| 3 | 162 |
| 4 | 166 |
| 5 | 177 |

Linear regression relates the dependent variable Y (sales) to the independent variable X (week) by the straight line Y = a + bX. The values of a and b that minimize the sum of squared deviations (least squares) are

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}, \qquad a = \frac{\sum Y - b\sum X}{n}$$

| Week (X) | Sales (Y) | XY | X² | Y² |
|---|---|---|---|---|
| 1 | 150 | 150 | 1 | 22500 |
| 2 | 157 | 314 | 4 | 24649 |
| 3 | 162 | 486 | 9 | 26244 |
| 4 | 166 | 664 | 16 | 27556 |
| 5 | 177 | 885 | 25 | 31329 |
| **Total** | **812** | **2499** | **55** | **132278** |

$$b = \frac{5(2499)-(15)(812)}{5(55)-(15)^2} = \frac{315}{50} = 6.30$$
$$a = \frac{812-6.30(15)}{5} = 143.50$$

The regression model is **Y = 143.5 + 6.3 X**. Sales increase by about 6.3 units every week.

Correlation coefficient r = 0.985, so r² = 0.970: 97.0 percent of the variation in sales is explained by the week number, a very close fit. The standard error of the estimate is 2.02 units.

Forecast for week 6: Y = 143.5 + 6.3(6) = 181.3 units; for week 7: 187.6 units.

**Answer:** Y = 143.5 + 6.3 X; the forecast for week 6 is 181.3 units.

| | | | | |
|---|---|---|---|---|
| **65** | Compute a 3-week moving average forecast for the arrival of medical clinic patients in week 4. The number of arrivals for the past 3 weeks was 400, 380 and 411 (weeks 1, 2 and 3). If the actual number of patient arrivals in week 4 is 415, what is the forecast error for week 4? Compute the forecast for week 5.<br>Source: Unit III \| SEE Feb/Mar 2025 (IM352IA), Q5a | 10 | 3 | 3 |

Simple moving average: the forecast for period t + 1, made at the end of period t, is the average of the n most recent demands:

$$F_{t+1} = \frac{D_t + D_{t-1} + \cdots + D_{t-n+1}}{n}$$

**(a) Forecast for week 4**

$$F_4 = \frac{411+380+400}{3} = \frac{1191}{3} = 397.0$$

**(b) Forecast error for week 4**

$$E_4 = D_4 - F_4 = 415 - 397.0 = 18$$

**(c) Forecast for week 5**

The forecast for week 5 uses the three most recent weeks, weeks 2 to 4.

$$F_5 = \frac{415+411+380}{3} = \frac{1206}{3} = 402.0$$

**Answer:** Forecast for week 4 = 397 patients; error = 18 patients (actual demand was higher); forecast for week 5 = 402 patients.

| | | | | |
|---|---|---|---|---|
| **66** | Arrivals of patients into a clinic have been noted for the past 3 weeks: 400, 380 and 411 (weeks 1, 2 and 3). It is now the end of week 3. Using exponential smoothing with $\alpha = 0.10$: (i) calculate the exponential smoothing forecast for week 4; (ii) what was the forecast error for week 4 if the actual demand turns out to be 415; (iii) what is the forecast for week 5?<br>Source: Unit III \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q7c | 06 | – | – |

Exponential smoothing: F(t+1) = $\alpha$ D(t) + (1 − $\alpha$) F(t). It needs an initial forecast; as in the textbook, the average of the demands of the first two weeks is taken: (400 + 380)/2 = 390 = $F_3$.

**(i) Forecast for week 4**

$$F_4 = 0.10(411) + 0.90(390) = 392.1$$

**(ii) Forecast error for week 4**

$$E_4 = D_4 - F_4 = 415 - 392.1 = 22.9$$

**(iii) Forecast for week 5**

$$F_5 = 0.10(415) + 0.90(392.1) = 394.4$$

The unrounded $F_4$ is used for $F_5$; only the final result is rounded.

**Answer:** $F_4$ = 392.1 (about 392 patients); $E_4$ = 22.9 (about 23 patients); $F_5$ = 394.4 (about 394 patients).

| | | | | |
|---|---|---|---|---|
| **67** | The manager of a pizza shop must forecast the weekly demand for the special pizzas so that he can order pizza shells weekly. Recent demand has been as follows: June 2: 50; June 9: 65; June 16: 52; June 23: 56; June 30: 55; July 7: 60. Forecast the demand for pizza for June 23 to July 14 by using the simple moving average method with $n = 3$. Then repeat the forecast by using the weighted moving average method with $n = 3$ and weights of 0.50, 0.30 and 0.20, with 0.50 applying to the most recent demand.<br>Source: Unit III \| SEE Feb/Mar 2025 (IM352IA), Q6a | 10 | 4 | 3 |

**Simple moving average (n = 3)**

Each forecast is the average of the three most recent weekly demands.

$$F_{\text{Jun 23}} = \frac{50+65+52}{3} = 55.67$$

$$F_{\text{Jun 30}} = \frac{65+52+56}{3} = 57.67$$

$$F_{\text{Jul 7}} = \frac{52+56+55}{3} = 54.33$$

$$F_{\text{Jul 14}} = \frac{56+55+60}{3} = 57.00$$

**Weighted moving average (n = 3; weights 0.50, 0.30, 0.20)**

The weight 0.50 is applied to the most recent demand, 0.30 to the one before it and 0.20 to the oldest of the three.

$$F_{\text{Jun 23}} = 0.5(52) + 0.3(65) + 0.2(50) = 55.50$$

$$F_{\text{Jun 30}} = 0.5(56) + 0.3(52) + 0.2(65) = 56.60$$

$$F_{\text{Jul 7}} = 0.5(55) + 0.3(56) + 0.2(52) = 54.70$$

$$F_{\text{Jul 14}} = 0.5(60) + 0.3(55) + 0.2(56) = 57.70$$

| Week | Demand | Simple moving average (n = 3) | Weighted moving average (0.5, 0.3, 0.2) |
|---|---|---|---|
| June 2 | 50 | | |
| June 9 | 65 | | |
| June 16 | 52 | | |
| June 23 | 56 | 55.67 | 55.50 |
| June 30 | 55 | 57.67 | 56.60 |
| July 7 | 60 | 54.33 | 54.70 |
| July 14 | ? | 57.00 | 57.70 |

**Answer:** Simple moving average: 55.67, 57.67, 54.33 and 57.00 pizzas for June 23, June 30, July 7 and July 14. Weighted moving average: 55.50, 56.60, 54.70 and 57.70 pizzas. The weighted forecasts respond more to the most recent demand.

| | | | | |
|---|---|---|---|---|
| **68** | PM Computer Services assembles customized personal computers from generic parts. The demand for the last 12 months is: Jan 37, Feb 40, Mar 41, Apr 37, May 45, Jun 51, Jul 43, Aug 47, Sep 56, Oct 52, Nov 55, Dec 54. (i) Use exponential smoothing with smoothing parameter $\alpha = 0.3$ to compute the demand forecast for January (Period 13). (ii) Use exponential smoothing with smoothing parameter $\alpha = 0.5$ to compute the demand forecast for January (Period 13).<br>Source: Unit III \| SEE Apr/May 2024 (21IM53), Q5a | 10 | 3 | 3 |

Exponential smoothing: F(t+1) = $\alpha$ D(t) + (1 − $\alpha$) F(t). The method needs an initial forecast; as in the textbook, the actual demand of the first period is used as the forecast for period 2 ($F_2 = D_1 = 37$), and the forecasts are then updated each period.

$$F_3 = 0.3(40) + 0.7(37) = 37.90 \quad (\alpha = 0.3)$$

$$F_3 = 0.5(40) + 0.5(37) = 38.50 \quad (\alpha = 0.5)$$

Continuing the same calculation for every period gives the table below.

| Period | Month | Demand | F ($\alpha$ = 0.3) | F ($\alpha$ = 0.5) |
|---|---|---|---|---|
| 1 | Jan | 37 | | |
| 2 | Feb | 40 | 37.00 | 37.00 |
| 3 | Mar | 41 | 37.90 | 38.50 |
| 4 | Apr | 37 | 38.83 | 39.75 |
| 5 | May | 45 | 38.28 | 38.38 |
| 6 | Jun | 51 | 40.30 | 41.69 |
| 7 | Jul | 43 | 43.51 | 46.34 |
| 8 | Aug | 47 | 43.36 | 44.67 |
| 9 | Sep | 56 | 44.45 | 45.84 |
| 10 | Oct | 52 | 47.91 | 50.92 |
| 11 | Nov | 55 | 49.14 | 51.46 |
| 12 | Dec | 54 | 50.90 | 53.23 |
| 13 | Jan (next) | | 51.83 | 53.61 |

$$F_{13} = 0.3(54) + 0.7(50.90) = 51.83$$

$$F_{13} = 0.5(54) + 0.5(53.23) = 53.61$$

**Answer:** Forecast for January (period 13): 51.8 units with $\alpha$ = 0.3 and 53.6 units with $\alpha$ = 0.5. The larger $\alpha$ follows the recent upward trend more closely, but because the series has a trend, both forecasts lag behind recent demand.

| | | | | |
|---|---|---|---|---|
| **69** | The Instant Paper Clip Office Supply Company wants to forecast the demand for deliveries during the next month. The orders for the past 10 months are: Jan 120, Feb 90, Mar 100, Apr 75, May 110, Jun 50, Jul 75, Aug 130, Sep 110, Oct 90. (a) Compute the monthly demand forecast for February through November using the naive method. (b) Compute the monthly demand forecast for April through November using a 3-month moving average. (c) Compute the monthly demand forecast for June through November using a 5-month moving average. (d) Compute the monthly demand forecast for April through November using a 3-month weighted moving average, with weights 0.5, 0.33 and 0.17 (heavier weights on the more recent months).<br>Source: Unit III \| SEE Apr/May 2024 (21IM53), Q6a | 10 | 3 | 3 |

Each forecast for month t uses only the demands of earlier months.

- **(a) Naive method:** the forecast for a month equals the demand of the previous month, F(t) = D(t−1).
- **(b) 3-month moving average:** F(t) = [D(t−1) + D(t−2) + D(t−3)]/3. For example, F(Apr) = (100 + 90 + 120)/3 = 103.33.
- **(c) 5-month moving average:** F(t) = average of the five previous months. For example, F(Jun) = (110 + 75 + 100 + 90 + 120)/5 = 99.00.
- **(d) 3-month weighted moving average** with weights 0.50 (most recent), 0.33 and 0.17. For example, F(Apr) = 0.50(100) + 0.33(90) + 0.17(120) = 100.10.

| Month | Orders | (a) Naive | (b) 3-month MA | (c) 5-month MA | (d) 3-month WMA |
|---|---|---|---|---|---|
| Jan | 120 | | | | |
| Feb | 90 | 120 | | | |
| Mar | 100 | 90 | | | |
| Apr | 75 | 100 | 103.33 | | 100.10 |
| May | 110 | 75 | 88.33 | | 85.80 |
| Jun | 50 | 110 | 95.00 | 99.00 | 96.75 |
| Jul | 75 | 50 | 78.33 | 85.00 | 74.05 |
| Aug | 130 | 75 | 78.33 | 82.00 | 72.70 |
| Sep | 110 | 130 | 85.00 | 88.00 | 98.25 |
| Oct | 90 | 110 | 105.00 | 95.00 | 110.65 |
| Nov | | 90 | 110.00 | 91.00 | 103.40 |

**Answer:** The forecasts for November are 90 (naive), 110.00 (3-month MA), 91.00 (5-month MA) and 103.40 (3-month WMA) orders.

| | | | | |
|---|---|---|---|---|
| **70** | The monthly demand for units manufactured by the Acme Rocket Company has been as follows.<br><br>(i) Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; $\alpha = 0.2$. (ii) Calculate the absolute percentage error for each month from June through December, and the MAD and MAPE of the forecast error as of the end of December. (iii) Calculate the tracking signal as of the end of December. What can you say about the performance of the forecasting method?<br><br>Source: Unit III \| CIE-II, 07/01/2025, Q3 (parts i to iii, 10 M); SEE Feb/Mar 2026 (IM352IA), Q5b (parts i, ii); SEE Jan/Feb 2021 (18IM54), Q7a (part i); SEE Fast Track Oct 2020 (16IM55/12IM62), Q7a (parts i, ii) | 10 | 3 | 3 |

| Month | Units | Month | Units |
|---|---|---|---|
| May | 100 | September | 105 |
| June | 80 | October | 110 |
| July | 110 | November | 125 |
| August | 115 | December | 120 |

Exponential smoothing with $\alpha = 0.2$: F(t+1) = 0.2 D(t) + 0.8 F(t), starting from the forecast for May, F(May) = 105.

**(i) Forecasts for June to January**

$$F_{\text{Jun}} = 0.2(100) + 0.8(105) = 104.00$$

$$F_{\text{Jul}} = 0.2(80) + 0.8(104.00) = 99.20$$

$$F_{\text{Aug}} = 0.2(110) + 0.8(99.20) = 101.36$$

$$F_{\text{Sep}} = 0.2(115) + 0.8(101.36) = 104.09$$

$$F_{\text{Oct}} = 0.2(105) + 0.8(104.09) = 104.27$$

$$F_{\text{Nov}} = 0.2(110) + 0.8(104.27) = 105.42$$

$$F_{\text{Dec}} = 0.2(125) + 0.8(105.42) = 109.33$$

$$F_{\text{Jan}} = 0.2(120) + 0.8(109.33) = 111.47$$

**(ii) Absolute percentage error, MAD and MAPE (June to December)**

| Month | Demand D | Forecast F | Error E | \|E\| | \|E\|/D (%) |
|---|---|---|---|---|---|
| June | 80 | 104.00 | -24.00 | 24.00 | 30.00 |
| July | 110 | 99.20 | 10.80 | 10.80 | 9.82 |
| August | 115 | 101.36 | 13.64 | 13.64 | 11.86 |
| September | 105 | 104.09 | 0.91 | 0.91 | 0.87 |
| October | 110 | 104.27 | 5.73 | 5.73 | 5.21 |
| November | 125 | 105.42 | 19.58 | 19.58 | 15.67 |
| December | 120 | 109.33 | 10.67 | 10.67 | 8.89 |
| **Total** | | | **37.33** | **85.33** | **82.31** |

$$MAD = \frac{\sum |E_t|}{n} = \frac{85.33}{7} = 12.19$$

$$MAPE = \frac{\sum (|E_t|/D_t)\,100}{n} = \frac{82.31}{7} = 11.76\%$$

**(iii) Tracking signal at the end of December**

$$CFE = \sum E_t = 37.33$$

$$\text{Tracking signal} = \frac{CFE}{MAD} = \frac{37.33}{12.19} = 3.06$$

The CFE is positive and equals about 3.1 MADs, which means the forecasts were consistently **below** the actual demand

(bias toward underestimating). The average forecast error (MAD) is about 12.2 units, or 11.8% of demand on average. The method is lagging behind a gently rising demand, as exponential smoothing does when there is a trend; if the tracking signal exceeds the control limits, a model with a trend (for example trend projection with regression) or a larger $\alpha$ should be tried.

**Answer:** Forecasts: Jun 104.0, Jul 99.2, Aug 101.4, Sep 104.1, Oct 104.3, Nov 105.4, Dec 109.3, Jan 111.5. MAD = 12.19, MAPE = 11.76%, tracking signal = 3.06.

| | | | | |
|---|---|---|---|---|
| **71** | A lathe manufacturer noted the sales invoices from January to August and found the sales in the months as 10, 11, 10, 11, 14, 15, 14 and 15 units respectively. Calculate the forecast for all the periods and his estimate of sales for September, (i) if he has followed a 4-month weighted moving average (with weights 0.4, 0.3, 0.2 and 0.1), (ii) if he has followed simple exponential smoothing (SES) with $\alpha = 0.2$ and his estimated sales for January is 11.<br>Source: Unit III \| CIE-II, 15/12/2025, Q3 (September estimate, 12 M); SEE Feb/Mar 2026 (IM352IA), Q6b (forecast for all periods, 08 M) | 12 | 4 | 3 |

**(i) Four-month weighted moving average (weights 0.4, 0.3, 0.2, 0.1; 0.4 on the latest month)**

F(t) = 0.4 D(t−1) + 0.3 D(t−2) + 0.2 D(t−3) + 0.1 D(t−4). The first forecast can be made for May, since four months of data are needed.

$$F_{\text{May}} = 0.4(11) + 0.3(10) + 0.2(11) + 0.1(10) = 10.60$$

$$F_{\text{Sep}} = 0.4(15) + 0.3(14) + 0.2(15) + 0.1(14) = 14.60$$

**(ii) Simple exponential smoothing ($\alpha$ = 0.2, F(Jan) = 11)**

F(t+1) = 0.2 D(t) + 0.8 F(t).

$$F_{\text{Feb}} = 0.2(10) + 0.8(11) = 10.80$$

$$F_{\text{Mar}} = 0.2(11) + 0.8(10.80) = 10.84$$

$$\ldots$$

$$F_{\text{Sep}} = 0.2(15) + 0.8(12.49) = 12.99$$

| Month | Sales | 4-month WMA forecast | SES forecast ($\alpha$ = 0.2) |
|---|---|---|---|
| January | 10 | | 11.00 |
| February | 11 | | 10.80 |
| March | 10 | | 10.84 |
| April | 11 | | 10.67 |
| May | 14 | 10.60 | 10.74 |
| June | 15 | 12.00 | 11.39 |
| July | 14 | 13.40 | 12.11 |
| August | 15 | 14.00 | 12.49 |
| September | | 14.60 | 12.99 |

**Answer:** Estimated sales for September: 14.6 units with the 4-month weighted moving average and 13.0 units with exponential smoothing ($\alpha$ = 0.2). The weighted moving average responds faster to the recent rise in sales.

| | | | | |
|---|---|---|---|---|
| **72** | Explain the importance of measuring forecast error in operations management. Discuss the managerial implications of each measure with suitable examples.<br>Source: Unit III \| CIE-II, 15/12/2025, Q1 | 08 | 2 | 3 |

### Why forecast error must be measured

- Forecasts almost always contain errors. Random error results from unpredictable factors; it can be reduced by selecting an appropriate model but never eliminated.
- Error measures show **how accurate** a technique is, help to **choose the best method** and the best parameters (n for the moving average, the weights, $\alpha$ for exponential smoothing), and give early warning when a method stops tracking demand.
- Forecast error for period t is E(t) = D(t) − F(t) (actual demand minus forecast). It is the basis of all the measures below.

### Error measures and their managerial implications

| Measure | Formula | Managerial implication |
|---|---|---|
| Cumulative sum of forecast errors (CFE), bias | CFE = ΣE | Measures the total error and the **bias**. A consistently positive CFE means the forecast is always too low; negative means too high. Bias is the worst kind of error because it systematically disrupts planning (stockouts or excess inventory); the aim is zero bias. |
| Average forecast error (mean bias) | $\bar{E}$ = CFE/n | The average direction and size of the bias per period. |
| Mean squared error (MSE) | MSE = ΣE²/n | Dispersion of the errors; large errors get far more weight because they are squared, so it is used when big misses are costly. |
| Standard deviation of the errors (s) | s = $\sqrt{[\Sigma(E - \bar{E})^2/(n-1)]}$ | Dispersion about the mean error; used to set safety capacity or stock for the forecast uncertainty. |
| Mean absolute deviation (MAD) | MAD = Σ\|E\|/n | The average size of the error in the units of demand, ignoring whether it is over or under. Easy to understand and widely used. |
| Mean absolute percent error (MAPE) | MAPE = [Σ(\|E\|/D)(100)]/n | Relates the error to the level of demand; the best measure for comparing the accuracy of items with different demand levels (different SKUs). |
| Tracking signal | CFE/MAD | The number of MADs in the CFE; a control-chart check on whether the method still tracks demand, so that the model can be revised. |

### Example (upholstered chairs, 8 months)

| Month t | Demand D | Forecast F | Error E | E² | \|E\| | \|E\|/D (%) |
|---|---|---|---|---|---|---|
| 1 | 200 | 225 | -25 | 625 | 25 | 12.5 |
| 2 | 240 | 220 | 20 | 400 | 20 | 8.3 |
| 3 | 300 | 285 | 15 | 225 | 15 | 5.0 |
| 4 | 270 | 290 | -20 | 400 | 20 | 7.4 |
| 5 | 230 | 250 | -20 | 400 | 20 | 8.7 |
| 6 | 260 | 240 | 20 | 400 | 20 | 7.7 |
| 7 | 210 | 250 | -40 | 1,600 | 40 | 19.0 |
| 8 | 275 | 240 | 35 | 1,225 | 35 | 12.7 |
| **Total** | | | **-15** | **5,275** | **195** | **81.3** |

$$CFE = -15, \quad \bar{E} = \frac{-15}{8} = -1.875$$

$$MSE = \frac{5{,}275}{8} = 659.4, \quad s = 27.4$$

$$MAD = \frac{195}{8} = 24.4, \quad MAPE = \frac{81.3}{8} = 10.2\%$$

- A CFE of -15 shows a slight bias to **overestimate** demand (average bias -1.88 units per month), so little corrective action is needed.

- A MAD of 24.4 means that the forecast was off by 24.4 chairs on average, and a MAPE of 10.2% means that the error was about 10 percent of demand; the standard deviation 27.4 is the spread of the errors, which can be used to plan the safety capacity.

| | | | | |
|---|---|---|---|---|
| **73** | Provide a note on the following types of error along with the formula used to compute the error: (i) mean absolute deviation, (ii) mean absolute percentage error.<br>Source: Unit III \| SEE Feb/Mar 2025 (IM352IA), Q5b | 06 | 2 | 3 |

### (i) Mean absolute deviation (MAD)

MAD is the mean of the absolute forecast errors over a series of time periods, without regard to whether the error was an overestimate or an underestimate. It measures the dispersion of forecast errors.

$$MAD = \frac{\sum |E_t|}{n}, \qquad E_t = D_t - F_t$$

A small MAD means that the forecast is typically close to the actual demand; a large MAD indicates the possibility of large errors. It is easy to understand and is in the same units as the demand. If errors are normally distributed with mean zero, s = 1.25 MAD (MAD ≈ 0.8 s), and MAD is also used to compute the tracking signal.

### (ii) Mean absolute percent error (MAPE)

MAPE relates the forecast error to the level of demand and is therefore useful for putting forecast performance in the proper perspective; for example, an absolute error of 100 is a larger percentage error when demand is 200 units than when it is 10,000 units.

$$MAPE = \frac{\sum \left(|E_t|/D_t\right)(100)}{n} \quad \text{(expressed as a percentage)}$$

MAPE is the best error measure to use when making comparisons between time series for different items (SKUs).

**Illustration (textbook data, 8 months):** MAD = 195/8 = 24.4 chairs and MAPE = 81.3/8 = 10.2%.

| | | | | |
|---|---|---|---|---|
| **74** | A firm uses three forecasting models: moving average, exponential smoothing and linear regression. Explain how forecast error is calculated and used to identify the most accurate forecasting model.<br>Source: Unit III \| SEE Feb/Mar 2026 (IM352IA), Q6a | 08 | 3 | 2 |

### How the forecast error is calculated

For each period the error is the actual demand minus the forecast, E(t) = D(t) − F(t). Each model is run on the same history file and the errors are summarized over all periods by CFE (bias), MSE and s, MAD and MAPE.

### How the errors identify the most accurate model

(1) **Minimize bias (CFE):** a good method has errors that cancel out (CFE close to zero); bias is the worst kind of error.
(2) **Minimize MAD, MAPE or MSE:** the model with the smallest dispersion of errors is the more accurate; MAPE is used to compare series of different size.
(3) **Maximize $r^2$** for regression models (fit of the line to the data).
(4) **Check managerial expectations** of the demand components (for example, a trend), and give weight to the errors in the **recent** periods.
(5) The parameters are also selected by errors: n for the moving average, the weights, $\alpha$ for exponential smoothing and the period from which the regression starts. Because a model that fits the past best is not necessarily best for the future, a **holdout sample** of recent periods is used as a final test, and the tracking signal is monitored afterwards.

**Illustration with the Acme Rocket data (August to December)**

| Measure | Naive | 3-month moving average | Exponential smoothing ($\alpha$ = 0.2) |
|---|---|---|---|
| CFE | 10.0 | 43.3 | 50.5 |
| MAD | 8.00 | 8.67 | 10.11 |
| MSE | 80.0 | 123.3 | 143.4 |
| MAPE (%) | 6.92 | 7.33 | 8.50 |

On these five periods the naive method has the smallest MAD (8.00) and MAPE (6.92%), so it would be preferred on the accuracy measures; the exponential smoothing and moving-average forecasts have a large positive CFE (43.3 and 50.5) because they lag behind the rising demand, i.e. they are biased. The ranking can change when other periods or parameter values are used, which is why the best-fitting model is re-checked on a holdout sample and its tracking signal is monitored. (When no single model is clearly best, an average of the forecasts, a combination forecast, is often more accurate than any one of them.)

| 75 | Examine the various factors affecting demand forecasting.<br>Source: Unit III \| SEE Apr/May 2024 (21IM53), Q6b | 06 | 2 | 3 |
|---|---|---|---|---|

The following factors, discussed in the textbook, affect how demand is forecast and how accurate the forecast is:

| Factor | Effect on forecasting |
|---|---|
| Demand patterns | Demand is a combination of horizontal, trend, seasonal and cyclical patterns plus random variation. The method must match the pattern; random variation can never be forecast and makes every forecast inaccurate. Cyclical patterns are the hardest because they depend on business cycles and product life cycles. |
| Availability of a history file | Quantitative methods need adequate historical data; for a new product or technology, or when past data are distorted by special events, judgment methods are the only practical method. |
| External (causal) factors | Promotional campaigns, pricing, economic conditions, competitors' actions and weather can be related to demand through causal methods (regression) or used to adjust forecasts judgmentally. |
| Level of aggregation | Forecasts for groups of items and for the year are more accurate than forecasts for individual items and short periods (errors for the total are typically below 5%); hence the two-tier system of families and SKUs. |
| Unit of measurement | Forecast in service or product units (or machine-hours) rather than dollars, and then convert. |
| Time horizon | Short-term forecasts use time-series methods; trend projection and causal methods suit longer horizons; stable demand allows larger n or smaller $\alpha$, dynamic demand needs smaller n or larger $\alpha$. |
| Process and information sharing | Forecasts depend on the quality of the forecasting process, on contextual knowledge of the stakeholders and on collaboration with customers and suppliers (CPFR); better collaboration and communication improve accuracy more than a more advanced technique. |

| | | | | |
|---|---|---|---|---|
| **76** | The forecast process itself, typically done on a monthly basis, consists of structured steps. These steps often are facilitated by someone who might be called a demand manager, forecast analyst or demand/supply planner. However, many other people are typically involved before the plan for the month is authorized. Identify and discuss those structured steps for the forecasting process.<br>Source: Unit III \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q7a | 08 | – | – |

Companies often prepare forecasts for hundreds or thousands of services or products every month. A typical forecasting process uses a history file of past demand (with notes explaining unusual demand, such as promotions and closeouts) plus information from salesforce estimates, outstanding bids, booked orders, market research, competitor behavior, economic outlook, new product introductions, pricing and promotions. The outputs are forecasts for several periods ahead (typically monthly, projected from six months to two years), rolled up to product families or broken down to SKUs.

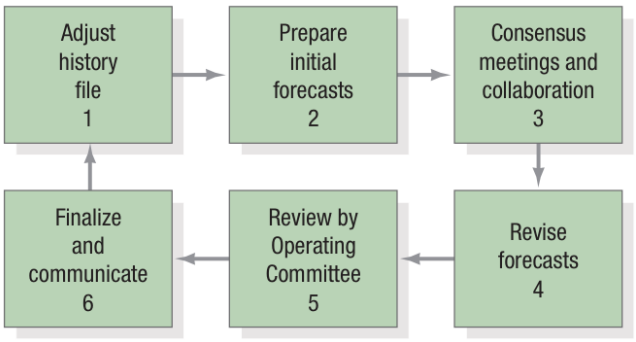



**Figure 38:** Steps of a typical forecasting process

(1) **Adjust the history file:** the cycle begins mid-month, just after the forecasts have been finalized and communicated. Update the history file with the actual demand at the end of the month and review the forecast accuracy.

(2) **Prepare initial forecasts:** use a forecasting software package and judgment; adjust the parameters to find models that fit the past demand well and still reflect the demand manager's judgment on irregular events and on future sales information from various sources.

(3) **Hold consensus meetings and collaboration:** meet the stakeholders (marketing, sales, supply chain planners, finance), make it easy for business units and field sales to give input, and use the Internet to get information from key customers and suppliers; the goal is a consensus forecast.

(4) **Revise the forecasts** using judgment, considering the inputs from the consensus meetings and the collaborative sources.

(5) **Review by the operating committee:** present the forecasts to the operating committee to reach a final set of forecasts that everybody agrees upon and will work to support.

(6) **Finalize and communicate** the forecasts, based on the committee's decisions, to the important stakeholders (supply chain planners are the biggest users).

Collaborative planning, forecasting and replenishment (CPFR) adds considerable collaboration with customers and suppliers, especially in step 3. Like all work, forecasting is a process that should be continually reviewed for improvement.

| | | | | |
|---|---|---|---|---|
| 77 | What is the Theory of Constraints (TOC)? Clearly outline its central ideas and explain the seven key principles of the Theory of Constraints with suitable examples. List any two benefits of applying the TOC to practical scenarios.<br>Source: Unit IV \| SEE Apr/May 2024 (21IM53), Q7a; SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q6a; SEE Fast Track Oct 2020 (16IM55/12IM62), Q6b; SEE Feb/Mar 2025 (IM352IA), Q7b (list of the 7 principles) | 08 | 3 | 3 |

### Meaning and central idea

The **theory of constraints (TOC)** is a systematic management approach that focuses on actively managing those constraints that impede a firm's progress toward its goal of maximizing profits and effectively using its resources. It was developed by Eli Goldratt and outlines a deliberate process for identifying and overcoming constraints.

The chief concept is that the **bottlenecks should be scheduled to maximize their throughput** of services or products while adhering to the promised completion dates. The process focuses not on the efficiency of individual processes but on the bottlenecks that constrain the system as a whole, and on making materials flow rapidly through the entire system so that profits rise, and inventory and workforce levels fall while the critical resources are still used effectively.

| Operational measure | TOC view | Relation to financial measures |
|---|---|---|
| Inventory (I) | All the money invested in purchasing things that the system intends to sell | A decrease in I raises net profit, ROI and cash flow |
| Throughput (T) | Rate at which the system generates money through sales | An increase in T raises net profit, ROI and cash flow |
| Operating expense (OE) | All the money the system spends to turn inventory into throughput | A decrease in OE raises net profit, ROI and cash flow |
| Utilization (U) | Ratio of average output rate to maximum capacity | An increase in U at the bottleneck raises profit |

### Seven key principles of the TOC (with examples)

Example used below: a rake plant where the welding station is loaded at 105% of capacity and every other process at 75%. Welding is the bottleneck.

(1) **Balance flow, not capacity.** The goal is a smooth flow of work at the bottleneck's pace; the rake line should be paced to the welder rather than building every station to the same capacity.

(2) **Maximizing the output and efficiency of every resource may not maximize the throughput of the entire system.** Running the blanking press at full speed only piles up rake heads in front of the welder.

(3) **An hour lost at a bottleneck is an hour lost for the whole system; an hour saved at a non-bottleneck is a mirage.** Any idle time at the welding station reduces the output of rakes, whereas the cleaning station saving an hour gains nothing.

(4) **Inventory is needed only in front of the bottlenecks (to keep them from sitting idle) and in front of assembly and shipping points (to protect customer schedules).** Building stock elsewhere should be avoided.

(5) **Work should be released into the system only as frequently as the bottlenecks need it.** Bottleneck flow should equal market demand; pacing everything to the slowest resource minimizes inventory and operating expense.

(6) **Activating a non-bottleneck resource (using it more, without raising throughput) is not the same as utilizing a bottleneck resource** (which increases throughput). Activation of non-bottlenecks cannot raise throughput or improve the financial measures.

(7) **Every capital investment must be viewed from the perspective of its global impact on throughput (T), inventory (I) and operating expense (OE).** A new blanking press that does not raise welding output adds only to I and OE.

### Practical application: the five steps

(1) Identify the system bottleneck(s).

(2) Exploit the bottleneck(s): schedule it to maximize its throughput, passing only good-quality parts to it.

(3) Subordinate all other decisions to step 2: non-bottlenecks are scheduled to support the bottleneck.

(4) Elevate the bottleneck(s): add a shift or a machine if it is still the constraint after the schedule improvements.

(5) Do not let inertia set in: the constraint may shift, so repeat steps 1 to 4.

### Benefits of applying the TOC (any two)

- It increases profit by raising throughput while reducing inventory and operating expense; at the U.S. Marine Corps Maintenance Center the repair cycle time fell from 167 days to 58 days, work-in-process fell from 550% to 140% of demand and the repair cost fell by 25 to 30 percent.
- It changes the mind-set of managers and employees: instead of focusing only on their own functions they see the big picture and where improvements in the whole system lie.

| | | | | |
|---|---|---|---|---|
| **78** | Explain how the Theory of Constraints can be applied to identify and manage bottlenecks in manufacturing and service processes. Provide examples of how identifying and addressing bottlenecks can improve the overall efficiency.<br>Source: Unit IV \| CIE-III, 27/01/2025, Q1; SEE Apr/May 2024 (21IM53), Q8b | 10 | 3 | 3 |

According to the TOC, performance is improved by finding the bottleneck, scheduling it so that it never starves, and subordinating everything else to it (identify, exploit, subordinate, elevate, repeat).

### 1. Identifying a bottleneck

A workstation is a bottleneck if (1) it has the **highest total time per unit processed**, or (2) it has the **highest average utilization and total workload**. Piles of material in front of a workstation, and the opinion of supervisors, also point to it.

#### Service process (loan approval at a bank)

Approved loans pass through: check documents (15 min) → categorize loans (20 min) → credit check (15 min) in parallel with data entry (12 min) → paperwork (10 min). Step 2, categorizing, has the highest time per loan, so it is the bottleneck:

$$\text{Capacity} = \tfrac{60}{20} = 3 \text{ loans per hour } \Rightarrow\ 3 \times 5 = 15 \text{ loans in a 5-hour day}$$

$$\text{Throughput time} = 15 + 20 + \max(15, 12) + 10 = 60 \text{ minutes}$$

Management can increase the flow only by increasing the capacity of step 2 (more staff or a faster method) up to the point where another step becomes the bottleneck.

#### Manufacturing process (batch process, Diablo Electronics)

Four products (A to D) are made on five workstations (V to Z), each available 2,400 minutes per week and setups are negligible. The load on a station is the sum of (time per unit × weekly demand):

| Workstation | Load from A | Load from B | Load from C | Load from D | Total load (min) |
|---|---|---|---|---|---|
| V | 1800 | 0 | 0 | 0 | **1,800** |
| W | 0 | 0 | 400 | 1500 | **1,900** |
| X | 600 | 1600 | 400 | 0 | **2,600** |
| Y | 600 | 800 | 400 | 500 | **2,300** |
| Z | 0 | 0 | 400 | 1000 | **1,400** |

Workstation X has the largest load (2,600 minutes against 2,400 available), so it is the bottleneck.

### 2. Managing the bottleneck

- **Keep the bottleneck busy:** monitor short-term schedules, keep a time buffer of work in front of it, remove idle time caused by delays elsewhere and feed it only good-quality parts.
- **Cut setup losses:** process many units per setup at the bottleneck (fewer setups mean less lost time).
- **Pace the non-bottlenecks to the bottleneck** and release material only as the bottleneck needs it (drum-buffer-rope).
- **Elevate the bottleneck:** if it is still the constraint, add a shift or more days per week, add or replace machines, or redesign the process (process reengineering or improvement).
- **Choose the product mix by the bottleneck** (contribution margin per bottleneck minute).

**Example of the effect:** at the Marine Corps Maintenance Center the real constraint was the scheduling policy (products were pushed on to the shop floor without regard to the status of the resources), not the capacity. A simplified drum-buffer-rope system cut the repair cycle time from 167 to 58 days and the repair cost by 25 to 30 percent. The throughput of the whole system rose without buying any equipment.

| 79 | How are bottlenecks removed (relieved) in manufacturing? Explain.<br>Source: Unit IV \| SEE Mar/Apr 2023 (18IM54), Q6a | 08 | – | – |
|---|---|---|---|---|

The key to relieving a bottleneck is to preserve its capacity (short term) and then expand it (long term).

**Short-term: preserve the bottleneck's capacity**

- Carefully monitor the short-term schedules and keep the bottleneck as busy as is practical.
- Minimize idle time at the bottleneck caused by delays elsewhere in the system and make sure it has all the resources it needs to stay busy.
- Process a **large number of units per setup** at the bottleneck compared with the non-critical operations, so that fewer setups mean less total time lost to setups (setup time is unproductive time).
- Schedule the bottleneck first and subordinate the other operations to it (drum-buffer-rope), protecting it with a time buffer.

**Long-term: expand the capacity of the bottleneck**

- Invest in **new equipment** and in brick-and-mortar **facility expansion**.
- Operate the bottleneck **more hours per week**: add employees and go from one shift to multiple shifts, or from 5 days to 6 or 7 days per week.
- Redesign the process by **process reengineering or process improvement**, or purchase additional machines or machines that handle more capacity.

Once the bottleneck is relieved, the constraint may shift to another resource, so the steps are repeated (do not let inertia set in).

| 80 | Explain how the Theory of Constraints is applied to product mix decisions.<br>Source: Unit IV \| SEE Feb/Mar 2026 (IM352IA), Q8b | 08 | 3 | 3 |
|---|---|---|---|---|

Managers are tempted to make the products with the highest contribution margin per unit (the **traditional method**). But the firm's actual throughput and profit depend more on the contribution margin earned **at the bottleneck** than on the margin of each product. The **bottleneck method** therefore ranks the products by the **contribution margin per minute of processing time at the bottleneck** and accepts orders in that sequence, up to the limit of demand and until the bottleneck capacity is used.

The textbook example is Diablo Electronics (products A to D, workstations V to Z, 2,400 minutes per week at each, bottleneck workstation X, labor \$3,600 and overhead \$8,500 per week, both fixed):

| | A | B | C | D |
|---|---|---|---|---|
| Price per unit (\$) | 75 | 72 | 45 | 38 |
| Raw material and purchased parts (\$) | 10 | 5 | 5 | 10 |
| Contribution margin (\$) | 65 | 67 | 40 | 28 |
| Time at bottleneck X (min) | 10 | 20 | 5 | 0 |
| Contribution margin per bottleneck minute (\$) | 6.50 | 3.35 | 8.00 | not defined |

**Decision rule 1: traditional method**

Sequence of products by contribution margin per unit: B, A, C, D. Allocating the resources in this order until X is used up gives the mix **60 A, 80 B, 40 C, 100 D**.

$$\text{Revenue} = 15,860 \quad \text{Materials} = 2,200$$

$$\text{Profit} = 15,860 - 2,200 - 3,600 - 8,500 = \$1,560$$

**Decision rule 2: bottleneck method**

Sequence by contribution margin per minute at X: D, C, A, B (D uses no time at X, so it is scheduled first). The resulting mix is **60 A, 70 B, 80 C, 100 D**.

$$\text{Revenue} = 16,940 \quad \text{Materials} = 2,350$$

$$\text{Profit} = 16,940 - 2,350 - 3,600 - 8,500 = \$2,490$$

**Answer:** The bottleneck method raises the weekly profit from $1,560 to $2,490, an increase of $930 (about 60 percent), because the product sequence is reversed and the bottleneck resource is used for the products that earn the most per bottleneck minute. (Linear programming could also be used to find the best mix.)

| | | | | |
|---|---|---|---|---|
| **81** | Given job processing data, identify the bottleneck machine and schedule the jobs to minimize idle time.<br>Source: Unit IV \| SEE Feb/Mar 2026 (IM352IA), Q9b | 08 | 3 | 4 |

| Machine | Available time | Required time |
|---|---|---|
| M1 | 40 | 35 |
| M2 | 40 | 45 |
| M3 | 40 | 30 |

### Step 1: identify the bottleneck

A workstation is the bottleneck if it has the highest utilization and workload. Utilization is the ratio of the load (required time) to the available time.

| Machine | Available time (h) | Required time (h) | Utilization = required/available | Idle time if all work is done (h) | Overload (h) |
|---|---|---|---|---|---|
| M1 | 40 | 35 | 87.5% | 5 | 0 |
| M2 | 40 | 45 | 112.5% | 0 | 5 |
| M3 | 40 | 30 | 75.0% | 10 | 0 |

Only **M2** requires more time (45 h) than is available (40 h); its utilization is 112.5%, the highest, so M2 is the bottleneck.

### Step 2: schedule the jobs (TOC steps: exploit and subordinate)

(1) **Exploit M2:** schedule the bottleneck first and keep it busy for all 40 available hours; no idle time may arise at M2 (put a time buffer of work in front of it, give it only good parts and cut its setups).
(2) **Subordinate M1 and M3:** release work to M1 only at the rate that M2 can take it. Since M2 can process only 40 of the 45 hours of work (88.9%), M1 needs only about $35 \times 0.889 = 31.1$ h and M3 about $30 \times 0.889 = 26.7$ h of the available 40 hours; the rest of their time is idle (and cannot raise throughput).
(3) **Elevate M2 (if the full demand must be met):** the 5-hour shortfall is covered by overtime, a second shift, subcontracting or a second machine at M2.

**Answer:** M2 is the bottleneck (utilization 112.5%). It is scheduled for all 40 hours with no idle time, M1 and M3 are paced to it (idle 5 h and 10 h when all required work is done), and the 5-hour overload at M2 is removed by adding capacity at M2.

| | | | | |
|---|---|---|---|---|
| **82** | Take a process that you encounter on a daily basis, such as the lunch cafeteria or the journey from your home to school or work, and identify the bottlenecks that limit the throughput of this process.<br>Source: Unit IV \| SEE Jan/Feb 2021 (18IM54), Q6a | 06 | – | – |

Take the **lunch process in a college cafeteria** (a front-office service process with a single line of students; one worker at each step, demand greater than capacity).

| Step | Activity | Time per student (min) |
|---|---|---|
| 1 | Enter the line and pick up a tray and cutlery | 0.5 |
| 2 | Choose and be served the food at the counter | 3.0 |
| 3 | Collect drinks and dessert | 1.0 |
| 4 | Pay at the cashier | 1.5 |
| 5 | Find a seat | 1.0 |

A step is the bottleneck if it has the highest time per customer. Here that is **step 2, the food counter (3 minutes per student)**:

$$\text{Capacity of the process} = \frac{60}{3} = 20 \text{ students per hour}$$

$$\text{Throughput time} = 0.5 + 3.0 + 1.0 + 1.5 + 1.0 = 7 \text{ minutes (without waiting)}$$

Every student must pass the counter, so the cafeteria cannot serve more than 20 students an hour however fast the cashier works, and a queue builds up in front of the counter. An hour saved at the cashier or at the drinks station would be a mirage (principle 3 of the TOC).

**How the bottleneck can be relieved**

- **Exploit:** keep the counter busy and prevent starvation (prepared food kept ready; staff breaks staggered).
- **Subordinate:** the other steps need not run faster than 20 students per hour.
- **Elevate:** add a second serving counter or worker, or serve the most popular items from a separate express counter; the cashier (capacity 40 per hour) then becomes the next constraint.
- **The same method applies to the journey to college:** the step with the longest time per trip (for example the traffic signal or bus change) limits the throughput of the whole journey.

| | | | | |
|---|---|---|---|---|
| **83** | The Dahlia Medical Center has 30 labour rooms, 15 combination labour and delivery rooms, 3 delivery rooms, and 1 special delivery room reserved for complicated births. All these facilities operate around the clock. Time spent in a labour room varies from hours to days, with an average of about a day. The average uncomplicated delivery requires about 1 hour in a delivery room. During an exceptionally busy 3-day period, 109 healthy babies were born at Dahlia Medical Center. Sixty babies were born in separate labour and delivery rooms, 45 were born in combined labour and delivery rooms, and only 4 babies required a labour room and the complicated-delivery room. Which of the facilities (labour rooms, combination labour and delivery rooms, or delivery rooms) had the greatest utilization rate?<br>Source: Unit IV \| SEE Jan/Feb 2021 (18IM54), Q6b | 10 | – | – |

Utilization is the ratio of the average output (here, the room-hours used) to the maximum capacity, expressed as a percentage:

$$\text{Utilization} = \frac{\text{Average output rate}}{\text{Maximum capacity}} \times 100\%$$

All facilities operate 24 hours a day, so each room provides 24 × 3 = 72 room-hours in the 3-day period. A baby delivered in separate rooms uses a labour room for about a day (24 h) and a delivery room for about 1 hour; a baby in a combination room uses that room for the labour and the delivery (24 + 1 = 25 h); the 4 complicated cases use a labour room and the special delivery room.

| Facility | Number of rooms | Capacity in 3 days (room-hours) | Room-hours used | Utilization |
|---|---|---|---|---|
| Labour rooms | 30 | 2160 | (60 + 4) × 24 = 1536 | 71.1% |
| Combination labour and delivery rooms | 15 | 1080 | 45 × (24 + 1) = 1125 | 104.2% |
| Delivery rooms | 3 | 216 | 60 × 1 = 60 | 27.8% |

$$\text{Labour rooms: } \frac{1536}{2160} = 71.1\%$$

$$\text{Combination rooms: } \frac{1125}{1080} = 104.2\%$$

$$\text{Delivery rooms: } \frac{60}{216} = 27.8\%$$

**Answer:** The combination labour and delivery rooms had the greatest utilization (about 104%); the labour rooms had about 71% and the delivery rooms only about 28%. The utilization above 100% in this exceptionally busy period shows that the combination rooms are the bottleneck of the facility.

| | | | | |
|---|---|---|---|---|
| **84** | Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for at least the next 3 months. The plant will operate 40 hours per week. (i) What should be the line's cycle time? (ii) What is the smallest number of workstations that she could hope for in designing the line for this cycle time? (iii) Suppose that she finds a solution that requires only 5 stations; what would be the line's efficiency?<br>Source: Unit IV \| SEE Feb/Mar 2025 (IM352IA), Q7a; SEE Mar/Apr 2023 (18IM54), Q6b; SEE Jan/Feb 2021 (18IM54), Q5a (parts i, ii); SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q6b | 10 | 4 | 3 |

### (i) Cycle time

The desired output rate is r = 2,400 units / 40 hours = 60 units per hour. The cycle time is the reciprocal of the desired hourly output rate:

$$c = \frac{1}{r} = \frac{1}{60} \text{ hour/unit} = 1 \text{ minute/unit} = 60 \text{ seconds/unit}$$

### (ii) Smallest number of workstations (theoretical minimum)

The Big Broadcaster's work elements (textbook data) are given below.

| Work element | Description | Time (s) | Immediate predecessor(s) |
|---|---|---|---|
| A | Bolt leg frame to hopper | 40 | — |
| B | Insert impeller shaft | 30 | A |
| C | Attach axle | 50 | A |
| D | Attach agitator | 40 | B |
| E | Attach drive wheel | 6 | B |
| F | Attach free wheel | 25 | C |
| G | Mount lower post | 15 | C |
| H | Attach controls | 20 | D, E |
| I | Mount nameplate | 18 | F, G |
| | **Total** | **244** | |

$$TM = \frac{t}{c} = \frac{244 \text{ seconds}}{60 \text{ seconds}} = 4.067 \Rightarrow 5 \text{ stations (rounded up)}$$

### (iii) Efficiency of a five-station solution

$$\text{Efficiency} = \frac{t}{nc}(100) = \frac{244}{5(60)}(100) = 81.3\%$$

Idle time per unit = nc − t = 300 − 244 = 56 seconds and the balance delay = 100 − 81.3 = 18.7%.

**Answer:** Cycle time = 60 seconds per spreader; smallest number of stations = 5; efficiency of the five-station line = 81.3%. If a five-station solution satisfying the precedence and cycle-time requirements is found, it is optimal, since five is the theoretical minimum.

| | | | | |
|---|---|---|---|---|
| **85** | Johnson Logs wants to set up a line to serve 60 customers per hour. The work elements and their precedence relationships are as follows: A (40 s, none); B (30 s, A); C (50 s, A); D (40 s, B); E (6 s, B); F (25 s, C); G (15 s, C); H (20 s, D and E); I (18 s, F and G); J (30 s, H and I); total 274 s. (i) What is the minimum number of stations? (ii) How many stations are required using the longest work element decision rule? (iii) Suppose that a solution requiring five stations is obtained; what is the efficiency?<br>Source: Unit IV \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q6a | 08 | – | – |

Desired output = 60 customers per hour, so the cycle time is c = 3600/60 = **60 seconds** per customer.

**(i) Minimum number of stations**

$$TM = \frac{t}{c} = \frac{274}{60} = 4.57 \Rightarrow 5 \text{ stations}$$

**(ii) Longest work element decision rule**

Stations are created one at a time. A work element is a candidate if all its predecessors have been assigned and adding it does not push the station time above the cycle time (60 s); among the candidates the **longest element** is chosen first, which fits the most difficult elements first and leaves the short ones to fill out the station.

| Station | Work elements assigned (in order) | Station time (s) | Idle time (s) |
|---|---|---|---|
| S1 | A (40) | 40 | 20 |
| S2 | C (50) | 50 | 10 |
| S3 | B (30), F (25) | 55 | 5 |
| S4 | D (40), G (15) | 55 | 5 |
| S5 | I (18), E (6), H (20) | 44 | 16 |
| S6 | J (30) | 30 | 30 |

The rule needs **6 stations**.

**(iii) Efficiency of a five-station solution**

$$\text{Efficiency} = \frac{t}{nc}(100) = \frac{274}{5(60)}(100) = 91.3\%$$

**Answer:** Minimum number of stations = 5; the longest-work-element rule gives 6 stations; if a five-station solution can be found, the efficiency is 91.3% (balance delay 8.7%).

| | | | | |
|---|---|---|---|---|
| **86** | In order to meet holiday demand, a pizza house requires a production line that is capable of producing 50 pizzas per week while operating only 40 hours per week. There are only 4 steps required to produce a single pizza, with respective processing times of 5 min, 5 min, 45 min and 15 min. (i) What should be the line's cycle time? (ii) What is the smallest number of workstations the pizza house could hope for in designing the line for this cycle time? (iii) Suppose that the pizza house finds a solution that requires only four stations; what would be the efficiency of this line?<br>Source: Unit IV \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q6b | 08 | – | – |

**(i) Cycle time**

$$r = \frac{50 \text{ pizzas}}{40 \text{ hours}} = 1.25 \text{ pizzas/hour}$$

$$c = \frac{1}{r} = \frac{1}{1.25} \text{ hour} = 48 \text{ minutes per pizza}$$

**(ii) Smallest number of workstations**

$$TM = \frac{t}{c} = \frac{5+5+45+15}{48} = \frac{70}{48} = 1.46 \Rightarrow 2 \text{ stations}$$

Because the steps follow each other in a fixed order and the 45-minute step cannot be combined with the next 15-minute

step within 48 minutes, a practical solution needs three stations (steps 1 and 2 = 10 min; step 3 = 45 min; step 4 = 15 min); two stations is the theoretical minimum.

**(iii) Efficiency with four stations**

$$\text{Efficiency} = \frac{t}{nc}(100) = \frac{70}{4(48)}(100) = 36.5\%$$

**Answer:** Cycle time = 48 minutes; smallest number of workstations = 2; efficiency of a four-station line = 36.5% (balance delay 63.5%).

| | | | | |
|---|---|---|---|---|
| **87** | Explain how the concept of dependent demand in material requirements planning is fundamental to resource planning.<br>Source: Unit IV \| CIE-II, 07/01/2025, Q5 | 10 | 2 | 3 |

Resource planning begins with the concept of **dependent demand**, which sets it apart from the independent-demand inventory techniques.

**Independent and dependent demand**

- **Independent demand** is the demand for a final product (for example, a bicycle) that is influenced only by market conditions; it must be forecast.
- **Dependent demand** is the demand for an item that occurs because the quantity required varies with the production plans for other items held in the firm's inventory. The demand for spokes, rims, handlebars, pedals and frames depends on the planned production of bicycles. Every bicycle needs two wheel rims, so 1,000 bicycles need 1,000 × 2 = 2,000 rims; forecasting techniques are not needed for the rims, they are calculated.

**Why it makes independent-demand systems unsuitable**

- Dependent demand is **lumpy**: even if customer demand for bicycles is continuous and uniform, assembly orders are placed in large lots (for example 1,000 bicycles every few days), and the rims are withdrawn in large quantities on only a few days. Reorder-point systems built for independent demand gave unsatisfactory results.
- The production decisions for the parent (assembly costs and capacity) **determine** the demand for the components, so the requirements for each component must be computed from the parent's schedule.
- A component can have **both** kinds of demand (rims needed for new bicycles, and sold as replacement parts); MRP handles both.

**How MRP uses the concept**

Material requirements planning (MRP) is a computerized information system developed specifically to manage dependent-demand inventory and to schedule replenishment orders. It uses the **parent–component relationships** in the bill of materials, the **master production schedule** of the end items, and the **inventory records** to explode the requirements of the end items level by level into time-phased requirements (gross requirements, planned receipts and planned order releases) for every subassembly, component and raw material. The result is the right quantity at the right time for every item, which in turn defines the resources (materials, workforce, equipment and cash) that the firm needs. This is why dependent demand is fundamental to resource planning.

| | | | | |
|---|---|---|---|---|
| **88** | List and explain the inputs and outputs of MRP with the help of an MRP explosion diagram.<br>Source: Unit IV \| Improvement CIE, 12/01/2026, Q2 (three major inputs); SEE Apr/May 2024 (21IM53), Q8a (inputs); SEE Mar/Apr 2023 (18IM54), Q8b (inputs and outputs) | 10 | 2 | 2 |

Material requirements planning (MRP) translates the master production schedule and other sources of demand into the requirements for all subassemblies, components and raw materials needed to make the parent items. This process is called the **MRP explosion**.

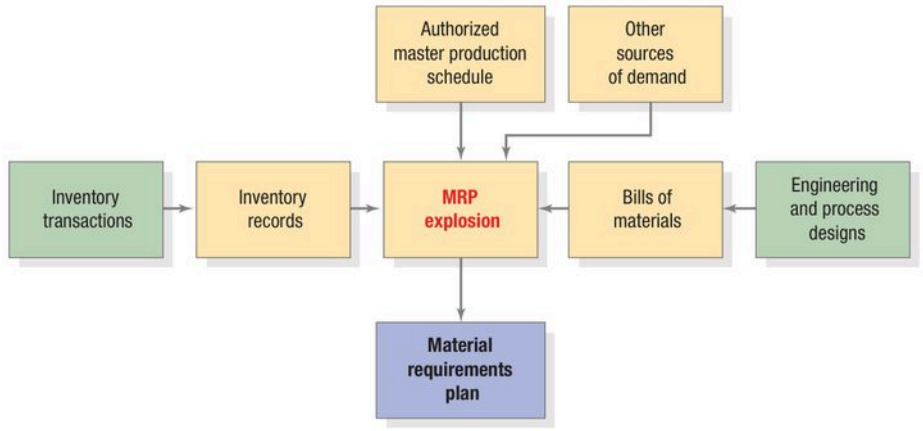



**Figure 39:** Material requirements plan inputs (MRP explosion)

### Inputs

(1) **Authorized master production schedule (MPS):** details how many end items will be produced in specified time periods; it breaks the sales and operations plan into product schedules. Other sources of demand, such as independent demand for replacement parts and maintenance items, are added.

(2) **Bill of materials (BOM):** the record of all components of an item, the parent–component relationships and the usage quantities, derived from the engineering and process designs. The replenishment schedule of a component follows from the schedules of its parents.

(3) **Inventory records:** show each item's lot-size policy, lead time and time-phased data (gross requirements, scheduled receipts, projected on-hand inventory, planned receipts, planned order releases), kept up to date by inventory transactions (releasing orders, receiving, withdrawals, cancellations, corrections).

### Outputs

- **Material requirements plan:** the planned order release schedule (timing and quantity) of each component, subassembly and purchased item.
- **Action notices:** computer-generated memos that alert planners to release new orders or adjust the due dates of scheduled receipts.
- **Priority reports:** such as dispatch lists and supplier schedules, based on the due dates of scheduled receipts.
- **Capacity reports:** capacity requirements planning (CRP), finite-capacity scheduling and input–output control, which project the workload at the workstations and show the bottlenecks.
- **Performance reports,** and the **manufacturing resources plan (MRP II),** which ties MRP to cost and price data and to the financial system (projected shipments, inventories, backlogs, cash flow and profit).

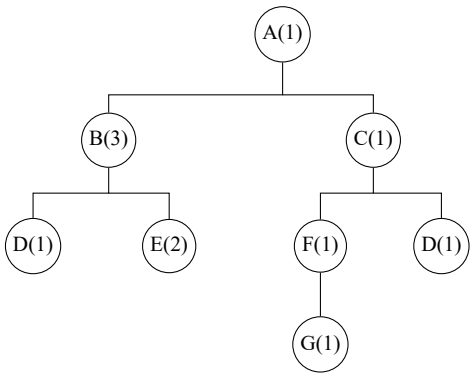
| **89** | Create a bill of materials for a ceiling fan. Provide the part number, quantity, unit of measure and description for each item.<br>Source: Unit IV \| SEE Feb/Mar 2025 (IM352IA), Q8a | 10 | 5 | 3 |
|---|---|---|---|---|

A **bill of materials (BOM)** is a record of all the components of an item, the parent–component relationships and the usage quantities derived from the engineering and process designs. The usage quantity is the number of units of a component needed to make one unit of its immediate parent. The ceiling fan is the end item (a parent but not a component); the motor, blade and down-rod assemblies are intermediate items (subassemblies); the remaining items are purchased items.

| Level | Part number | Description | Quantity per parent | Unit of measure |
|---|---|---|---|---|
| 0 | CF-1000 | Ceiling fan (complete, end item) | 1 | EA |
| 1 | CF-1100 | Motor assembly | 1 | EA |
| 2 | CF-1110 | Stator with winding | 1 | EA |
| 2 | CF-1120 | Rotor | 1 | EA |
| 2 | CF-1130 | Ball bearing | 2 | EA |
| 2 | CF-1140 | Motor housing (top and bottom) | 1 | SET |
| 2 | CF-1150 | Capacitor | 1 | EA |

| | | | | |
|---|---|---|---|---|
| 1 | CF-1200 | Blade assembly | 1 | EA |
| 2 | CF-1210 | Blade | 4 | EA |
| 2 | CF-1220 | Blade arm (bracket) | 4 | EA |
| 2 | CF-1230 | Screw M5 × 12 mm | 16 | EA |
| 1 | CF-1300 | Down-rod and canopy assembly | 1 | EA |
| 2 | CF-1310 | Down-rod | 1 | EA |
| 2 | CF-1320 | Canopy | 1 | EA |
| 2 | CF-1330 | Hanging bracket | 1 | EA |
| 1 | CF-1400 | Speed regulator (switch) | 1 | EA |
| 1 | CF-1500 | Wiring harness | 1 | EA |
| 1 | CF-1600 | Hardware and packing kit | 1 | KIT |

EA = each. The quantities are for one unit of the immediate parent, for example one blade assembly uses 4 blades, so one fan needs 1 × 4 = 4 blades. Total requirements for any production quantity follow by multiplying the usage quantities down the levels (MRP explosion).

| | | | | |
|---|---|---|---|---|
| **90** | Refer to the bill of materials for product A shown below. If there is no existing inventory and no scheduled receipts, how many units of items G, E and D must be purchased to produce five units of end item A?<br>  <br><br>Source: Unit IV \| CIE-II, 07/01/2025, Q4 | 10 | 4 | 3 |

The requirement of a component is found by multiplying the usage quantities along the path from the end item down to the component (the usage quantities are shown in parentheses in the BOM). With no inventory and no scheduled receipts, the requirements equal the gross requirements.

| Item | Calculation | Units needed |
|---|---|---|
| B | 3 per A × 5 units of A | 15 |
| C | 1 per A × 5 units of A | 5 |
| E | 2 per B × 15 units of B (= 2 × 3 × 5) | 30 |
| D (under B) | 1 per B × 15 units of B (= 1 × 3 × 5) | 15 |
| D (under C) | 1 per C × 5 units of C (= 1 × 1 × 5) | 5 |
| **D (total)** | D is used in both B and C: 15 + 5 | **20** |
| F | 1 per C × 5 units of C | 5 |
| **G** | 1 per F × 5 units of F (= 1 × 1 × 1 × 5) | **5** |

**Answer:** To produce 5 units of A, purchase 5 units of G, 30 units of E and 20 units of D.

| 91 | Explain the MRP explosion process with a neat example.<br>Source: Unit IV \| SEE Feb/Mar 2026 (IM352IA), Q8a; SEE Fast Track Oct 2020 (16IM55/12IM62), Q8c | 08 | 3 | 3 |
|---|---|---|---|---|

MRP **explodes** the master production schedule and the other sources of demand into the requirements of all the subassemblies, components and raw materials that the firm needs to produce the parent items. The explosion proceeds level by level down the bill of materials. The gross requirements of an item come from three sources: (1) the MPS for immediate parents that are end items; (2) the **planned order releases** (not the gross requirements, scheduled receipts or planned receipts) of the immediate parents below the MPS level; (3) any other requirements not originating in the MPS, such as replacement parts.

For each item the inventory record is completed with the formulas

$$\text{Projected on-hand}_t = \text{Projected on-hand}_{t-1} + \text{Scheduled or planned receipts}_t - \text{Gross requirements}_t$$

A planned receipt is scheduled in the week in which the projected on-hand inventory would become negative (size from the lot-sizing rule), and the planned order release is the receipt week minus the lead time.

### Example (textbook): seat subassembly C and its components

Seat subassembly C (one seat frame H, one seat cushion I; the frame needs four seat-frame boards J). Records: C: FOQ 230, lead time 2 weeks, 37 on hand, scheduled receipt of 230 in week 1, gross requirements from the MPS of the chairs; seat frame: FOQ 300, 1 week, 40 on hand, scheduled receipt of 300 in week 2; seat cushion: lot-for-lot, 1 week, none on hand; seat-frame boards: FOQ 1,500, 1 week, 200 on hand.

| **Item: C seat subassembly (FOQ 230, lead time 2 weeks)** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** |
|---|---|---|---|---|---|---|---|---|
| Gross requirements | 150 | | | 120 | | 150 | 120 | |
| Scheduled receipts | 230 | | | | | | | |
| Projected on-hand inventory | 117 | 117 | 117 | 227 | 227 | 77 | 187 | 187 |
| Planned receipts | | | | 230 | | | 230 | |
| Planned order releases | | 230 | | | 230 | | | |

The planned order releases of C (230 units in weeks 2 and 5) become the **gross requirements** of the seat frame and the seat cushion.

| **Item: H seat frame (FOQ 300, lead time 1 week)** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** |
|---|---|---|---|---|---|---|---|---|
| Gross requirements | | 230 | | | 230 | | | |
| Scheduled receipts | | 300 | | | | | | |
| Projected on-hand inventory | 40 | 110 | 110 | 110 | 180 | 180 | 180 | 180 |
| Planned receipts | | | | | 300 | | | |
| Planned order releases | | | | 300 | | | | |

| **Item: I seat cushion (L4L, lead time 1 week)** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** |
|---|---|---|---|---|---|---|---|---|
| Gross requirements | | 230 | | | 230 | | | |
| Scheduled receipts | | | | | | | | |
| Projected on-hand inventory | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Planned receipts | | 230 | | | 230 | | | |
| Planned order releases | 230 | | | 230 | | | | |

The seat frame's planned order release of 300 in week 4 gives the gross requirement of the seat-frame boards: $4 \times 300 = 1{,}200$ boards in week 4.

| **Item: J seat-frame boards (FOQ 1,500, lead time 1 week)** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** |
|---|---|---|---|---|---|---|---|---|
| Gross requirements | | | | 1200 | | | | |
| Scheduled receipts | | | | | | | | |
| Projected on-hand inventory | 200 | 200 | 200 | 500 | 500 | 500 | 500 | 500 |
| Planned receipts | | | | 1500 | | | | |
| Planned order releases | | | 1500 | | | | | |

**Answer:** Orders to be released: 300 seat frames in week 4, seat cushions of 230 units in weeks 1 and 4, and 1,500 seat-frame boards in week 3 (the seat subassembly orders of 230 in weeks 2 and 5 drive these requirements). If the plan is updated weekly, only the order due in the current week is actually released; the later ones can be revised in the next explosion.

| | | | | |
|---|---|---|---|---|
| **92** | What is meant by a lot-sizing rule in the context of MRP explosion? Differentiate between the fixed order quantity rule and the periodic order quantity rule.<br>Source: Unit IV \| SEE Feb/Mar 2025 (IM352IA), Q8b | 06 | 2 | 3 |

A **lot-sizing rule** determines the timing and the size of the order quantities in the inventory record. A rule must be assigned to each item before the planned receipts and planned order releases can be computed. It matters because it determines the number of setups (orders) and the inventory holding cost of the item. Three rules are used: fixed order quantity (FOQ), periodic order quantity (POQ) and lot-for-lot (L4L).

| Basis | Fixed order quantity (FOQ) | Periodic order quantity (POQ) |
|---|---|---|
| Rule | The same order quantity is issued each time an order is placed (set by equipment capacity, truckload, quantity discount, minimum purchase quantity or the EOQ). | A different order quantity for each order, but orders are issued at predetermined time intervals of P periods (for example every 3 weeks). |
| Order size | Fixed; if one week's gross requirement is larger than the FOQ, the lot is increased (a multiple of the FOQ). | Total gross requirements of P weeks less the projected on-hand inventory at the end of the previous week, i.e. exactly P weeks' requirement. |
| Timing | Irregular: whenever the projected inventory would become negative | Planned for the week of shortage; the lot covers the following P weeks |
| Inventory | Creates remnants and a high average inventory | Lower average inventory because the lot matches requirements |
| Example (seat subassembly, weeks 4 to 8) | Average on-hand = 181 units | Average on-hand (P = 3) = 60 units |

Lot-for-lot is the special case of POQ with P = 1; its average inventory is 0 units, but it places the largest number of orders (setups).

| | | | | |
|---|---|---|---|---|
| **93** | The MPS start quantities for product A call for the assembly department to begin final assembly according to the following schedule: 100 units in week 2; 200 units in week 4; 120 units in week 6; 180 units in week 7; and 60 units in week 8. Develop a material requirements plan for the next 8 weeks for items B, C and D. The BOM for A is: A (lead time 2 weeks) is made from 1 B (lead time 1 week) and 2 C (lead time 2 weeks); each C is made from 1 D (lead time 3 weeks). The inventory records are: B: lot-sizing rule POQ (P = 3), no scheduled receipts, 20 units on hand; C: lot-for-lot, scheduled receipt of 200 units in week 1, none on hand; D: fixed order quantity of 500 units, no scheduled receipts, 425 units on hand.<br>Source: Unit IV \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q8b | 08 | – | – |

The MPS start quantities of A are the planned order releases of A, so they give the gross requirements of its components in the same weeks: B needs 1 per A and C needs 2 per A. The gross requirements of D come from the **planned order releases of C** (1 D per C).

### Item C (lot-for-lot, lead time 2 weeks, scheduled receipt 200 in week 1)

| Item: C (L4L, lead time 2 weeks) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Gross requirements | | 200 | | 400 | | 240 | 360 | 120 |
| Scheduled receipts | 200 | | | | | | | |
| Projected on-hand inventory | 200 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Planned receipts | | | | 400 | | 240 | 360 | 120 |
| Planned order releases | | 400 | | 240 | 360 | 120 | | |

### Item B (POQ with P = 3 weeks, lead time 1 week, 20 on hand)

The first shortage occurs in week 2 (20 − 100 = −80). The POQ lot is the gross requirements of weeks 2 to 4 less the inventory at the end of week 1: (100 + 0 + 200) − 20 = 280 units. The next shortage is in week 6: (120 + 180 + 60) − 0 = 360 units.

| Item: B (POQ, P = 3, lead time 1 week) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Gross requirements | | 100 | | 200 | | 120 | 180 | 60 |
| Scheduled receipts | | | | | | | | |
| Projected on-hand inventory | 20 | 200 | 200 | 0 | 0 | 240 | 60 | 0 |
| Planned receipts | | 280 | | | | 360 | | |
| Planned order releases | 280 | | | | 360 | | | |

### Item D (FOQ 500, lead time 3 weeks, 425 on hand)

| Item: D (FOQ 500, lead time 3 weeks) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Gross requirements | | 400 | | 240 | 360 | 120 | | |
| Scheduled receipts | | | | | | | | |
| Projected on-hand inventory | 425 | 25 | 25 | 285 | 425 | 305 | 305 | 305 |
| Planned receipts | | | | 500 | 500 | | | |
| Planned order releases | 500 | 500 | | | | | | |

**Answer:** Planned order releases: B: 280 in week 1, 360 in week 5; C: 400 in week 2, 240 in week 4, 360 in week 5, 120 in week 6; D: 500 in week 1, 500 in week 2.

| 94 | What are the key challenges associated with ERP implementation? How can organizations mitigate these challenges?<br>Source: Unit IV \| SEE Apr/May 2024 (21IM53), Q7b | 08 | 4 | 4 |
|---|---|---|---|---|

**Enterprise resource planning (ERP)** systems are large integrated information systems that support many enterprise processes and data storage needs. They revolve around a single comprehensive database that is available across the enterprise; a transaction entered in one application automatically updates the related information in the other applications (manufacturing, supply chain, accounting and finance, human resources, sales and marketing, customer service).

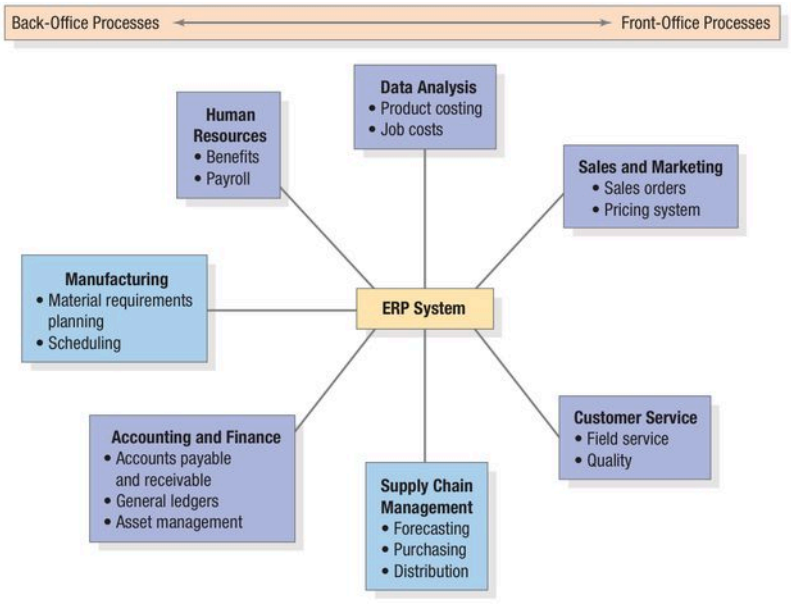



**Figure 40:** ERP application modules

| Key challenge | How organizations can mitigate it |
|---|---|
| **High cost and complexity.** Firms can spend excessive amounts on ERP systems that are complex to use and costly to manage, especially when they customize the software extensively. | Keep the implementation simple, work with a small number of software vendors and use standardized systems rather than customizing them extensively (the greatest rewards go to such firms). |

| | |
|---|---|
| **Process redesign.** Designing an ERP system requires careful analysis of the major processes; redundancies and convoluted information flows must be reengineered before the firm can enjoy the benefits of an integrated system. | Analyze and, where necessary, completely reengineer the processes before or during the implementation; decide which applications need to be integrated. |
| **Legacy systems.** Older systems must be coordinated with the new software; applications left out do not share information with the ERP system. | Design the ERP to interface with the existing legacy systems, and decide carefully which applications to integrate. |
| **Training and roll-out risk.** Users must learn the screens and menus, and errors in the roll-out can disrupt operations (Cadbury suffered a build-up of excess inventory of chocolate bars and took a 12-million-pound hit to profits after the roll-out of a new SAP-based ERP system). | Train the users on the screens they need to do their jobs, and use a **sequenced implementation** of the modules (as Dow Corning did with SAP, linking key processes from order generation to production planning, warehousing, delivery and billing). |

Companies that manage these challenges gain a view of their operations as a whole, real-time operating information across functions, streamlined data flows and better coordination of plants and processes with market requirements.

| | | | | |
|---|---|---|---|---|
| **95** | Evaluate the resource planning approaches for service providers.<br>Source: Unit IV \| SEE Feb/Mar 2026 (IM352IA), Q7a | 08 | 4 | 5 |

Service providers must plan their resources just as manufacturers do. However, unlike finished goods, **services cannot be inventoried**; they must be provided on demand, so resource planning for services focuses on maintaining the **capacity** to serve the customers. Two ideas of MRP are applied:

### 1. Dependent demand for services

Dependent demand is extended to include the demand for resources that is driven by the forecasts of customer requests for services or by plans for activities in support of the services. Examples:

- **Restaurant:** each meal ordered creates a need for food items, plates, staff (chef, servers) and equipment; forecasts of the demand for each type of meal give the needs; daily specials improve the forecasts.
- **Airline:** each scheduled flight needs beverages, snacks, fuel, labor (pilots, attendants) and equipment (plane, gate); the airline explodes its master schedule of flights.
- **Hospital:** admission appointments form a master schedule that is exploded into medicines, linens, staff and equipment (operating room, tools), while ensuring that key equipment and personnel are not overcommitted.
- **Hotel:** reservations plus forecast walk-ins give the master schedule; the number of rooms cannot be adjusted easily, so hotels try to raise utilization by group rates and promotions.

### 2. Bill of resources (BOR)

The service analogy to the bill of materials is the **bill of resources**, a record of the service firm's parent–component relationships and of all the materials, equipment time, staff and other resources associated with them, including the usage quantities. A BOR can be as complex as a manufacturer's BOM (for example, the treatment of an aneurysm has seven levels from preoperative testing to discharge with a lead time of 12.2 days). After the master schedule is complete, the BOR tells what resources are needed, how much and when. Cash is also planned through the master schedule and accounts receivable and payable.

**Evaluation**

| Strengths | Limitations |
|---|---|
| The master schedule (appointments, reservations, flights) allows resource needs to be exploded in advance, as in manufacturing. | Services cannot be stocked, so any mismatch of resources shows up at once as idle capacity or poor service. |
| The BOR gives time-phased requirements of staff, equipment and supplies, which supports workforce scheduling and cash planning. | Key resources (such as rooms and specialized staff) are fixed in the short term, and overbooking or no-shows disturb the plan. |
| Reports analogous to MRP II can be given to the functional managers (finance, purchasing, staffing). | Forecasts of customer requests are uncertain and a BOR can be as complex as a manufacturer's BOM. |

| | | | | |
|---|---|---|---|---|
| **96** | Consider the master flight schedule of a major airline, such as Air New Zealand. Discuss the ways in which it is analogous to a master production schedule for a manufacturer.<br>Source: Unit IV \| SEE Jan/Feb 2021 (18IM54), Q7b | 06 | – | – |

The **master production schedule (MPS)** details how many end items will be produced within specified periods of time. It breaks the sales and operations plan into specific product schedules, and the MRP system explodes it into the requirements of the components and raw materials. The airline's **master schedule of flights** does the same job for a service provider:

| Feature | Manufacturer's MPS | Airline's master flight schedule |
|---|---|---|
| What is scheduled | End items (products) by period | Flights (routes, departure and arrival times) by day and week |
| Derived from | Sales and operations plan, forecasts and booked customer orders | The airline's plan of services, forecast passengers and booked reservations |
| Explosion | Bill of materials gives components, subassemblies and raw materials | Bill of resources: each flight needs fuel, food and drink, pilots and flight attendants, an aircraft and an airport gate; the airline can explode its master schedule to determine them |
| Planning information | Available-to-promise tells marketing what can be promised | Seats available can be promised to new bookings (reservations) |
| Resource limits | The MPS must respect machine, labor, storage and capital capacities; it is revised until feasible | Aircraft, crews and gates are limited; crews cannot be overcommitted |
| Stability | The near-term portion is frozen because changes are costly | Flights are fixed in the short term; last-minute changes (weather, breakdowns) are costly |
| Control | Authorized MPS drives plant and supplier activity | The schedule drives crew rosters, maintenance and airport services |

As in manufacturing, the quality of the plan decides how well capacity is used: expensive equipment such as an aircraft earns nothing when it is idle, and on-time schedule performance is critical for customer satisfaction.

| 97 | Briefly discuss the different types of operations planning and scheduling across the organization.<br>Source: Unit V \| SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q8a | 08 | – | – |
|---|---|---|---|---|

Operations planning and scheduling is the process of **balancing supply with demand**, from the aggregate level down to the short-term scheduling level. It begins with plans that take a macro (big-picture) view, aggregated over product families, workforce and time, and ends with detailed schedules ready for immediate action. The types of plans are:

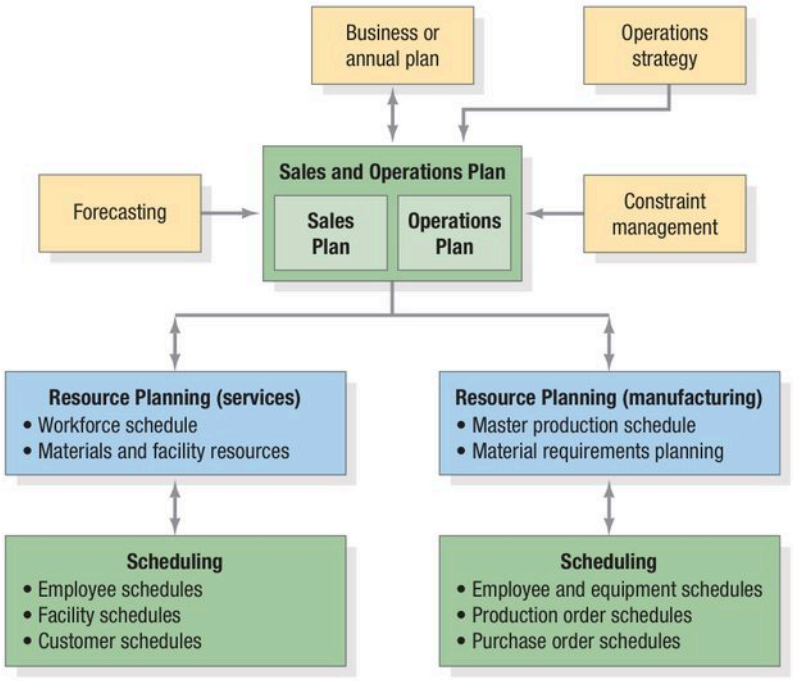



**Figure 41:** The relationship of sales and operations plans and schedules to other plans

| Type of plan | Description |
|---|---|
| **Business plan** (for-profit firms) or **annual plan** (not-for-profit organizations) | A financial assessment of the near future (1 or 2 years): a projected statement of income, costs and profits, with budgets, a projected balance sheet and cash-flow statement. It unifies the plans of operations, finance, sales and marketing and gives the direction and objectives for the sales and operations plan. |
| **Sales and operations plan (S&OP)** | A plan of future aggregate resource levels so that supply is in balance with demand; it states production rates, workforce levels and inventory holdings consistent with the demand forecasts and capacity constraints. It is time-phased (months or quarters, about one year or more). |
| **Production plan** (manufacturing) | The S&OP of a manufacturing firm, centred on production rates and inventory holdings by product family. |
| **Staffing plan** (services) | The S&OP of a service firm, centred on staffing and other human-resource-related factors (number and types of employees). |
| **Resource plan** | An intermediate step between the S&OP and scheduling; it determines the requirements for materials and other resources at a more detailed level: for manufacturers the master production schedule and material requirements planning, for services the workforce schedule and the materials and facility resources. |
| **Schedule** | The lowest level: a detailed plan that allocates resources over shorter time horizons to accomplish specific tasks, such as employee schedules, facility schedules and customer schedules (services), and employee and equipment schedules, production-order and purchase-order schedules (manufacturing). |

Information flows both ways: top-down (broad to detailed) and bottom-up (detailed to broad). If an S&OP cannot be developed to meet the business plan with the existing resources, the business plan may need adjustment; if a feasible master production schedule or workforce schedule cannot be developed, the S&OP may need adjustment. The process is dynamic, with periodic (usually monthly) revisions.

**Aggregation:** planning begins at the aggregate level, along three dimensions: services or products (product families), workforce (a single aggregate group when workers are flexible) and time (months, quarters or seasons).

| | | | | |
|---|---|---|---|---|
| **98** | Represent in the form of a block diagram the managerial inputs that go into developing the sales and operations plan, from the functional areas of operations, distribution and marketing, materials, accounting and finance, engineering and human resources. Discuss the supply options for operations planning and scheduling.<br>Source: Unit V \| SEE Fast Track Jul 2019 (16IM55/12IM62), Q8a | 08 | – | – |

### Managerial inputs to the sales and operations plan

Consensus among the firm's departments is needed so that the plan is a good one and also doable. Inputs are sought from each functional area:

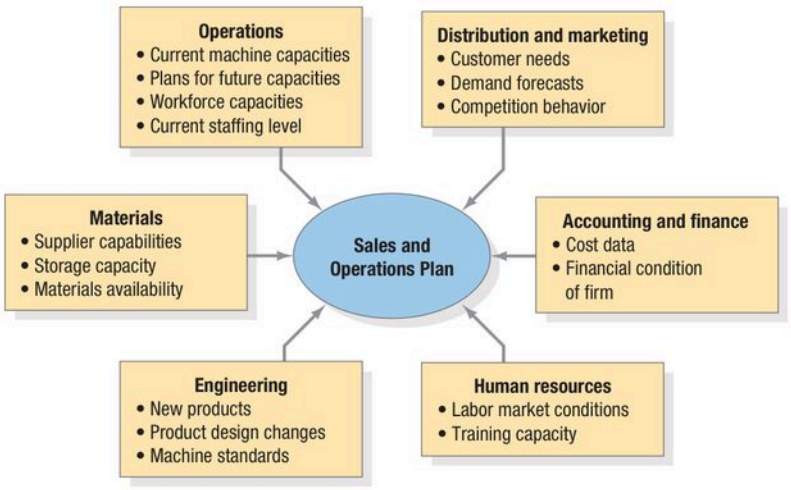



**Figure 42:** Managerial inputs from functional areas to sales and operations plans

### Supply options for operations planning and scheduling

Given the demand forecasts (as modified by demand management), operations managers develop a plan to meet the demand from the following supply options:

| Supply option | Explanation |
|---|---|
| **Anticipation inventory** | Stock built in light-demand periods to absorb uneven demand (for example air conditioners); costly to hold, not possible for customized products, and services cannot be stocked. |
| **Workforce adjustment** (hiring and layoffs) | Attractive when the labor is unskilled or semi-skilled and the labor pool is large; limited by the training capacity and union contracts; costs of hiring and layoff. |
| **Workforce utilization** (overtime and undertime) | Overtime (typically 150% of the regular pay) supplies output that cannot be completed in regular time; undertime (paid or unpaid) is the idle regular-time capacity when demand is low. |
| **Part-time workers** | Paid only for the hours worked at the peak times; no fringe benefits; common for low-skill jobs. |
| **Subcontractors** | Overcome short-term capacity shortages in services, components, subassemblies or whole products. |
| **Vacation planning (shutdown)** | Slack periods are used for vacations or plant shutdowns, and vacations are discouraged in peak periods. |
| **Backorders and stockouts** | A last resort of lowering service: a back order is filled later, a stockout loses the order. |

The **demand options** that even out the load are complementary products, promotional pricing, appointments and reservations, revenue management, backlogs, backorders and stockouts. Two basic strategies (chase and level) are starting points, and the best plan is usually a mixed strategy that uses all the supply options.

| | | | | |
|---|---|---|---|---|
| **99** | Clearly bring out the distinction between the two basic planning strategies of chase strategy and level strategy.<br>Source: Unit V \| SEE Fast Track Oct 2020 (16IM55/12IM62), Q8a | 04 | – | – |

| **Basis** | **Chase strategy** | **Level strategy** |
|---|---|---|
| Meaning | Hiring and laying off employees to match the demand forecast over the planning horizon. | Keeping the workforce constant (except possibly at the start of the horizon) and varying its utilization to match the demand forecast. |
| Supply options used | Hiring and layoffs (regular-time capacity varies with demand). | Overtime, undertime (paid or unpaid) and vacation planning. |
| Inventory | No inventory investment, overtime or undertime is needed. | Workforce size can be chosen large (little overtime, much undertime) or small (much overtime in the peaks). |
| Drawbacks | Cost of continually adjusting the workforce level, potential alienation of the workforce, and loss of productivity and quality due to constant change. | Overtime is expensive and strains the workforce and quality; undertime wastes paid time. |
| Example | A distribution center staffing part-time pickers exactly to the forecast, with many hirings and layoffs (cost $173,500 in the textbook example). | The same center with a constant staff of 15 part-time workers, using overtime in the peak (cost $164,000). |

The two pure strategies used alone usually do not give the best plan. The best is usually a **mixed strategy** that considers the full range of supply options (anticipation inventory, part-time workers, subcontractors, backorders and stockouts).

| | | | | |
|---|---|---|---|---|
| **100** | The Mowry machine shop still has 5 jobs to be processed as of 8 AM today (day 23) at its bottleneck operation. The day when the order arrived, the processing time and the promised due dates are given in the following table. The jobs are listed in the order of arrival. | 08 | – | – |

| Job | Time since order arrived (days ago) | Processing time (days) | Due date (days from now) |
|---|---|---|---|
| A | 13 | 8 | 36 |
| B | 15 | 4 | 42 |
| C | 17 | 4 | 39 |
| D | 22 | 3 | 53 |
| E | 24 | 10 | 45 |

(i) Develop separate schedules by using the FCFS and EDD rules. Compare the schedules on the basis of average flow time and average days past due. (ii) Which rule gives the best schedule, in your judgment? Why?

Source: Unit V | SEE Fast Track Jul 2019 (16IM55/12IM62), Q8b

**FCFS** (first-come, first-served): the job that arrived first (the largest number of days since arrival) is processed first. **EDD** (earliest due date): the job with the earliest due date is processed next. For each job (all available at time 0):

$$\text{Finish time} = \text{Start time} + \text{Processing time}, \quad \text{Flow time} = \text{Finish time} + \text{Time since the order arrived}, \quad \text{Days past due} = \max(0, \text{Finish} - \text{Due date})$$

**(i) FCFS schedule: E – D – C – B – A**

| Job | Start | Processing time | Finish | Due date | Days past due | Days since arrival | Flow time |
|---|---|---|---|---|---|---|---|
| E | 0 | 10 | 10 | 45 | 0 | 24 | 34 |
| D | 10 | 3 | 13 | 53 | 0 | 22 | 35 |
| C | 13 | 4 | 17 | 39 | 0 | 17 | 34 |
| B | 17 | 4 | 21 | 42 | 0 | 15 | 36 |
| A | 21 | 8 | 29 | 36 | 0 | 13 | 42 |

$$\text{Average flow time} = \frac{181}{5} = 36.2 \text{ days}$$

$$\text{Average days past due} = \frac{0}{5} = 0.0 \text{ days}$$

**EDD schedule: A – C – B – E – D**

| Job | Start | Processing time | Finish | Due date | Days past due | Days since arrival | Flow time |
|---|---|---|---|---|---|---|---|
| A | 0 | 8 | 8 | 36 | 0 | 13 | 21 |
| C | 8 | 4 | 12 | 39 | 0 | 17 | 29 |
| B | 12 | 4 | 16 | 42 | 0 | 15 | 31 |
| E | 16 | 10 | 26 | 45 | 0 | 24 | 50 |
| D | 26 | 3 | 29 | 53 | 0 | 22 | 51 |

$$\text{Average flow time} = \frac{182}{5} = 36.4 \text{ days}$$

$$\text{Average days past due} = \frac{0}{5} = 0.0 \text{ days}$$

| Performance measure | FCFS | EDD |
|---|---|---|
| Average flow time (days) | 36.2 | 36.4 |
| Average days past due | 0.0 | 0.0 |

**(ii) Which rule is better?**

Neither schedule makes any job late, so the average past due is zero for both rules. FCFS gives the slightly lower average flow time (36.2 against 36.4 days), and it is also the fairer rule, since every job is treated equally. EDD does not improve either measure for this data set, because the jobs that arrived first also happen to have the earliest finish requirements. Neither rule guarantees an optimal schedule; a schedule that does well on one measure may do poorly on another. If the average flow time is the most important measure, the shortest processing time (SPT) rule gives an even lower value (32.0 days, with no job past due), but it delays the long job E, which has been waiting the longest.

**Answer:** FCFS (sequence E D C B A): average flow time 36.2 days, average days past due 0.0. EDD (sequence A C B E D): average flow time 36.4 days, average days past due 0.0. FCFS is judged the better schedule, since it has the lower flow time and no job is late.

**101** A company manufactures wooden decors. Management schedules overtime every weekend to reduce the backlogs on the most popular models. The automatic routing machine is used to cut certain types of edges on the desktops. The following orders need to be scheduled for the routing machine. | 08 | – | –

| Order | Time since order arrived (hours ago) | Estimated machine time (hours) | Due date (hours from now) |
|---|---|---|---|
| 1 | 1 | 9 | 20 |
| 2 | 0 | 7 | 21 |
| 3 | 6 | 8 | 12 |
| 4 | 5 | 3 | 8 |
| 5 | 3 | 12 | 18 |

The due dates reflect the needs of the order to be at its next operation. (i) Develop separate schedules by using the FCFS and EDD rules. Compare the schedules on the basis of average flow time and average past-due hours. (ii) Comment on the performance of these two rules relative to these measures.

Source: Unit V | SEE Fast Track Oct 2020 (16IM55/12IM62), Q8b

Flow time = finish time + hours since the order arrived; past due hours = max(0, finish time − due date). All the orders are available for processing at time 0.

**(i) FCFS schedule: orders 3 – 4 – 5 – 1 – 2**

| Order | Start | Machine time | Finish | Due date | Hours past due | Hours since arrival | Flow time |
|---|---|---|---|---|---|---|---|
| 3 | 0 | 8 | 8 | 12 | 0 | 6 | 14 |
| 4 | 8 | 3 | 11 | 8 | 3 | 5 | 16 |
| 5 | 11 | 12 | 23 | 18 | 5 | 3 | 26 |
| 1 | 23 | 9 | 32 | 20 | 12 | 1 | 33 |
| 2 | 32 | 7 | 39 | 21 | 18 | 0 | 39 |

$$\text{Average flow time} = \frac{128}{5} = 25.6 \text{ hours}$$

$$\text{Average past due} = \frac{38}{5} = 7.6 \text{ hours}$$

**EDD schedule: orders 4 – 3 – 5 – 1 – 2**

| Order | Start | Machine time | Finish | Due date | Hours past due | Hours since arrival | Flow time |
|---|---|---|---|---|---|---|---|
| 4 | 0 | 3 | 3 | 8 | 0 | 5 | 8 |
| 3 | 3 | 8 | 11 | 12 | 0 | 6 | 17 |
| 5 | 11 | 12 | 23 | 18 | 5 | 3 | 26 |
| 1 | 23 | 9 | 32 | 20 | 12 | 1 | 33 |
| 2 | 32 | 7 | 39 | 21 | 18 | 0 | 39 |

$$\text{Average flow time} = \frac{123}{5} = 24.6 \text{ hours}$$

$$\text{Average past due} = \frac{35}{5} = 7.0 \text{ hours}$$

| Performance measure | FCFS | EDD |
|---|---|---|
| Average flow time (hours) | 25.6 | 24.6 |
| Average past due (hours) | 7.6 | 7.0 |

**(ii) Comment**

The EDD rule performs better than FCFS on both measures: its average flow time is 24.6 hours against 25.6 hours, and the average past-due time is 7.0 hours against 7.6 hours. EDD gives priority to the orders that are needed soonest at their next operation, so order 4 (due in 8 hours) is no longer late and order 3 meets its due date, while FCFS processes the orders

in the order of arrival regardless of how urgent they are. Even so, three orders (5, 1 and 2) are still past due under both rules because the total load (39 hours) is large compared with the due dates; managers would use overtime or expediting, and, as the textbook points out, neither rule guarantees an optimal schedule and a schedule that does well on one measure may do poorly on another.

**Answer:** FCFS: average flow time 25.6 h and average past due 7.6 h. EDD: average flow time 24.6 h and average past due 7.0 h. EDD is better on both measures.

| | | | | |
|---|---|---|---|---|
| **102** | Consider the following single machine scheduling problem.<br>Job $i$: 1 \| 2 \| 3 \| 4 \| 5<br>Processing time $t_j$: 15 \| 4 \| 5 \| 14 \| 8<br>Find the optimal sequence which will minimize the mean flow time and also obtain the mean flow time.<br>Source: Unit V \| SEE Feb/Mar 2025 (IM352IA), Q9a | 10 | 3 | 4 |

For jobs at a single workstation, all available at time 0, the flow time of a job equals its completion time. The shortest processing time (SPT) rule, which processes the job requiring the shortest processing time next, **minimizes the mean flow time**. The jobs are therefore arranged in increasing order of processing time.

| Position | Job | Processing time | Completion (flow) time |
|---|---|---|---|
| 1 | 2 | 4 | 4 |
| 2 | 3 | 5 | 9 |
| 3 | 5 | 8 | 17 |
| 4 | 4 | 14 | 31 |
| 5 | 1 | 15 | 46 |
| | | **Total** | **107** |

$$\text{Mean flow time} = \frac{\sum \text{flow times}}{n} = \frac{107}{5} = 21.4$$

For comparison, the job order 1-2-3-4-5 gives a total flow time of 142 (mean 28.4), which is larger.

**Answer:** Optimal sequence (SPT): 2 – 3 – 5 – 4 – 1; the mean flow time is 21.4 time units.

* * *